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Breathing and Exchange of Gases

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1. How many of the following statements are correct?

(a) O_2 is utilised by the organisms to indirectly break down nutrient molecules.

(b) Process of exchange of CO_2 from the atmosphere with O_2 produced by the cells is called breathing.

(c) Mammals have an ill developed respiratory system

(d) We have a pair of external nostrils, which opens out side above the upper lips.

(1) Only one (2) Only two

(3) Three (4) All are correct

BBG249

2. Match the columns and choose the correct answer from the options given below :–

Column-I (Animal)		Column II (Respiratory organ)	
A	Sponge	i	Lungs
B	Insects	ii	Gills
C	Aquatic arthropoda	iii	Absent
D	Fish	iv	Trachea
E	Mammal		

(1) A-i B-ii, C-iii D-iv E-i

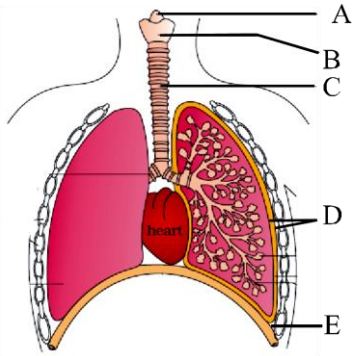
(2) A-iii B-iv, C-i D-ii E-i

(3) A-iii B-iv, C-ii D-ii E-i

(4) A-iii B-ii, C-iv D-i E-ii

BBG250

3. Select the wrong statement about the diagram given below :–



(1) 'B' is a cartilaginous box which helps in sound production.

(2) 'A' prevent the entry of food into the larynx.

(3) 'C' is a straight tube extending upto the left thoracic cavity, which divides at the level of 5th thoracic vertebrae.

(4) 'D' is double layered and a fluid is present in between them

BBG251

4. Arrange the following steps of respiration in a proper sequence :–

(a) Breathing

(b) Diffusion of gases across alveolar membrane

(c) Transport of gases by the blood

(d) Diffusion of O_2 and CO_2 between blood and tissues

(e) Uitilisation of O_2 by the cells for catabolic reactions and resultant release of CO_2

(1) a, b, c, d, e

(2) a, e, b, c, d

(3) a, c, b, d, e

(4) a, b, d, e, c

BBG252

5. Which of the following is not a function of conducting part of respiratory tract ?

(1) Transport of atmospheric air to alveoli

(2) Humidification

(3) Brings the air to body temperature

(4) Site of actual diffusion of O_2 and CO_2 between blood and atmospheric air

BBG253

6. Which of the following is not a property of alveoli ?

(1) irregular thin walled

(2) highly vascularised

(3) supported by incomplete rings

(4) enormous in number

BBG254

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7. If O_2 concentration in tissue is almost as high as at the respiratory surface :-
(1) Oxyhaemoglobin would dissociate to supply O_2 to the tissue
(2) Haemoglobin would combine with more O_2 at respiratory surface
(3) Oxyhaemoglobin would not dissociate to supply O_2 to the tissue
(4) CO_2 will interfere the O_2 transport

BBG255

8. The blood leaving the lungs has all its haemoglobin oxygenated and gives up O_2 to the tissue because :-
(1) Tissue can absorb O_2 from oxyhaemoglobin
(2) O_2 concentration is lower at tissue level than in the lungs
(3) Oxyhaemoglobin undergoes reduction
(4) None of these

BBG256

9. Which is wrong with respect to expiration ?
(1) Ribs and sternum to original position
(2) Air expelled from lungs
(3) Volume of thorax decrease
(4) Diaphragm contracted

BBG257

10. Diffusion of gases occurs in the alveolar region only and not in the other parts of respiratory system. Why ?
(1) Alveoli are very few in number, very thin, irregular walled and highly vascularised
(2) Alveoli are very large in number, very thin, irregular walled and poorly vascularised
(3) Alveoli are very large in number, very thin, irregular walled and highly vascularised
(4) Alveoli are very large in number, very thick regular walled and poorly vascularised.

BBG258

11. The thoracic chamber is formed dorsally by (A) ventrally by (B), laterally by (C) and on the lower side by (D).
Identify (A), (B), (C) and (D) structures in the above para :–

	A	B	C	D
(1)	sternum	ribs	ribs	vertebral column
(2)	vertebral column	sternum	ribs	dome shaped diaphragm
(3)	vertebral column	ribs	sternum	dome shaped diaphragm
(4)	sternum	vertebral column	ribs	dome shaped diaphragm

BBG259

12. Which of the following represents the correct combinations ?

	A	B
(1)	Exchange part of respiratory tract	Brings the air to the body temperature
(2)	Conducting part of respiratory tract	Site of actual diffusion of O_2 and CO_2
(3)	Inspiration	initiated by contraction of diaphragm which increases volume of thoracic chamber in the antero-posterior axis
(4)	Lungs	are situated in the abdominal chamber which is anatomically an air tight chamber

BBG260

13. If the CO_2 concentration in the blood increases, the breathing rate will :-
(1) Increase (2) Decrease
(3) Stop (4) No affect

BBG261

14. How O_2 and CO_2 get transported through blood?
(1) With the help of RBCs and blood plasma
(2) With the help of RBCs and WBCs
(3) With the help of WBCs and blood serum
(4) With the help of platelets and corpuscles

BBG262

15. Hb most strongly combines with :-
(1) CO_2 (2) CO
(3) O_2 (4) O_3

BBG263

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16. Quantity of CO_2 transported in the form of bicarbonates by blood is around :-
(1) 70% (2) 85%
(3) 2% (4) 98%
BBG264

17. If total O_2 transported by blood from lungs to cell in a minute is 1000 ml. Then the volume of O_2 transported by blood found dissolved in plasma is
(1) 970 ml (2) 30 ml
(3) 1000 ml (4) Nil
BBG265

18. Increase CO_2 & temperature will cause the O_2/Hb. dissociation curve to shift :-
(1) Towards left
(2) Towards right
(3) Become more sigmoid
(4) Flattening of curve
BBG266

19. Maximum CO_2 is transported from tissues to lungs in the form of :–
(1) Dissolved form in plasma.
(2) Carbamino compounds.
(3) Bicarbonates.
(4) Carboxyhaemoglobin.
BBG267

20. Which of the following does not occur during transport of CO_2 in man ?
(1) Formation of Carboxy haemoglobin
(2) Chloride shift.
(3) Formation of carbonic acid.
(4) Haldane effect.
BBG268

21. If the addition of a particular substances in the blood of a person actually causes a rise in the capacity of the person haemoglobin to combine with oxygen. Then on which side the O_2/Hb curve will shift and what the substance possibly can be –
(1) Rightwards, CO_2
(2) Leftwards, fetal Haemoglobin
(3) Upwards, sodium
(4) Rightwards, 2,3, DPG
BBG269

22. The chloride content of RBC will be higher than that of plasma in –
(1) Systemic arteries and pulmonary veins
(2) Systemic veins and pulmonary arteries
(3) Systemic and pulmonary arteries
(4) Systemic and pulmonary veins
BBG270

23. Under a given oxygen concentration in blood, dissociation of oxyhaemoglobin will increase if:
(1) pH of blood rises
(2) pH of blood falls
(3) CO_2 concentration is blood falls
(4) Free fatty acid concentration in blood falls
BBG271

24. Find out the amount of O_2 which is delivered by 100 ml of oxygenated blood to the tissue under normal physiological conditions -
(1) 50 ml (2) 20 ml (3) 5 ml (4) 25 ml
BBG272

25. Which of the following statement is incorrect?
(1) Nearly 20-25 percent of CO_2 is transported by RBC's.
(2) About 3 percent of CO_2 is carried in a dissolved state through plasma.
(3) About 77 percent of CO_2 is transported by plasma.
(4) About 97 percent of O_2 is transported by RBC's.
BBG273

26. The nerve impulses which stimulate the intercostal muscles and diaphragm to permit breathing, originate in the :-
(1) Cerebellum
(2) Pons
(3) Hypothalamus
(4) Medulla oblongata
BBG274

27. What is true about regulation of respiration -
(1) Respiratory rhythm centre lie on pons region.
(2) The role of oxygen in the regulation of respiratory rhythm is quite insignificant.
(3) Chemo-sensitive area can recognise changes in O_2
(4) Neural signals from cerebellum can increase the duration of inspiration.
BBG275

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28. Identify false statement :-
(1) Dead space air is 150 ml
(2) Air filled in bronchial tree is the dead space air
(3) Bronchial tree collapses during expiration
(4) Respiratory tree is inflated during inspiration

BBG276

29. Tidal air in mammalian lungs is :-
(1) The total maximum air that can be driven into-the lungs
(2) The air that normally goes in and comes out of the lungs during breathing
(3) The air that is left in the lungs after normal expiration
(4) The air that can be expelled out forcibly after the normal expiration

BBG277

30. In "emphysema" :-
(1) Trachea is constricted
(2) Diaphragm and costal muscles do not contract
(3) Many alveoli collapse together to form large chambers because of the destruction of alveolar wall
(4) O_2 absorption decreases with haemoglobin

BBG278

31. What is the total vital capacity of man :-
(1) About 4500 cc
(2) About 5500 cc
(3) About 2000 cc
(4) About 2200 cc

BBG279

32. Choose the correct equation :-
(1) Total capacity of lungs=Vital capacity + Tidal air
(2) Total capacity of lungs = Vital capacity + Residual air
(3) Vital capacity of lungs = Tidal air + complementary air
(4) Total capacity of lungs = Tidal air + Complementary air + Supplementary air

BBG280

33. Dead space air is :-
(1) The amount of air remaining in the alveoli
(2) The amount of air left behind in lungs at the end of deep expiration
(3) The amount of air taken in and out
(4) The air left in the bronchial tree

BBG281

34. After maximum inspiration when a person exhales out the maximum possible volume of air, it is called
(1) Residual volume
(2) Functional residual capacity
(3) Vital capacity
(4) Total lung capacity

BBG282

35. If in a person functional residual capacity is 2300 ml, residual volume is 1200 ml and inspiratory capacity is 3500 ml then find out its vital capacity-
(1) 5800 ml (2) 7000 ml
(3) 4600 ml (4) 2300 ml

BBG283

36. Which steps are involved in respiration ?
(i) Pulmonary ventilation
(ii) Diffusion of gases
(iii) Transport of gases by the blood
(iv)Utilisation of O_2 by the cells
(1) (i), (ii), (iii), (iv) (2) (i), (ii), (iii)
(3) (i), (ii), (iv) (4) (ii), (iii)

BBG284

37. Which of the following is not part of respiratory tree ?
(1) Atria
(2) Terminal bronchiole
(3) Respiratory bronchiole
(4) Alveolar sac

BBG285

38. Which one of the following has smallest diameter (Narrowest) ?
(1) Terminal bronchioles
(2) Respiratory bronchioles
(3) Alveolar duct
(4) Atria

BBG286

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39. Inspiration can occur when :-
(1) Intra pulmonary pressure (IPP) is less than atmospheric pressure
(2) IPP is more than atmospheric pressure
(3) IPP is equal to atmospheric pressure
(4) All of the above

BBG287

40. Which muscle contracts during normal expiration ?
(1) Diaphragm (2) EICM
(3) IICM (4) None of these

BBG288

41. Which of the following pulmonary volume can't be measured by spirometer directly ?
(1) Tidal volume
(2) Vital capacity
(3) Inspiratory capacity
(4) Residual volume

BBG289

42. Partial pressure of oxygen in deoxygenated blood is :-
(1) more than that in the oxygenated blood
(2) more than that in tissue under normal condition
(3) less than that in systemic venous blood
(4) equal to that in pulmonary arterial blood

BBG290

43. Gases exchange at tissue level occurs by :-
(1) facilitated diffusion
(2) $Na^+ - K^+$ pump
(3) against partial pressure gradient
(4) simple diffusion

BBG291

44. Which factors are favourable for formation of oxyhaemoglobin ?
(a) High acidity
(b) Low Pco_2
(c) Lesser $[H^+]$
(d) Low Temperature
(1) a and b (2) a, b and c
(3) b and c (4) b, c and d

BBG292

45. In the alveoli, dissociation of CO_2 from carbaminohaemoglobin occur, due to :-
(1) pCO_2 is high pO_2 is low
(2) pCO_2 is low pO_2 is high
(3) pCO_2 and pO_2 are high
(4) pCO_2 and pO_2 are low

BBG293

46. Which of the following cause increased strength of inspiratory as well as expiratory signal ?
(1) Decreased Pco_2 in arterial blood
(2) Decreased Pco_2 in venous blood
(3) Decreased $[H^+]$ in arterial blood
(4) Decreased P_H of arterial blood

BBG294

47. Which of the following statement is correct ?
(1) Solubility of CO_2 is 20-25 times higher than that of O_2.
(2) Weak pneumotaxic signals results complete filling of lungs.
(3) High concentration of hydrogen ions in arterial blood stimulate inspiratory centre.
(4) All are correct

BBG295

48. Match the column I with column II.

Column I		Column II	
A	Tracheal tubes	I	Arachnida
B	Gills	II	Frog
C	Moist skin	III	Cockroach
D	Book lungs	IV	Fishes

	A	B	C	D
(1)	III	IV	II	I
(2)	III	I	IV	II
(3)	IV	III	II	I
(4)	IV	II	I	III

BBG296

49. Spiracles are part of respiratory passage in :-
(1) Insects (2) Frog
(3) Human (4) Sponges

BBG297

50. What is the function of pleural fluid ?
(1) reduces friction on the lung surface
(2) protection from external shocks
(3) provide moisture
(4) all of the above

BBG298

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51. Which of the following structures/chemical prevent collapse of bronchioles/passage of conduction zone and exchange zone respectively ?
(1) C-shape hyaline rings; Lecithin
(2) PSCCGE; YFCT sheath
(3) C-Shaped elastin rings; Lecithin
(4) C-Shaped hyaline rings; YFCT sheath

BBG299

52. Expiration can occur if :-
(a) intrapulmonary pressure is higher than the atmospheric pressure.
(b) intrapulmonary pressure and atmospheric pressure are equal.
(c) intrapulmonary pressure is less than atmospheric pressure.
(d) muscles of diaphragm relax.
(1) a, d (2) b, c (3) c, d (4) a, b

BBG300

53. True about thoracic cavity/cage is :-
(1) An air tight chamber
(2) Formed dorsally by sternum
(3) Formed laterally by ribs
(4) Both (1) and (3)

BBG301

54. Identify the type of pulmonary volume/ capacity on the basis of quantity of air present in the lungs given below.
(I) 1100 ml to 1200 ml
(II) 500 ml
(III) 5000 ml to 6000 ml
Choose the correct option ?

	I	II	III
(1)	Vital capacity	FRC	Residual volume
(2)	Residual volume	Tidal volume	TLC
(3)	Expiratory capacity	Inspiratory capacity	Residual volume
(4)	Tidal volume	IRV	ERV

BBG302

55. Additional volume of air, a person can inspire by a forcible inspiration is :-
(1) average 1000-1100 ml
(2) IRV
(3) IRV + TV
(4) IRV (avg. 1000-1100 ml)

BBG303

56. After deep inspiration, capacity of maximum expiration of lung is called :-
(1) TV (Tidal volume)
(2) FRC + RV
(3) IRV + TV
(4) VC (Vital capacity)

BBG304

57. Partial pressure of oxygen and carbondioxide (in mm of Hg) in deoxygenated blood is ? (Respectively)
(1) 40; 45 (2) 95; 40
(3) 95; 45 (4) 104; 40

BBG305

58. During respiration, exchange of gases occurs:
(1) between blood and tissues
(2) at alveoli level only
(3) between alveoli and blood
(4) both (1) and (3)

BBG306

59. Chronic disorder in which alveolar walls are damaged ?
(1) asthma
(2) emphysema
(3) pneumonia
(4) occupational respiratory disorders

BBG307

60. How many of the following statements are correct ?
(a) Human beings have a significant ability to maintain and moderate the respiratory rhythm to suit the demands of the body tissues.
(b) Pneumotaxic centre can moderate the functions of the respiratory rhythm centre.
(c) A chemosensitive area is situated adjacent to the rhythm centre which is highly sensitive to CO_2 and hydrogen ions.
(d) The role of oxygen in the regulation of respiratory rhythm is quite insignificant
(1) only a (2) only b
(3) only a and b (4) all

BBG308

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61. **Assertion:** The amount of CO_2 that can diffuse through the diffusion membrane per unit difference in partial pressure is much higher when compared to that of O_2.
Reason: The solubility of CO_2 is 20 to 25 times higher than that of O_2.
(1) Assertion is true but the reason is false.
(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(4) Both assertion and reason are false
BBG309

62. **Assertion:** Functional residual capacity is equal to ERV+RV.
Reason: Expiratory capacity is equal to TV+IRV.
(1) Both assertion and reason are false
(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) Assertion is true but the reason is false.
(4) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
BBG310

63. **Assertion:** Oxygen dissociation curve is sigmoid.
Reason: Affinity of oxygen for Hb increases with addition of each molecule of oxygen one after another.
(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Assertion is true but the reason is false.
(3) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(4) Both assertion and reason are false
BBG311

64. **Assertion:** Oxygen is transported mainly as oxyhaemoglobin in human.
Reason: CO_2 is transported mainly as carbamino - haemoglobin in human.
(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both assertion and reason are false
(3) Assertion is true but the reason is false.
(4) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
BBG312

65. **Assertion:** During swallowing food can't enter in larynx.
Reason: During swallowing open of larynx glottis is covered by epiglottis
(1) Both assertion and reason are false
(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(4) Assertion is true but the reason is false
BBG313

66. **Assertion:** External nostril to terminal bronchioles constitutes conducting as well as respiratory part of respiratory tract.
Reason: It help in conduction of atmospheric air to the alveoli as well as diffusion of O_2 and CO_2 between blood and atmospheric air.
(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both assertion and reason are false.
(3) Assertion is true but the reason is false.
(4) Both the assertion and reason are true but the reason is not a correct explanation of the assertion
BBG314

67. **Assertion:** Human breathing is negative pressure breathing.
Reason: Air drawn in lungs because of pressure less than atmospheric pressure in lungs during inspiration
(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) Assertion is true but the reason is false.
(4) Both assertion and reason are false
BBG315

68. **Assertion:** CO_2 produced by cells have to be released out of animal body.
Reason: CO_2 is harmful gas.
(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both assertion and reason are false
(3) Assertion is true but the reason is false.
(4) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.

BBG316

69. **Assertion:** Inspiration can occur if intrapulmonary pressure is less than the atmospheric pressure.
Reason: Inspiration is initiated by the relaxation of diaphragm & external intercoastal muscle.
(1) Assertion is true but the reason is false.
(2) Assertion and Reason both are true and Reason is the correct explanation of the Assertion.
(3) Assertion and Reason both are true but Reason is not the correct explanation of the Assertion.
(4) Assertion is false but Reason is true.

BBG317

70. **Assertion:** Emphysema is a chronic disorder in which respiratory surface is decreased.
Reason: One of the major cause of this is cigarette smoking.
(1) Assertion is true but the reason is false.
(2) Assertion is false but Reason is true.
(3) Assertion and Reason both are true but Reason is not the correct explanation of the Assertion.
(4) Assertion and Reason both are true and Reason is the correct explanation of the Assertion.

BBG318

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	3	3	1	4	3	3	2	4	3	2	3	1	1	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	1	2	2	3	1	2	2	2	3	2	4	2	3	2	3
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	1	2	4	3	3	1	2	4	1	4	4	4	4	4	2
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	4	4	1	1	4	1	1	4	2	2	4	1	4	2	4
Question	61	62	63	64	65	66	67	68	69	70					
Answer	3	3	1	3	3	2	1	1	1	3					

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Body Fluids and Circulation

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1. RBC's are nucleated in :-
(1) Rat (2) Cat
(3) Frog (4) Rabbit
BFC420

2. Concave surface of mammalian RBC's is helpful in :-
(1) Increasing surface area
(2) Reducing surface tension
(3) Formation of haemoglobin
(4) providing less space for haemoglobin
BFC421

3. You are required to draw blood from a patient and to keep it in a test tube for analysis of blood corpuscles and plasma you are also provided with the following four types of test tubes. Which of them you will not use for this purpose?
(1) chilled test tube
(2) test tube containing heparin
(3) test tube containing sodium oxalate
(4) test tube containing calcium bicarbonate
BFC422

4. If RBC's are kept in hypotonic solution, they will:-
(1) Stick together
(2) remain normal
(3) shrink and collapse
(4) increase in size and burst
BFC423

5. Which of the following statements is false for mammalian RBC's ?
(1) Antigens are found on the surface of RBC
(2) Mature RBC have very large nucleus
(3) Membrane of RBC is called donnan's membrane
(4) Mature RBC lacks nucleus but immature RBC has nucleus
BFC424

6. RBC's of mammals are without nucleus because they :-
(1) do not need it
(2) are not true cells
(3) lose it during their formation/maturation
(4) do not have nucleus from the beginning
BFC425

7. Match the columns and choose the correct answer from the options :-

	Column-A (Type of WBC)		Column-B (Shape of nucleus)
a	neutrophils	i	kidney shaped
b	eosinophils	ii	s-shaped
c	basophils	iii	3 to 5 lobes
d	monocytes	iv	2 lobes
		v	Disc shaped

(1) a-iii, b-iv, c-ii, d-i (2) a-iii, b-iv, c-v, d-i
(3) a-iii, b-ii, c-v, d-i (4) a-iii, b-ii, c-i, d-iv
BFC426

8. Which of the following cations are required for the conversion of prothrombin into active thrombin by thromboplastin ?
(1) Fe^{+2} (2) Ca^{+2} (3) Cu^{+2} (4) Fe^{3+}
BFC427

9. Match the following :-

	Column-A (Type of WBC)		Column-B (Shape of nucleus)
A	Basophils	1	Phagocytosis
B	Neutrophils	2	Inflammation
C	Plasma cells	3	Blood clotting
D	Thrombocytes	4	Antibodies

(1) A=2, B=1, C=4, D=3
(2) A=2, B=1, C=3, D=4
(3) A=1, B=2, C=4, D=3
(4) A=4, B=1, C=2, D=3
BFC428

10. Which of the following statements are wrong?
(i) Leucocytes disintegrate in the spleen and liver
(ii) RBC, WBC and blood platelets are produced by bone marrow
(iii) Neutrophils causes protein destruction
(iv) The important function of lymphocytes is to produce antibodies
(1) (i) and (ii) only (2) (i) and (iv) only
(3) (i) and (iii) only (4) (ii) and (iii) only
BFC429

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11. Fibrin is formed from fibrinogen due to the activity of :-
(1) Heparin (2) Prothrombin
(3) Active thrombin (4) Antiprothrombin
BFC430

12. The difference between blood and lymph :-
(1) Blood has WBCs while lymph has RBCs
(2) Blood has dissolved salt while lymph has no cells
(3) Blood has RBCs and WBCs while lymph has no cells
(4) Blood has both RBCs and WBCs while lymph has only WBCs
BFC431

13. The important function of lymph is to -
(1) transport oxygen to the brain
(2) return RBCs to the lymph nodes
(3) return interstitial fluid to the blood
(4) transport carbon dioxide to the lungs
BFC432

14. Serum is ?
(1) Blood without corpuscles
(2) Blood without fibrinogen
(3) Also called as plasma
(4) Blood without fibrinogen and corpuscles
BFC433

15. Conduction of impulse is slowest in :–
(1) SA node (2) AV node
(3) Purkinje fibre (4) Bundle of His
BFC434

16. Mitral valve is present at :-
(1) Left AV foramen
(2) Right AV foramen
(3) Between Aorta & Left ventricle
(4) Between right ventricle and pulmonary Artery
BFC435

17. Purkinje fibres is :-
(1) Muscle fibre distributed through out heart walls
(2) Muscular fibres found only in ventricle wall
(3) Neuromuscular fibres distributed throughout heart
(4) Nerve fibre distributed in ventricles
BFC436

18. The correct sequence of conduction pathway in heart is :–
(1) SA node → Bundle of His → AV node → Purkinje fibre
(2) SA node → AV node → Bundle of His → Purkinje fibre
(3) AV node → SA node → Purkinje fibre → Bundle of His
(4) Papillary muscles → SA node → Purkinje fibre → AV node
BFC437

19. Systemic heart comprises of :–
(1) Left auricle & right auricle.
(2) Left ventricle & right ventricle.
(3) Right auricle & right ventricle.
(4) Left auricle & left ventricle.
BFC438

20. In heart, automatic rhythmic impulse generation capability is present in the :-
(1) S.A. node (2) A.V. node
(3) Bundle of His (4) All of the above
BFC439

21. Stimulation of sympathetic nerves of the heart accelerates it through :-
(1) Release of acetylcholine
(2) Inhibition of nor-adrenalin
(3) Release of nor-adrenalin
(4) Inhibition of renin
BFC440

22. Coronary sinus :–
(1) Drains impure blood from pulmonary veins to left auricle.
(2) Receives impure blood from heart musculature via coronary veins.
(3) Drains pure blood in right auricle
(4) More than one options are correct
BFC441

23. On the basis of number of cusps. Which of the following valves is the odd one :-
(1) Right AV valve
(2) Mitral valve
(3) Aortic semilunar valve
(4) Pulmonary semilunar valve
BFC442

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24. The papillary muscles :-
(1) are found in the atria
(2) are attached to chordae tendineae
(3) contract to close the foramen ovale
(4) are attached to the semilunar valves

BFC443

25. Ventricle is thick walled as compared to atrium because :-
(1) it is to receive blood from the atria
(2) it is present on posterior side
(3) it is to pump blood
(4) none of the above

BFC444

26. **Statement 1:** Parasympathetic neural signals release nor adrenaline to modulate rate of heart beat.
Statement 2 : Cardiac activity can be altered by intervention of both nervous and endocrine system.
(1) Both statement 1 and 2 are correct.
(2) Both statement 1 and 2 are incorrect.
(3) Statement 1 is correct but statement 2 is incorrect.
(4) Statement 1 is incorrect but statement 2 is correct.

BFC445

27. During ventricular systole :-
(1) Oxygenated blood is pumped into the aorta and deoxygenated blood is pumped into the pulmonary artery.
(2) Oxygenated blood is pumped into the pulmonary artery and deoxygenated blood is pumped into the aorta
(3) Oxygenated blood is pumped into the aorta and deoxygenated blood is pumped into the pulmonary vein
(4) Oxygenated blood is pumped into the pulmonary vein and deoxygenated blood is pumped into the pulmonary artery

BFC446

28. All of the following are true except :-
(1) Trabeculae carnae, the muscular ridges of ventricles are also known as columnae carnae
(2) Musculi pectinati are present in ventricular walls
(3) There are three papillary muscles in right ventricle and two in left ventricle
(4) Papillary muscles are attached to AV valves through chordae tendinae

BFC447

29. First heart sound is :–
(1) Lubb ; Ventricular systolic.
(2) Lubb ; Auricular systolic
(3) Dubb ; ventricular systolic
(4) Dubb ; Auricular systolic

BFC448

30. If B.P. of a person is 160/100 the pulse pressure is :–
(1) 160 mm. Hg (2) 100 mm. Hg
(3) 60 mm. Hg (4) 40 mm. Hg

BFC449

31. Maximum amount of blood is transferred from atrium to ventricle in cardiac cycle during
(1) Atrial systole
(2) Ventricular systole
(3) Atrial diastole
(4) Joint diastole

BFC450

32. Amount of blood pumped by right ventricle per minute is
(1) 70 ml.
(2) 9% of the total cardiac output
(3) 5 litre
(4) 10 litre

BFC451

33. 'Dub' sound in heart beat is produced when
(1) Tricuspid valve opens.
(2) Mitral valve closes.
(3) Mitral valve opens.
(4) Semilunar valves of the two arches close.

BFC452

34. The amount of blood that passively flows from atria to ventricles is (in man) :–
(1) 25–30% (2) 35–45%
(3) 70–75% (4) 15–20%

BFC453

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35. Which of the following set of events occurs in time interval after the end of second heart sound and before onset of first heart sound:-
(1) Auricular diastole, ventricular systole
(2) Auricular diastole, ventricular diastole
(3) Auricular diastole, ventricular diastole, auricular systole
(4) Auricular diastole, ventricular diastole, auricular systole & ventricular systole

BFC454

36. The Cardiac output of a person whose pulse rate is 60/minute and stroke volume 70 ml is-
(1) 6 lit./mt (2) 5 lit./mt
(3) 4.2 lit./mt (4) 5.5 lit./mt

BFC455

37. In a human cardiac cycle which of the following statement is false :-
(1) The time taken for one cardiac cycle decreases in Tachycardia
(2) The time interval for joint Diastole in heart is about 0.4 seconds
(3) The time interval for joint systole in heart is about 0.4 second
(4) Left ventricle takes 0.5 seconds to complete one diastole

BFC456

38. During diastole auricles receive :-
(1) Oxygenated blood
(2) Deoxygenated blood
(3) Mixed blood
(4) Venous blood

BFC457

39. In human heart, maximum systolic pressure is developed/exerted by
(1) Right Auricle
(2) Left Auricle
(3) Right Ventricle
(4) Left Ventricle

BFC458

40. Decrease in pH of blood will cause heart rate to:
(1) Decrease
(2) Increase
(3) No change
(4) First increase then decrease

BFC459

41. Observe the given diagram and choose the incorrect option regarding it.



(1) Provides protection against parasitic infection.
(2) Can be stained by dyes like methylene blue.
(3) Has bilobed nucleus.
(4) Cytoplasm is granular.

BFC460

42. Systolic pressure is higher than diastolic pressure because :-
(1) Arteries are contracting during systole
(2) Blood is pumped with a pressure in the arteries by the heart during systole but not during diastole
(3) Arteries resist during systole only
(4) Volume of blood is higher in systole than that of diastole in the heart

BFC461

43. Fall in blood pressure due to loss of blood is soon restored because the :-
(1) Blood vessels dilate
(2) Blood cells decrease in number
(3) Heart beat is increased
(4) Heart beat is decreased

BFC462

44. Hardening of arteries & loss in their elasticity associated with increase in age is called :
(1) Heart attack
(2) Angina
(3) Arteriosclerosis
(4) Thrombosis

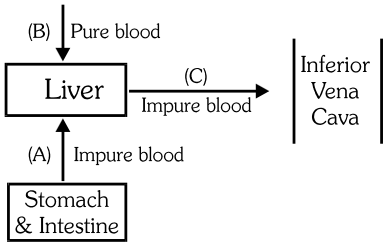
BFC463

45. What is true about vein :-
(1) All veins carry deoxygenated blood except coronary vein
(2) All veins carry oxygenated blood except pulmonary vein
(3) They carry blood from organs towards heart
(4) They carry blood from heart towards the organs

BFC464

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46. Please observe the diagram below



Here vessel A, B and C are –

(1) A – Hepatic vein, B – Portal vein, C – Hepatic artery

(2) A – Portal vein, B – Hepatic artery, C – Hepatic vein

(3) A – Hepatic artery, B – Hepatic vein, C – Portal vein

(4) A – Portal vein, B – Hepatic vein, C – Hepatic artery

BFC465

47. Internal elastic lamina is present in the walls of:

(1) Arteries (2) Veins

(3) Heart (4) All

BFC466

48. Glucose from the digestive tract is transported to liver by :-

(1) Hepatic vein (2) Portal vein

(3) Renal vein (4) Aorta

BFC467

49. Which vein carries blood away from an organ and ends in another organ instead of vena cava or right atrium :–

(1) Hepatic vein (2) Renal vein

(3) Portal vein (4) Coronary vein

BFC468

50. Observe the given column I and II for correct match and choose the correct option.

	Column-I		**Column-II**
(A)		(i)	Number of QRS in a given time period
(B)		(ii)	Depolarization of atria
(C)		(iii)	Depolarization of ventricles
(D)	Heart Rate	(iv)	Repolarization of ventricles

Options:

(1) A-(ii), B-(i), C-(iv), D-(iii)

(2) A-(iv), B-(i), C-(ii), D-(iii)

(3) A-(iv), B-(iii), C-(ii), D-(i)

(4) A-(ii), B-(iii), C-(iv), D-(i)

BFC469

51. What will be the immediate effect if arteries supplying blood to heart musculature (coronary arteries) are completely blocked:-

(1) Heart will stop beating

(2) Heart will beat slowly and then stop

(3) Heart will beat normally or even faster but heart muscles will start dying

(4) Heart will still be normal as new capillaries will open up

BFC470

52. The vessel carrying minimum quantity of excretory waste product in blood is :–

(1) Hepatic vein (2) Renal vein

(3) Pulmonary vein (4) Pulmonary artery

BFC471

53. A vein differs from the artery in having :-

(1) Narrow lumen

(2) Thick muscular wall

(3) Valves to control direction of flow

(4) Dark pigmented wall

BFC472

54. Match the blood vessels of human heart listed under column–I with the functions given under column–II: Choose the answer which gives the correct combination of the alphabets of the two columns.

	Column–I (Blood vessel)		**Column–II (Function)**
A	Superior vena cava	P	Carries deoxygenated blood to lungs
B	Inferior vena cava	Q	Carries oxygenated blood to lungs
C	Pulmonary artery	R	Brings deoxygenated blood from lower parts of the body to the right atrium

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D	Pulmonary vein	S	Brings oxygenated blood to the left atrium
		T	Brings deoxygenated blood from upper parts of the body into the right atrium

(1) A = T, B = P, C = R, D = Q
(2) A = T, B = R, C = P, D = S
(3) A = S, B = T, C = R, D = P
(4) A = T, B = P, C = Q, D = R

BFC473

55. An artificial pacemaker is implanted subcutaneously and connected to heart in patients –
(1) Having 90% blockage in the main coronary arteries
(2) Having very high blood pressure
(3) With irregularity in heart rhythm
(4) Suffering with arteriosclerosis

BFC474

56. Which of the following does not carry deoxygenated blood:
(1) Pulmonary artery (2) Coronary vein
(3) Portal vein (4) Hepatic artery

BFC475

57. P wave of ECG indicates :
(i) Activation of SA nodes
(ii) Depolarization of atrial muscles
(iii) Spread of excitation from AV nodes to purkinje fibres
(iv) Repolarisation of atria and deporization of ventricles
(1) i and ii are correct
(2) ii and iv are correct
(3) i and iii are correct
(4) i, ii and iii are correct

BFC476

58. QRS is related to :
(1) SA node activation
(2) Atrial relaxation
(3) Atrial contraction
(4) Ventricular contraction

BFC477

59. In ECG, T wave represents :
(1) Diastole of atria
(2) Systole of ventricles
(3) Repolarisation of ventricles
(4) Diastole of atria and ventricles

BFC478

60. Which is a correct pair of an ECG with respect to cardiac event ?
(1) P wave – depolarization of the ventricles
(2) P wave – depolarization of the AV nodes
(3) QRS wave – depolarization of the ventricles
(4) T-wave – repolarization of the atria

BFC479

61. Which statement is true for haemoglobin ?
(1) Healthy aduct has 12-16 gm of Hb in every 100 ml of blood.
(2) Hb is iron containing protein complex.
(3) It play an important role in transport of respiratory gases.
(4) All are true

BFC480

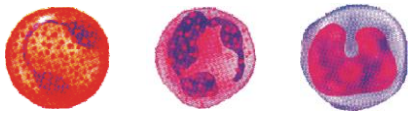
62. Globulin protein of plasma mainly involved in?
(1) BCOP
(2) Blood clotting
(3) Transport of gases
(4) Antibody formation

BFC481

63. Which WBC is associated with resistance of infection and allergic reactions ?
(1) Eosinophils (2) Monocytes
(3) Neutrophils (4) Lymphocyte

BFC482

64. Identify (A, B, C) blood cells in the following figures (respectively) ?



(A) (B) (C)

	A	B	C
(1)	Monocyte	Neutrophil	Eosinophil
(2)	Neutrophil	Eosinophil	Monocyte
(3)	Eosinophil	Neutrophil	Monocyte
(4)	Neutrophil	Monocyte	Eosinophil

BFC483

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65. Erythroblastosis foetalis condition is observed when :-
(1) Rh +ve pregnant mother have Rh –ve foetus
(2) Rh –ve pregnant mother have Rh +ve foetus
(3) Rh +ve pregnant mother have Rh +ve foetus
(4) Both in (2) and (3)

BFC484

66. Persons with 'O-' blood group have :-
(1) (A, B, Rh) antigens on RBC
(2) (a, b, Rh) antibodies in plasma always
(3) absence of (A, B, Rh) antigens on RBC and presence of (a, b, Rh) antibodies in plasma always
(4) presence of (a, b) antibodies in plasma

BFC485

67. A special system of blood vessles present for circulation of blood to and from the cardiac musculature is called as :-
(1) Hepatic portal system
(2) Hypophyseal portal system
(3) Coronary system
(4) Cutaneous circulation

BFC486

68. Closed circulatory system is more advantageous as compared to open circulatory system because :-
(1) flow of fluid can be more precisely regulated
(2) oxygenated and deoxygenated blood are always present in different compartments
(3) cell are not in direct contact with blood
(4) less loss of energy

BFC487

69. Which of the following blood vessel contain oxygenated blood only ?
(a) Pulmonary artery
(b) Renal artery
(c) Umbilical vein
(d) Hepatic portal vein
(1) a, b (2) b only
(3) b, c (4) b, c, d

BFC488

70. When SA node is stimulated by sympathatic nerve than the :-
(1) rate of heart beat and cardiac output decreases
(2) rate of heart beat decreases and cardiac output increases
(3) rate of heart beat and cardiac output increases
(4) rate of heart beat increases and cardiac output decreases

BFC489

71. Why atria and ventricles cann't contract simultaneously ?
(1) There is delay of signal at AV-node
(2) speed of conduction is slowest in ventricles
(3) More muscle mass in ventricles
(4) delay of signal at SA-node

BFC490

72. Which of the following statement is wrong ?
(1) Cardiac output of a person varies according to his physiological demand.
(2) Stroke volume of heart is almost equal to the amount of blood transferred from left atrium to left ventricle.
(3) Stroke volume is amount of blood pumped by the left ventricle in the systemic aorta in one beat.
(4) End systolic volume is about 50 ml.

BFC491

73. If the cardiac output of a person is 5 lit and his breathing rate is 18 per min. The amount of blood that will be oxygenated in the lungs per min will be :-
(1) 5000 ml (2) 4900 ml
(3) 70 ml (4) 9000 ml

BFC492

74. Which statement is correct ?
(a) Cardiac output of an athelete is higher than an ordinary man.
(b) Body has the ability to alter the stroke volume as well as the heart rate.
(c) Stoke volume multiplied by the heart rate gives the cardiac output.
(1) a and b (2) a, b and c
(3) b and c (4) a and c

BFC493

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75. Which of the following wave in ECG represents electrical excitation of the atria ?
(1) QRS complex (2) P-wave
(3) T-wave (4) S-wave

BFC494

76. Acute chest pain due to insufficient oxygen supply to heart muscle is :-
(1) Myocardial infarction (2) Angina
(3) Atherosclerosis (4) Emphysema

BFC495

77. True about vein is ?
(1) All veins carry deoxygenated blood except coronary vein.
(2) All veins carry oxygenated blood except pulmonary vein.
(3) They carry blood from organs, towards heart.
(4) They carry blood from heart towards the organs.

BFC496

78. Which of the following vertebrate organ receive only oxygenated blood ?
(a) Heart (b) Lungs (c) Liver
(d) Spleen (e) Gills
(1) a, c, d (2) c, d
(3) a, c, e (4) only d

BFC497

79. Which one is incorrect regarding the composition of blood plasma ?
(A) Albumin = 4%
(B) Organic compounds = 10%
(C) H_2O = 90%
(D) Inorganic compounds = 0.9%
(E) Globulin = 5%
(F) Prothrombin = 0.3%
(1) A, E (2) E, F (3) B, E (4) B, D

BFC498

80. Match column-I with column-II and select the correct option from below :-

Column I		Column II	
(a)	Eosinophil	(i)	Coagulation
(b)	RBC	(ii)	Universal recipient
(c)	AB group	(iii)	Resist infection
(d)	Platelet	(iv)	Contraction of heart
(e)	Systole	(v)	Gas transport

(1) b-v, d-iii, e-iv (2) c-ii, d-i, e-iv
(3) a-iii, b-v, c-i (4) a-v, c-i, e-iv

BFC499

81. Erythroblastosis foetalis condition occurs :-
(1) Mother Rh(+ve) and Foetus Rh (+ve)
(2) Mother Rh(–ve) and Foetus Rh (–ve)
(3) Mother Rh(+ve) and Foetus Rh (–ve)
(4) Mother Rh(–ve) and Foetus Rh (+ve)

BFC500

82. Which of the following inactivate heparin during blood clotting ?
(1) Thromboplastin
(2) Thrombokinase
(3) Prothrombin
(4) Fibrin

BFC501

83. How many of the following animals have double circulation.
Rabbit, Rat, Frog, Dolphin, Lizard, Whale, Peacock, Pigeon, Fish (Catla), Butterfly.
(1) Five (2) Eight
(3) Seven (4) Six

BFC502

84. If a person have two chambers (one atrium and one ventricle) with the vena cava entering the atrium and the aorta leaving the ventricle then which of the following statement would be correct ?
(1) no oxygen would reach the cells
(2) no blood would reach the head and neck
(3) the blood would be unable to carry food to the cells
(4) there would be no blood circulation

BFC503

85. Which of the following is mismatch regarding human heart ?
(1) Pacemaker – SA node
(2) Four chambered with right systemic arch
(3) Aortic semilunar valve – systemic aorta
(4) Tricuspid valve – between right atrium and right ventricle

BFC504

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86. The action potential reaches the AV node from the SA node, there is delay of the action potential. This delay is important because :-
(1) It allows atria to rest
(2) It allows a stronger right atrial contraction
(3) It allows ventricles to receive all the blood from the atria.
(4) It allows right atria to receive the blood from vena cava.

BFC505

87. All the components of the conduction system can generate an action potential for the contraction of heart muscle, but the SA node act as pacemaker because :-
(1) all the components of the heart cannot conduct the action potential.
(2) only SA node is auto excitable and autorhythmic.
(3) the SA node has a higher inherent rate of depolarisation.
(4) the SA node has a lower inherent rate of depolarisation.

BFC506

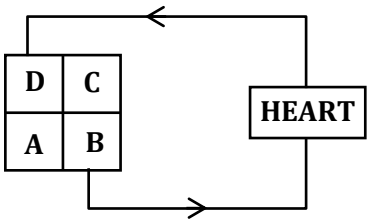
88. Given below are four statements regarding human blood vascular system ?
(a) there is no role of calcium ions in blood clotting.
(b) cardiac output is the amount of blood pumped by each ventricle in one second.
(c) arteries are thin walled and have narrow human as compared to veins.
(d) the stage of cardiac cycle where all the four chambers are in relaxed stage is called joint diastole.
How many of the above statements are correct?
(1) One (2) Two (3) Three (4) Four

BFC507

89. When heart muscle is suddenly damaged by an inadequate blood supply. This condition is called :-
(1) Angina pectoris (2) Heart failure
(3) Heart attack (4) Heart arrest

BFC508

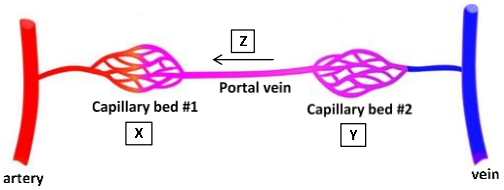
90.



Observe the given diagram and choose the option mentioning the correct type of circulation.
(1) Systemic circulation
(2) Double circulation
(3) Coronary circulation
(4) Pulmonary circulation

BFC509

91.



Observe the given and diagram choose the option with all correct labels.
(1) X-Intestines, Z-hepatic vein, Y-Liver
(2) X-liver, Z-Intestinal vein, Y-Intestine
(3) X-Liver, Z-Hypophyseal portal vein, Y-Intestines
(4) X-Liver, Z-Hepatic portal vein, Y-Intestine

BFC510

92. **Assertion :** Person with blood group O can donate blood to receivers with blood group A, B, AB and O.
Reason : Person with O blood group possess both antibody a and antibody b in their plasma.
(1) Both (A) & (R) are true and (R) is the correct explanation of (A).
(2) (A) is true but (R) is false.
(3) Both (A) and (R) is incorrect.
(4) Both (A) & (R) are true and (R) is not the correct explanation of (A).

BFC511

93. **Statement I :** Arteries do not collapse when empty.
Statement II : Blood flows with a low pressure in veins.
(1) Statement I and II both are correct.
(2) Statement I is correct but II is incorrect.
(3) Statement II is correct but I is incorrect.
(4) Both statement I and II are incorrect.

BFC512

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94. **Assertion :** AV valve do not evert during ventricular systole.

Reason : They are bound to papillary muscles by chordae tendinae.

(1) Both (A) & (R) are true and (R) is the correct explanation of (A).

(2) (A) is true but (R) is false.

(3) Both (A) and (R) is false.

(4) Both (A) & (R) are true and (R) is not the correct explanation of (A).

BFC513

95. Match column I with column II and choose the correct option.

	Column I		Column II
A.	Bicuspid valve	i.	Prevent back flow from aorta to left ventricle.
B.	Tricuspid valve	ii.	Prevent back flow of blood from left ventricle to left atrium
C.	SL valve	iii.	Receives deoxygenated blood only
D.	Superior vena cava	iv.	Prevents backflow of blood from right ventricle to right atrium

(1) A-iv, B-ii, C-i, D-iii

(2) A-iv, B-ii, C-iii, D-iv

(3) A-ii, B-iv, C-iii, D-i

(4) A-ii, B-iv, C-i, D-iii

BFC514

96. Match column I with column II and choose the correct option.

	Column I		Column II
A.	Lymph	i.	Plasma without large protein but having WBCs
B.	Serum	ii.	Contains antibodies against blood grouping antigens
C.	Plasma	iii.	Plasma without clotting factors
D.	Venous blood	iv.	Is present in pulmonary artery

(1) A-iv, B-iii, C-i, D-ii

(2) A-i, B-iii, C-ii, D-iv

(3) A-iv, B-ii, C-iii, D-i

(4) A-i, B-iv, C-iii, D-ii

BFC515

97. **Assertion :** SA node is known as the pacesetter of the heart.

Reason : SA node generates maximum number of action potential.

(1) Both (A) & (R) are correct and (R) is the correct explanation of (A).

(2) (A) is correct but (R) is not correct.

(3) (A) is not correct but (R) is correct.

(4) Both (A) & (R) are correct but (R) is not the correct explanation of (A).

BFC516

98. Match the column I with column II and choose the correct option.

	Column-I Blood groups		Column-II Can donate blood to
(i)	A	(p)	B, AB
(ii)	AB	(q)	O
(iii)	B	(r)	A, AB
(iv)	O	(s)	AB

Options:

(1) i-r, ii-s, iii-p, iv-q

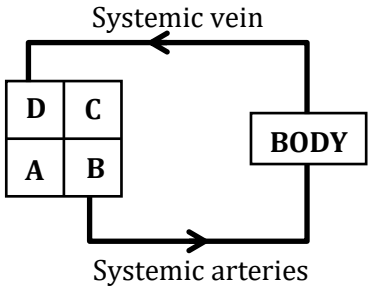
(2) i-s, ii-q, iii-r, iv-p

(3) i-p, ii-r, iii-q, iv-s

(4) i-p, ii-r, iii-s, iv-q

BFC517

99. Observe the given diagram choose the correct labelling.



(1) A — Right atrium

(2) B — Left ventricle

(3) C — Right ventricle

(4) D — Left atrium

BFC518

100. **Assertion:** QRS complex represent the depolarization of atria.
Reason: Beginning of the T-wave marks the end of systole.
(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) The assertion is true but the reason is false.
(4) Both the assertion and reason are false.

BFC519

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	1	4	4	2	3	1	2	1	3	3	4	3	4	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	1	2	2	4	4	3	2	2	2	3	4	1	2	1	3
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	4	3	4	3	3	3	3	4	4	2	2	2	3	3	3
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	2	1	2	3	3	3	2	3	2	3	4	1	4	3	3
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	4	4	1	3	2	4	3	1	3	3	1	2	1	2	2
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	2	3	4	3	2	4	2	4	1	2	3	3	1	3	3
Question	91	92	93	94	95	96	97	98	99	100					
Answer	4	4	1	1	4	2	3	1	2	4					

Locomotion and Movement-I

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1. An example of gliding joint is :
(1) Humerus and glenoid cavity
(2) Femur, tibia and fibula
(3) Occipital condyle and odontoid process
(4) Zygapophyses of adjacent vertebrae
BLC231

2. Function of long bones in adult mammals :
(1) Support only
(2) Support and produce RBC s only
(3) Support and produce WBCs only
(4) Support and produce RBC s and WBCs
BLC232

3. Which of the following vertebrae are fused ?
(1) Cervical (2) Sacral
(3) Lumbar (4) Thoracic
BLC233

4. Which of the following represents the correct order of vertebral regions from superior to inferior?
(I) Sacrum
(II) Thoracic
(III) Cervical
(IV) Lumbar
(V) Coccyx
(1) I -II-III-IV-V (2) II-IV-I-III-V
(3) IV-I-II-V-I (4) III-II-IV-I-V
BLC234

5. Hyoid bone is located :-
(1) at the roof of the buccal cavity.
(2) at the base of the buccal cavity.
(3) in front of the skull.
(4) behind the skull.
BLC235

6. Part of the body having a single pair of bones is called :-
(1) pelvic girdle (2) external ear
(3) wrist (4) lower jaw
BLC236

7. Which of the following is a single U shaped bone, present at the base of the buccal cavity and it is also included in the skull ?
(1) Hyoid (2) Malleus
(3) Sacrum (4) Scapula
BLC237

8. What is not true about human skull ?
(1) It is dicondylic
(2) It includes 6 ear ossicles
(3) It includes 22 facial bones
(4) Hyoid is included in skull bones.
BLC238

9. Which one of the following is the correct description of certain part of a normal human skeleton ?
(1) Parietal bone and the temporal bone of the skull are joined by fibrous joint
(2) First vertebra is axis which articulates with the occipital condyles.
(3) The 9th and 10th pairs of ribs are called the floating ribs.
(4) Glenoid cavity is a depression to which the thigh bone articulates.
BLC239

10. Select the correct statement with respect to skeletal system in humans :-
(1) Accumulation of uric acid crystals in joints causes their inflammation.
(2) The vertebral column has 10 thoracic vertebrae
(3) The joint between adjacent vertebrae is fibrous joint.
(4) The decreased level of progesterone causes osteoporosis in old people.
BLC240

11. The following pairs of the human skeletal parts are correctly matched with their respective category except for :-

	Pairs of skeletal parts	Category
(1)	Sternum and ribs	Axial skeleton
(2)	Clavicle and glenoid cavity	Pelvic girdle
(3)	Humerus and ulna	Appendicular skeleton
(4)	Malleus and shapes	Ear ossicles

BLC241

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12. Match column I (types of synovial joints) with column II (bones involved) and choose the correct option :-

Column-I (Type of synovial joint)		Column-II (Bone involved)	
(A)	Ball and socket joint	I	Carpal and metacarpal of thumb
(B)	Hinge joint	II	Humerus and pectoral girdle
(C)	Pivot joint	III	Knee
(D)	Saddle joint	IV	Atlas and axis

(1) A-I, B-II, C-III, D-IV
(2) A-II, B-III, C-IV, D-I
(3) A-III, B-I, C-IV, D-II
(4) A-IV, B-III, C-II, D-I

BLC242

13. Which of the following shows the correct characteristics and an example of a synovial joint in humans ?

	Characteristics	Example
(1)	Fluid filled between two joints provides cushion	Skull bones
(2)	Fluid filled synovial cavity between two bones	Joint between atlas and axis
(3)	Lymph filled between two bones limited movement	Gliding joint between carpals
(4)	Fluid cartilage between two bones, limited movement	Knee joint

BLC243

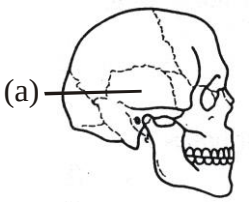
14. Which of the following option shows the correctly matched bones (given in column-I) with its pair (given in column II) ?

Column I		Column II	
(1)	Carpals	I	Bones that form the fingers and toes
(2)	Tarsal	II	Bones that form wrist
(3)	Phalanges	III	Bones that form the of the hands
(4)	Metacarpals	IV	Bones that form the ankles

(1) A-II, B-IV, C-I, D-III
(2) A-I, B-II, C-III, D-IV
(3) A-III, B-II, C-IV, D-I
(4) A-IV, B-I, C-III, D-II

BLC244

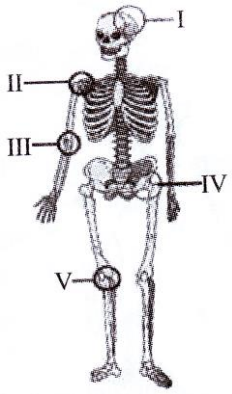
15. In the given diagram of skull, what does "a" represent ?



(1) Frontal bone (2) Temporal bone
(3) Occipital bonze (4) Parietal bone

BLC245

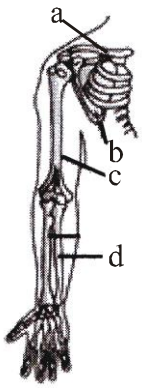
16. The given diagram of human skeleton system shows different types of joints which are marked as I, II, III, IV and V. Identify ball-socket type of joint



(1) I and II (2) II and IV
(3) III and IV (4) IV and V

BLC246

17. The given figure shows right pectoral girdle and upper arm. Few parts are marked as a, b, c and d.



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Which of the following options shows the correct labelling of marked parts ?

	a	b	c	d
(1)	Clavicle	Scapula	Humerus	Ulna
(2)	Scapula	Clavicle	Radius	Humerus
(3)	Sacrum	Scapula	Ulna	Tibia
(4)	Radius	Clavicle	Scapula	Humerus

BLC247

18. Which of the following is not the function of skeleton ?
(1) It allows movement.
(2) It supports the body.
(3) It connects muscle to joints.
(4) It protects the internal part of the body.

BLC248

19. "X" is a large triangular flat bone situated in the dorsal part of the thorax between "Y" and the seventh ribs. Identify "X" and "Y".
(1) X-Patella ; Y- Third
(2) X-Clavicle ; Y- Eight
(3) X-Scapula ; Y- Sixth
(4) X-Scapula ; Y- Second

BLC249

20. Which of the following pair shows the correct characteristics with an example of a synovial joint in humans ?
(1) Hinge joint - Between vertebrae
(2) Gliding joint - between carpals
(3) Cartilaginous joint - Skull bones
(4) Fibrous joint - Between phalanges

BLC250

21. The skull is composed of two sets of bones : cranial and facial, that total to __A__ bones. Cranial bones are __B__ whereas facial bones are __C__ in number. __D__U shaped bone called hyoid is present at the base of the buccal cavity and is also included in the skull.

	A	B	C	D
(1)	8	22	14	1
(2)	22	8	14	1
(3)	22	8	14	2
(4)	29	8	14	1

BLC251

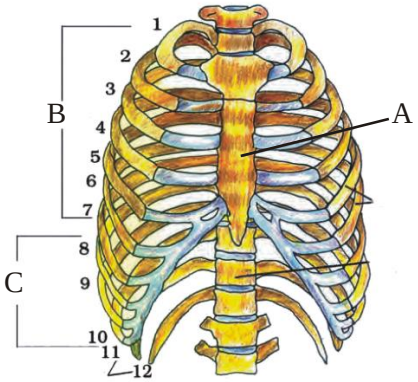
22. Vertebrochondral ribs of humans :-
(1) Are represented by 9th, 10th and 11th rib pairs
(2) Don't articulate directly with the sternum
(3) Join to the 7th and 11th rib
(4) Are represented by 11th and 12th rib pairs

BLC252

23. Limbs in humans are not used for :-
(1) Change in body postures
(2) Locomotion
(3) Movement of food in alimentary canal
(4) Both (1) and (3)

BLC253

24. Identify the parts labelled as A, B, C of a rib cage:-



	a	b	c
(1)	Sternum	True ribs	Floating ribs
(2)	Sternum	True ribs	False ribs
(3)	Vertebral column	False ribs	True ribs
(4)	Vertebral column	True ribs	False ribs

BLC254

25. Match the following :-

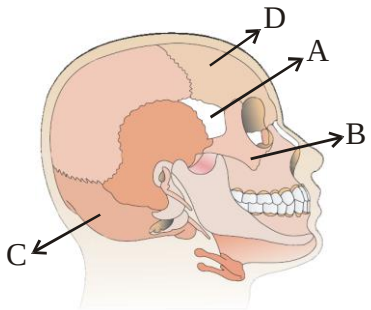
Column I		Column II	
(i)	Fibrous joint	(a)	Between two adjacent vertebrae
(ii)	Cartilaginous joint	(b)	Between humerus and pectoral girdle
(iii)	Pivot joint	(c)	Sutures
(iv)	Ball and socket joint	(d)	Between atlas and axis joint

(1) iv-d, iii-b, i-c, ii-a
(2) i-a, ii-c, iii-d, iv-b
(3) i-c, ii-a, iii-d, iv-b
(4) i-a, ii-d, iii-b, iv-c

BLC255

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26. Given below the diagrammatic view of human skull in which certain bones are labelled by A, B, C and D. Find out the correct option in which all the alphabetise are correctly labelled :-



Option :-

	a	b	c	D
(1)	Zygomatic bone	Sphenoid bone	Temporal bone	Frontal bone
(2)	Sphenoid bone	Maxilla bone	Occipital bone	Parietal bone
(3)	Sphenoid bone	Zygomatic bone	Occipital bone	Frontal bone
(4)	Zygomatic bone	Maxilla bone	Temporal bone	Parietal bone

BLC256

27. Read the following statements A-D

A. Our vertebral column is formed by 26 serially arranged units called vertebrae and is ventrally placed.

B. Vertebral column extends from the base of the skull and constitutes the main frame work of the trunk.

C. Each vertebra has a central hollow portion (neural canal) through which the spinal cord passes.

D. First vertebra is axis and it articulates with the occipital condyles

How many of the above statements are correct?

(1) Four (2) Three (3) Two (4) One

BLC257

28. Read the following statements (A to D) :-

A. There are 12 pairs of ribs

B. Each rib is a thin flat bone connected ventrally to the vertebral column and dorsally to the sternum

C. First seven pairs of ribs are called vertebrochondral ribs

D. The 8th, 9th and 10th pairs of ribs do not articulate directly with the sternum but join the seventh rib with the help of hyaline cartilage.

How many of the above statements are correct?

(1) Four (2) Three

(3) Two (4) One

BLC258

29. Read the following statements and select the correct option with regard to pelvic girdle ?

(A) It consists of two coxal bones.

(B) Each coxal bones is formed by the fusion of ilium, ischium and pubis.

(C) Ilium bone has a cavity called glenoid cavity.

(D) The two halves of the pelvic girdle meet dorsally to form the pubic symphysis.

Option :-

(1) Statements A, B and C correct while D is incorrect

(2) Statements A and B are correct while C and D are incorrect

(3) Statement A and B are incorrect while C and D are correct

(4) Only statement A is correct while B, C and D are incorrect

BLC259

30. Select the correct option :-

	Column -A	Column -B
(1)	Between humerus and pectoral girdle	Hinge joint
(2)	Between atlas and axis	Pivot joint
(3)	Between carpals	Saddle joint
(4)	Between the carpal and metacarpal of thumb	Gliding joint

BLC260

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31. **Assertion:** Bone have very hard matrix.
Reason: Matrix of bone contain calcium salt in it.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.
BLC261

32. **Assertion:** Osteoporosis is characterised by increase in bone mass.
Reason: Its common cause is increased level of oestrogen.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.
BLC262

33. **Assertion:** Skeleton helps in the blood cell formation.
Reason: Blood flows through skeleton.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.
BLC263

34. **Assertion:** Ball and socket joints are the most mobile joints.
Reason: Synovial fluid is present here in synovial cavity such arrangement allows considerable movement.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.
BLC264

35. **Assertion:** Skeletal muscles are striated muscles
Reason: They have striped appearance under the microscope
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.
BLC265

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	4	2	4	2	1	1	3	1	1	2	2	2	1	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	1	3	4	2	2	2	3	2	3	3	3	3	2	2
Question	31	32	33	34	35										
Answer	1	4	3	1	1										

Locomotion and Movement-II

1. Which of the following is the regulatory muscle protein having double helix structure?
(1) Troponin (2) Actin
(3) Ossein (4) Tropomyosin
BLC266

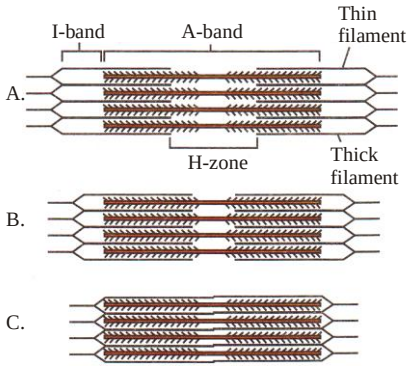
2. Find out incorrect statement:
(1) All locomotions are movement, but all movements are not locomotions
(2) Red muscle fibres having more mitochondria and myoglobin
(3) Skeletal muscles are voluntary and striated
(4) Cardiac muscles are unstriated, branched and involuntary.
BLC267

3. In the resting muscle fibre, tropomyosin covers:
(1) Ca-binding sites on troponin
(2) Actin binding sites on myosin
(3) Myosin binding sites on actin
(4) Ca-binding sites on actin
BLC268

4. Store house of calcium in muscle is ?
(1) Sarcomere
(2) Sarcosome
(3) Sarcoplasmic reticulum
(4) Lysosome
BLC269

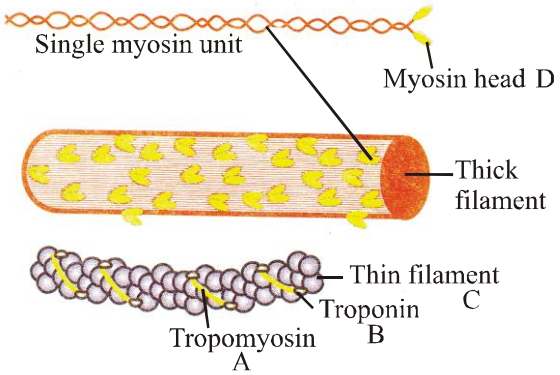
5. Cardiac muscle fibres are -
(1) Striated and voluntary
(2) Unstriated and voluntary
(3) Striated and involuntary
(4) Unstriated and involuntary
BLC270

6. In following diagram which is showing complete contraction :-



(1) A (2) B
(3) C (4) All of these
BLC271

7. In following diagram which protein functions as enzyme ?



(1) A (2) B (3) C (4) D
BLC272

8. Which one is correct for white muscle fibre:
(1) More mitochondria
(2) Less myoglobin
(3) More O_2 transportation
(4) More myoglobin
BLC273

9. ATPase enzyme, needed for muscle contraction is located in :
(1) Troponin (2) Myosin
(3) Actin (4) Tropomyosin
BLC274

10. Which one is not possible during contraction:
(1) I band remains same
(2) A band remains same
(3) Sarcomere length decrease
(4) H-zone disappears
BLC275

11. During the maximum muscle contraction, length of which of the following remains unchanged?
(1) Isotropic band
(2) Anisotropic band
(3) M–Band
(4) Length of Sarcomere
BLC276

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12. Which muscles are primarily involved in locomotory actions and changes of body postures:
(1) Cardiac muscles (2) Smooth muscles
(3) Skeletal muscles (4) All of the above
BLC277

13. Points which are correct with respect to cardiac muscle?
(i) Have dark and light bands
(ii) Involuntory in nature
(iii) Similar to smooth muscle in structure
(iv) Muscle fibres are branched, cylindrical
(v) get fatigued after a long period.
(1) (i), (ii), (iii) (2) (i), (ii), (iv)
(3) (iii), (iv), (v) (4) (i), (iii), (v)
BLC278

14. The distance or length of the myofibril between two adjacent Z-bands is called
(1) Sarcomere (2) Sarcolemma
(3) Muscle fibre (4) Sarcoplasm
BLC279

15. In a relaxed fibril, H-zone, a lighter region of low density can be seen in the centre of
(1) Anisotropic or A-band
(2) Isotropic or I-band
(3) Z-band
(4) Both in A and I-band
BLC280

16. Which is true for muscle contraction: -
(1) Sarcolemma becomes permeable to Ca^{++} ions
(2) Sarcolemma becomes permeable to Na^{+} ions
(3) Sarcolemma becomes impermeable to Na^{+} ions
(4) Concentration of Ca^{++} ions is reduced in sarcoplasm.
BLC281

17. Low ionized calcium level in the extra cellular fluid increases the permeability of neuronal membranes to sodium ion, causing a progressive depolarisation, which increases the possibility of action potential. This condition causes: -
(1) Gout (2) Myasthenia gravis
(3) Tetany in muscles (4) Tetanus in muscles
BLC282

18. Which point is incorrect about *Myasthenia gravis* disease: -
(1) Neuromuscular junction is affected
(2) Auto-immune disorder
(3) Secretion from nerve terminal is reduced
(4) Can lead to fatigue, weakening and paralysis of skeletal muscle.
BLC283

19. During contraction in skeletal muscle which reaction takes place in myofibril: -

	Light band	Dark band	H-Zone
(1)	Become smaller	Become smaller	Become small
(2)	Become smaller	Remains unchanged	Remains unchanged
(3)	Become smaller	Remains unchanged	Disappear
(4)	Remains unchanged	Become smaller	Become smaller

BLC284

20. During muscle contraction
(1) Chemical energy is changed into electrical energy
(2) Chemical energy is changed into mechanical energy
(3) Chemical energy is changed into physical energy
(4) Mechanical energy is changed into chemical energy
BLC285

21. Consider the following four statements (A-D) and select the option which includes all the correct ones only.
(A) Each organised skeletal muscle is made of a number of muscle bundles or fasciculi
(B) Each muscle fibre has many parallely arranged myofibrils.
(C) Each myofibril contains many serially arranged units called sarcomere.
(D) Each sarcomere has a central "A"–band and two half "I" bands.
Options :-
(1) Statements (B), (C) and (D)
(2) Statements (A) and (B)
(3) Statements (C) and (D)
(4) Statements (A), (B), (C), (D)
BLC286

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22. Which of the following structure is **correctly** organised from large to small?
(1) Muscle, Muscle fiber, Myofibril, Sarcomeres, Filaments.
(2) Muscle, muscle fibre, Sarcomere, Filaments Myofibril.
(3) Muscle, Filaments, Myofibrils, Sarocomere, Muscle fibre
(4) Filaments, Sarcomere, Myofibril, Muscle fibre, Muscle.

BLC287

23. In the alimentary canal and reproductive tract which type of muscles are present ?
(1) Striated, involuntary
(2) Striated, voluntary
(3) Unstriated, voluntary
(4) Unstriated, involuntary

BLC288

24. How many points are **correct** about ***visceral muscles*** :-
(i) Located in the inner walls of hollow visceral organs of the body
(ii) They do not exhibit any striations
(iii) They are primarily involved in locomotary actions and changes of body postures
(iv) Their activities are not under the voluntary control of the nervous system
(v) Contraction slow but for long period
(vi) Their muscle fibre is short, Spindle shape, branched, Uninucleated
(1) Two (2) Three (3) Four (4) Five

BLC289

25. In the rest state, a subunit of Troponin masks:
(1) Active binding sites for actin on the myosin filaments
(2) Active binding sites for myosin on the myosin filaments
(3) Active binding sites for myosin on the actin filaments
(4) Active binding sites for actin on the actin filaments

BLC290

26. The role of calcium in muscle contraction is :-
(1) to break the cross-bridges as a cofactor in the hydrolysis of ATP.
(2) to bind with troponin, changing its shape so that the active site on actin filament is exposed.
(3) to transmit the action potential across the neuromuscular junction.
(4) to spread the action potential through the T tubules.

BLC291

27. Which of the following statements about cross-bridges is false ?
(1) They are composed of myosin
(2) They bind to ATP after they attach to actin.
(3) They contain a ATPase.
(4) They split ATP before they attach to actin.

BLC292

28. Collagenous connective tissue layer holding the muscle bundles together is :-
(1) Sacroplasm (2) Pleura
(3) Fascia (4) Sarcolemma

BLC293

29. Which of the following statements are **correct**?
(i) Muscle contraction is initiated by a signal sent by motor neuron of CNS .
(ii) During muscle contraction, isotropic band gets elongated.
(iii) The dark band contains only myosin filament.
(iv) A motor neuron along with the muscle fibres connected to it constitute a motor unit.
(1) (i) and (ii) are correct
(2) (ii) and (iii) are correct
(3) (iii) and (iv) are correct
(4) (i) and (iv) are correct

BLC294

30. **Assertion :** Red muscle fibres contain plenty of myoglobin.
Reason : They are called aerobic muscles.
(1) Assertion and Reason both are incorrect
(2) Assertion is correct and Reason is incorrect
(3) Assertion and Reason both are correct; Reason is the correct explanation of Assertion
(4) Assertion and Reason both are correct; Reason is not the correct explanation of Assertion

BLC295

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31. **Assertion :** The 8th, 9th and 10th pair of ribs join the seventh rib.
Reason : 8th, 9th and 10th pair of ribs are called vertebrochondral ribs
(1) Assertion and Reason both are incorrect
(2) Assertion and Reason both are correct; Reason is the correct explanation of Assertion
(3) Assertion and Reason both are correct and Reason is the correct explanation of Assertion
(4) Assertion is incorrect whereas Reason is correct

BLC296

32. **Assertion:** Smooth muscles are involuntary muscles.
Reason: Their activities are not under voluntary control of the nervous system.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.

BLC297

33. **Assertion:** Ca^{2+} ion plays important role in muscle contraction.
Reason: Calcium ion binds to subunit of troponin on actin filament and removes the masking of active sites for myosin.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.

BLC298

34. **Assertion:** Gout is inflammation of muscles.
Reason: It occur due to deposition of urea in muscles.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.

BLC299

35. **Assertion:** White muscle fibres have very less quantity of myoglobin.
Reason: White muscle fibres have high number of sarcoplasmic reticulum.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.

BLC300

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	4	3	3	3	3	4	2	2	1	2	3	2	1	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	3	3	3	2	4	1	4	3	3	2	2	3	4	4
Question	31	32	33	34	35										
Answer	3	1	1	4	2										

Excretory Products and Their Elimination

1. The forms of nitrogenous waste excreted by animals is /are :
(1) ammonia (2) urea
(3) uric acid (4) All
BEE262

2. The most toxic form of nitrogenous waste is
(1) Ammonia (2) urea
(3) uric acid (4) amino acid
BEE263

3. Which of the following nitrogenous waste requires minimum and maximum amount of water respectively for its removal from the body ?
(1) urea, uric acid
(2) uric acid, ammonia
(3) ammonia, uric acid
(4) urea, ammonia
BEE264

4. Process of excretion of ammonia is called
(1) ammonotelism (2) ureotelism
(3) uricotelism (4) amminotelism
BEE265

5. Ammonia produced by metabolism gets converted into __A__ in the organ __B__.
(1) A = uric acid, B = stomach
(2) A = urea, B = Liver
(3) A = uric acid, B = kidney
(4) A = ammonia, B = liver
BEE266

6. Which of the following statements is incorrect with respect to human kidney?
(1) The peripheral region is the cortex and central is medulla
(2) Bowman's capsules are present in the cortex region
(3) Blood enters glomerulus through efferent arteriole
(4) The concave part of kidney consist of a notch called hilum
BEE267

7. Which of the following options is incorrect ?
(1) Protozoa — excretory organs are absent but excretion takes place by body surface
(2) Platyhelminthes — excretory organs are flame cells
(3) Earthworm — excretory organs are protonephridia
(4) Frog — excretory organs are kidney
BEE268

8. Kidneys are situated between the levels of ___A___ and ___B___ vertebrae
(1) A = first cervical, B = last thoracic
(2) A = last thoracic, B = third lumbar
(3) A = Last cervical, B = third Lumbar
(4) A = Last thoracic, B = First sacral
BEE269

9. The medulla is divided into few conical masses known as :
(1) calyces
(2) renal pelvis
(3) medullary pyramids
(4) hilum
BEE270

10. The cortex extends in between the medullary pyramids known as
(1) columns of Bertini
(2) Bowman's capsule
(3) proximal convoluted tubule
(4) Renal papilla
BEE271

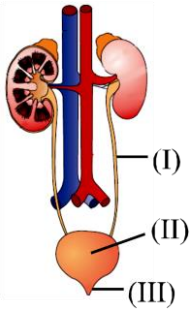
11. Almost all the aquatic animals excrete ammonia as the nitrogenous waste product. Which of the following statement is not in agreement with this situation?
(1) ammonia is easily soluble in water
(2) ammonia is insoluble in water
(3) ammonia is highly toxic and needs to be eliminated as and when formed
(4) ammonia gets converted into a less toxic form called urea
BEE272

12. Glomerulus is a tuft of capillaries formed by :
(1) efferent arteriole
(2) afferent arteriole
(3) Bowman's capsule
(4) PCT
BEE273

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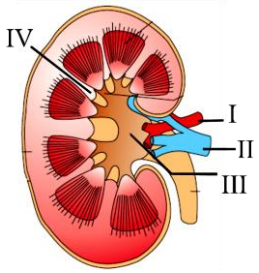
13. Which is the correct order for the path taken by urine after it leaves the kidney ?



(1) I → II → III (2) III → I → II
(3) II → III → I (4) II → I → III

BEE274

14. The given figure shows the longitudinal section of kidney with few structures labeled as I, II, III and IV. Identify renal vein in the given figure :-



(1) I (2) II (3) III (4) IV

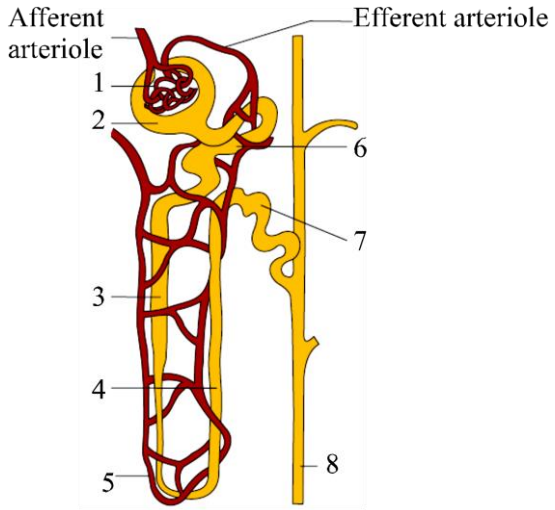
BEE275

15. In a mammalian kidney, Bowmans capsule occurs in ___A_____ whereas loop of Henle are situated in ___B____.

(1) A = renal cortex, B = renal medulla
(2) A = renal medulla, B = renal cortex
(3) A = renal cortex, B = renal pelvis
(4) A = renal medulla, B = renal pelvis

BEE276

16. Observe the diagram and answer the question asked :-

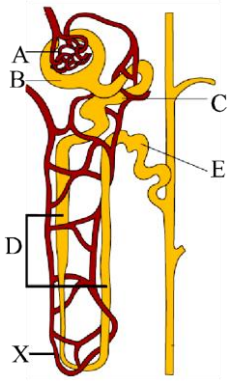


Which part is lined by simple cuboidal brush bordered epithelium to increase the surface area for reabsorption ?

(1) 3 , descending limb of loop of Henle
(2) 5 , Vasa recta
(3) 6 , Proximal convoluted tubule
(4) 7 , collecting duct

BEE277

17. The label X in the given diagram represents _____that function in _____ :-



(1) vasa recta — reabsorption of water and concentration of urine
(2) Henle's loop — filtration of plasma leaving the blood
(3) vasa recta — filtration of plasma leaving the blood
(4) Henle's loop — reabsorption of water, minerals and digestive end products

BEE278

18. Which of the following is the correct match with respect to function of human nephron?

(1) afferent arteriole — carries the blood away from the glomerulus towards renal vein
(2) podocytes — create minute spaces (slit pores) for the filtration of blood into the Bowman's capsule
(3) Henle's loop — reabsorption of the major substances from the glomerular filtrate
(4) Distal convoluted tubule — reabsorption of K^+ ions into the surrounding blood capillaries

BEE279

19. Which of the following is not a constituent of nephric filtrate?

(1) glucose (2) amino acids
(3) fatty acids (4) Proteins

BEE280

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20. Total number of layers in filtration membrane are :
(1) one (2) two
(3) three (4) four

BEE281

21. On an average, _____ml of blood is filtered by kidney per minute.
(1) 1100 - 1200 (2) 1500 - 1600
(3) 2000 - 2200 (4) 4500 - 4600

BEE282

22. In which segment of nephron conditional reabsorption of Na^+ and water takes place ?
(1) PCT
(2) DCT
(3) Collecting duct
(4) Loop of Henle

BEE283

23. Glomerular filtration rate (GFR) in a healthy individual is approximately
(1) 100 ml/minute, i.e., 180 litres per day.
(2) 125 ml/minute, i.e., 180 litres per day.
(3) 120 ml/minute, i.e., 100 litres per day.
(4) 130 ml/minute, i.e., 120 litres per day.

BEE284

24. Which of the following statements are true?
(a) Filterate is hypotonic in DCT
(b) when the filterate passes into the collecting duct, it becomes hypotonic
(c) filterate is isotonic in PCT
(d) filterate becomes more and more hypotonic as it passes through the loop of Henle
(1) a and d (2) a, b and c
(3) b and d (4) a and c

BEE285

25. Which of the following hormone is secreted from kidney ?
(1) ANF (2) Erythropoietin
(3) Rennin (4) Aldosterone

BEE286

26. Which of the following parts of the nephron given in column A is correctly matched with their functions given in column B ?

	Column-A		Column-B
I	Proximal convoluted tubule	a	Sodium is actively reabsorbed in this region
ii	Distal convoluted tubule	b	Sodium and water are reabsorbed under the influence of hormone in this region
iii	Descending limb	c	Primary site of glucose and amino acid reabsorption
iv	Ascending limb	d	Reabsorbtion of water

	i	ii	iii	iv
(1)	a	b	c	d
(2)	b	c	d	a
(3)	c	b	d	a
(4)	d	c	a	b

BEE287

27. ANF is released in response to the increase in blood volume and blood pressure. Which of the following is not the function of ANF?
(1) stimulates aldosterone secretion
(2) inhibit the secretion of rennin from JGA
(3) stimulates salt loss in urine
(4) inhibit sodium reabsorption from renal tubule

BEE288

28. Which of the following is a correct match of a specific part of a human nephron with its function ?
(1) Renin — enzyme that catalyses the formation of angiotensin I
(2) Aldosterone — regulates water reabsorption at the distal convoluted tubule
(3) Antidiuretic hormone — It is a powerful vasoconstrictor that stimulates the secretion of aldosterone
(4) Angiotensin II — promotes reabsorption of sodium at proximal convoluted tubule

BEE289

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29. Select the correct match ?

	Column I		Column II
A	Fall in GFR	i	Activate the JG cells to release renin
B	Angiotensin II	ii	Increase the glomerular blood pressure and thereby GFR
C	Renin	iii	Carries out the conversion of angiotensinogen in blood to angiotensin I
D	Aldosterone	iv	Causes reabsorption of Na^+ and water from the distal parts of the tubule. This also leads to an increase in blood pressure and GFR
E	An excessive loss of fluid from the body	v	Activates osmoreceptors which stimulates the hypothalamus

	A	B	C	D	E
(1)	i	ii	iii	iv	v
(2)	iii	v	ii	iv	i
(3)	iii	i	v	ii	iv
(4)	v	iv	iii	i	ii

BEE290

30. Volume of urine is regulated by
(1) aldosterone
(2) aldosterone and ADH
(3) aldosterone, ADH and testosterone
(4) ADH alone

BEE291

31. Under normal conditions which one is completely reabsorbed in the renal tubule?
(1) Salts (2) Urea
(3) Glucose (4) Uric acid

BEE292

32. (A) causes reabsorption of (B) and water from the distal parts of the tubule. This also leads to an increase in (C) and glomerular filtration rate. Identify (A), (B) and (C):

	A	B	C
(1)	ADH	Sodium ion	Blood pressure
(2)	ADH	Potassium ion	Blood pressure
(3)	Aldosterone	Sodium ion	Blood pressure
(4)	Aldosterone	Potassium ion	Ionic concentration

BEE293

33. Which of the following statement is correct?
(1) Vasa recta is not present in cortical nephrons.
(2) Maximum number of nephrons in kidney are juxtamedullary type.
(3) DCT of many nephrons open into collecting tubule.
(4) All of the above

BEE294

34. Which one of the following is metabolic waste of protein metabolism?
(1) Urea, ammonia and CO_2
(2) Urea, ammonia and glucose
(3) Urea, ammonia and fatty acid
(4) Urea, nitrogen and O_2

BEE295

35. Which of the following group of animals are ammoniotelic in nature?
(1) Many bony fishes, amphibians, Insects.
(2) Marine fishes, amphibians, aquatic insects
(3) Mammals, birds, reptiles
(4) Many bony fishes, aquatic amphibians, aquatic insects.

BEE296

36. Which of the following statements is false?
I. Outer cortex and inner medulla are the two zones in kidney.
II. Medulla is divided into renal pyramids.
III. Pyramid projects into calyx.
IV. Inward extension of cortex between the pyramids is called renal column of Bertini.
(1) I and IV (2) II and IV
(3) IV (4) None

BEE297

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37. Which of the following statement is not correct with respect to human kidney?
(1) The peripheral region is called cortex and central medulla.
(2) Malpighian capsules are present in the cortex region.
(3) Blood enters glomerulus through efferent arterioles.
(4) The concave part of kidney is called hilus.
BEE298

38. Which one of the following is not a part of a renal pyramid?
(1) Convoluted tubules
(2) Collecting ducts
(3) Loops Henle's
(4) Peritubular capillaries
BEE299

39. The main function of Henle's is loop is :-
(1) Conservation of water
(2) Filtration of blood
(3) Passage of urine
(4) Formation of urine
BEE300

40. The vessel taking blood (containing nitrogenous wastes) into the Bowman's capsule is known as :-
(1) Afferent arteriole
(2) Efferent arteriole
(3) Renal artery
(4) Renal vein
BEE301

41. Ureter develops from funnel like structure called :-
(1) Hilum (2) Renal pelvis
(3) Major calyx (4) Minor calyx
BEE302

42. Substances like amino acids and glucose in the filterate are reabsorbed __a__ in the tubular epithelial cells whereas the nitrogenous wastes are absorbed by __b__ transport :-
(1) a-actively b-passive
(2) a-passively b-active
(3) a-actively b-active
(4) a-passively b-passive
BEE303

43. Which one of the following statements in regard to the excretion by the human kidneys is correct?
(1) Ascending limb of Loop of Henle is impermeable to electrolytes.
(2) Descending limb of Loop of Henle is impermeable to water.
(3) Distal convoluted tubule is incapable of reabsorbing HCO_3^-.
(4) Nearly 99 per cent of the glomerular filtrate is reabsorbed by the renal tubules.
BEE304

44. Which statement is wrong ?
(1) The counter current mechanism changes the isotonic glomerular filtrate into a hypertonic urine by increasing salt concentration around the nephron and collecting tubule.
(2) The wall of collecting tubule is permeable to water whereas ascending limb is impermeable to water.
(3) The absorption of water in DCT is facultative.
(4) As the filtrate passes through the ascending limb, sodium is transported passively in medulla.
BEE305

45. The CNS passes on motor message to initiate the __a__ of smooth muscles of the urinary bladder and simultaneous ___b___ of the urethral sphincter causing the release of the urine.
(1) a-contraction b-relaxation
(2) a-relaxation b-contraction
(3) a-relaxation b-relaxation
(4) a-contraction b-contraction
BEE306

46. ANF mechanism, acts as a check on the :-
(1) Renin-angiotensin mechanism
(2) Couter-current mechanism
(3) JGA to secrete erythropoietin
(4) Micturition reflex
BEE307

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47. Which of the following is not associated with micturition reflex?
(1) Relaxation of urethral sphincter
(2) Contraction of detrusor muscle
(3) Activation of stretch receptors in wall of ureters
(4) Neural mechanism govern micturition reflex

BEE308

48. Which of the following statements are correct?
(i) Glucose has high threshold value.
(ii) Urine is concentrated in Henle's loop.
(iii) Haemodialyser removes urea, uric acid, glucose and proteins.
(iv) In glomerulus, urea, uric acid, water, glucose and plasma proteins are filtered out.
(1) (i), (iii) and (iv)
(2) (ii), (iii) and (iv)
(3) (i) and (ii)
(4) (i) and (iii)

BEE309

49. Match the excretory functions given in column-I with the parts of the excretory system in column-II. Choose the correct combination from the given options.

	Column – I (Function)		Column – II (Parts of excretory systems)
A.	Ultra filtration	I.	Henle's loop
B.	Concentration of urine	II.	Ureter
C.	Transport of urine	III	Urinary bladder
D.	Storage of urine	IV	Malpighian corpuscle
		V	Proximal convoluted tubule

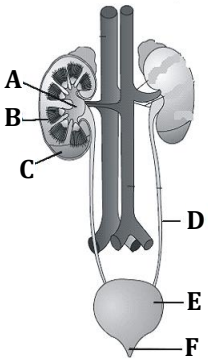
(1) A – IV; B – I; C – II; D – III
(2) A – IV; B – III; C – II; D – I
(3) A – V; B – IV; C – I; D – III
(4) A – V; B – IV; C – I; D – II

BEE310

50. Detrusor muscles are present mainly in :-
(1) Coat of kidney
(2) Muscular coat of urinary bladder
(3) Renal fasica of kidney
(4) Cortex of kidney

BEE311

51. Identify A, B, C and D, E and F in the given diagram ?



(1) A = Pelvis, B = Cortex, C = Medulla, D=Urethra, E=Urinary bladder, F=Ureter
(2) A = Pelvis, B = Medulla, C = Cortex, D=Urethra, E=Urinary bladder, F=Ureter
(3) A = Pelvis, B = Medulla, C = Cortex, D=Ureter, E=Urinary bladder, F=Urethra
(4) A = Pelvis, B = Cortex, C = Medulla, D=Ureter, E=Urinary bladder, F=Urethra

BEE312

52. Match the disorders given in column-I with their feature given in column-II and choose the correct option.

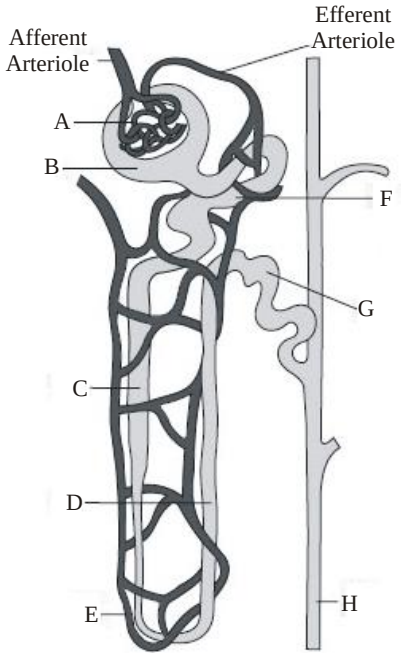
	Column – I (Disorders)		Column – II (Feature)
A.	Uremia	I.	Excess of protein in urine
B.	Hematuria	II.	Presence of high ketone bodies in urine
C.	Ketonuria in urine	III	Presence of blood cells
D.	Glycosuria urine	IV	Presence of glucose in urine
E.	Proteinuria	V	Excess of urea in blood

(1) A - V; B -III; C - II; D - IV; E - I
(2) A - IV; B - V; C - III; D - II; E - I
(3) A - V; B -III; C - IV; D - II; E - I
(4) A - III; B -V; C - II; D - I; E - IV

BEE313

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53. The given figure of nephron showing blood vessels and duct. Select the right answer



Which part are lined by simple squamous epithelium?

(1) B and E (2) B and G
(3) B and C (4) B and H

BEE314

54. Which of the following blood vessel contain minimum amount of urea?

(1) dorsal aorta (2) renal vein
(3) renal artery (4) hepatic artery

BEE315

55. Which statement is true regarding reabsorption.

(1) Nearly 99 percent of the filtrate has to be reabsorbed by the renal tubules.
(2) Nearly all of the essential nutrients are reabsorbed by PCT.
(3) DCT is also capable of reabsorption of HCO_3^-
(4) All the above

BEE316

56. Human urine contains

(1) 95% water, 2.6% urea, 2% salts and some uric acid.
(2) 99% water and 1% urea.
(3) 92% water and 8% salts.
(4) 90% water, 8% uric acid and 2% proteins.

BEE317

57. Find out the **correct** statement :

(1) An increase in body fluid volume stimulate the ADH release.
(2) A fall in glomerular blood flow can activate the JG cells to release renin.
(3) Angiotensin-II, being a powerful vasodialator, decreases the glomerular blood pressure.
(4) Decrease in blood flow to the atria of the heart can cause the release of atrial natriuretic factor (ANF)

BEE318

58. Select the correct option with the reference of hemodialysis

(1) when nitrogenous waste concentration is high in blood then nitrogenous waste are removed by hemodialysis
(2) anticoagulant like heparin used during hemodialysis
(3) Dilaysing fluid consist of all components of plasma except N_2 waste
(4) all of the above

BEE319

59. Which of the following not takes place when there is a fall in glomerular filteration rate?

(1) posterior pituitary to release vasopressin
(2) juxtaglomerular cells to release renin
(3) adrenal medulla to release adrenaline
(4) adrenal cortex to release aldosterone

BEE320

60. Which are the following group of hormones participate in the regulation of the renal function?

(1) ADH, TSH and ANF
(2) PCT, TSH and ANF
(3) ADH, DCT and Aldosterone
(4) ADH, ANF and Aldosterone

BEE321

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61. Ammonia which is produced by metabolism is converted into _____ in the liver of mammals and released into _____ which is filtered and _____ out by kidney

(1) A–Uric acid, B–Blood, C–excreted

(2) A–Urea, B–Blood, C–excreted

(3) A–Amino acid, B–Blood, C–excreted

(4) A–Sugar, B–Blood, C–excreted

BEE322

62. Which of the following is incorrect about human kidney?

(1) Kidney is covered by tough capsule

(2) Kidney is divided into cortex and medulla on the outer side

(3) The cortex is extended in between the medullary pyramid and the renal column of bertini.

(4) Kidney is situated close to the dorsal inner wall of abdominal cavity.

BEE323

63. Select the incorrect statement from the following:

(1) The DCTs of many nephrons opens into a straight tube called collecting duct.

(2) In cortical nephrons (majority), the loop of Henle is too short and extended only very little in medulla.

(3) In juxta medullary nephrons (minority), the loop of Henle is very long and runs deeply into medulla.

(4) Vasa recta is not a part of peritubular network.

BEE324

64. Blood vessel draining the glomerulus in a mammalian nephron is called

(1) Afferent arteriole and is narrower than the vessel entering it.

(2) Efferent venule and is narrower than the vessel entering it.

(3) Efferent arteriole and is narrower than the vessel entering it.

(4) Renal artery and is wider than the vessel entering it.

BEE325

65. The glomerular filtration rate would be decreased by

(1) Constriction of the efferent arteriole

(2) An increase in afferent arteriolar pressure

(3) Compression of the renal capsule

(4) An increase in the renal blood flow

BEE326

66. Urine formation mainly involves the process of:

(1) Ultrafiltration

(2) Selective reabsorption

(3) Secretion

(4) All of these

BEE327

67. Choose the correct statement about absorption in renal tubules from the following:

(1) Glucose, amino acids and Na+ reabsorbed actively.

(2) Nitrogenous waste are absorbed by passive transport.

(3) 70–80 per cent of electrolyte and water are absorbed in PCT.

(4) All the above

BEE328

68. Which of the following is incorrect about PCT?

(1) Lined with simple cuboidal brush border epithelium.

(2) All essential nutrient and 70 to 80 per cent of the electrolyte and water are reabsorbed here.

(3) It helps in the PH maintenance of body fluid by the selective secretion of H^+ ion and by the absorption of HCO_3^-.

(4) It does not help in the maintenance of ionic balance of body fluid.

BEE329

69. The effect of antidiuretic hormone (ADH) on the kidney is to increase the
(1) Excretion of water
(2) Excretion of Na'
(3) Permeability of the distal nephron to water
(4) Glomerular filtration rate

BEE330

70. Arrange the following steps in order
(1) Excessive loss of fluid
(2) Stimulation of osmoreceptor
(3) Stimulation of Hypothalamus
(4) Release of ADH or Vasopressin
(5) ADH facilitate water reabsorption from distal tubules
(6) Increase in body fluid switch off osmoreceptor and suppress the release of ADH.
(1) 1, 2, 3, 4, 5, 6
(2) 1, 3, 2, 4, 5, 6
(3) 6, 1, 2, 3, 4, 5
(4) 2, 3, 4, 1, 5, 6

BEE331

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	1	2	1	2	3	3	2	3	1	2	2	1	2	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	1	2	4	3	1	2	2	4	2	3	1	1	1	2
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	3	3	1	1	4	4	3	1	1	1	2	1	4	4	1
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	1	3	3	1	2	3	1	1	2	4	1	2	4	3	4
Question	61	62	63	64	65	66	67	68	69	70					
Answer	2	2	4	3	3	4	4	4	3	1					

Neural Control and Coordination

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1. Match List I with List II.

	List I		List II
(a)	Node of Ranvier	(i)	Multipolar
(b)	Synaptic knob	(ii)	Unipolar
(c)	Neuron found in cerebral cortex	(iii)	Gaps between two adjacent myelin sheath
(d)	Neuron found in embryonic stage	(iv)	Bulb-like structure which possess vesicles containing chemicals

(1) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)
(2) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)
(3) (a)-(i), (b)-(iv), (c)-(iii), (d)-(ii)
(4) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)

BNC266

2. Find out the correct match :-

	Column-I		Column-II
(A)	Dendron	(i)	Protein synthesis
(B)	Nissl's granule	(ii)	Saltatory conduction
(C)	Neurofibril	(iii)	Afferent part
(D)	Myelinated Axon	(iv)	Efferent part
(E)	Nerve fibre	(v)	Helps in internal conduction

(1) A-iii, B-i, C-ii, D-iv, E-v
(2) A-iv, B-i, C-ii, D-iii, E-v
(3) A-iv, B-v, C-i, D-ii, E-iii
(4) A-iii, B-i, C-v, D-ii, E-iv

BNC267

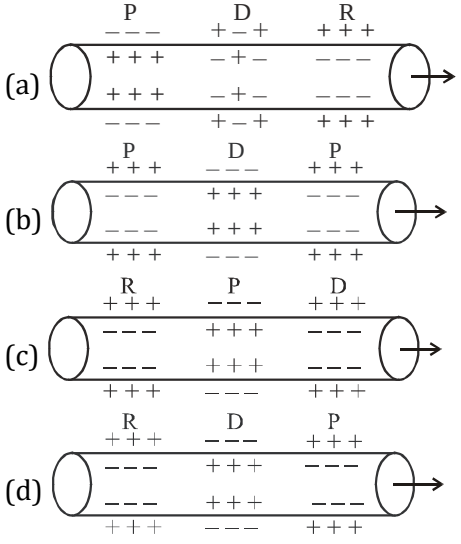
3. Transmission of nerve impulse or action potential is not possible in those neurons which are :-
(1) Discontinuous myelinated with node of Ranvier.
(2) Continuous myelinated without node of ranvier.
(3) Non myelinated axon
(4) Transmission is possible in myelinated and not myelinated neuron both

BNC268

4. **Statement I :** The neural system provides an organised network of point to point connections for a quick coordination.
Statement II: The cell body contains cytoplasm with typical cell organelles and certain granular bodies called Nissl's granules.
Statement III: Dendrites transmit impulses away from the cell body.
Statement IV: Transmission of an impulse across electric synapses is very similar to impulse conduction along a single axon.
(1) Only two
(2) Only three
(3) Only one
(4) All are correct

BNC269

5. Find out the correct labelling in following diagrammatic representation :-
(R = Repolarisation, P = Polarisation,
D = Depolarisation)

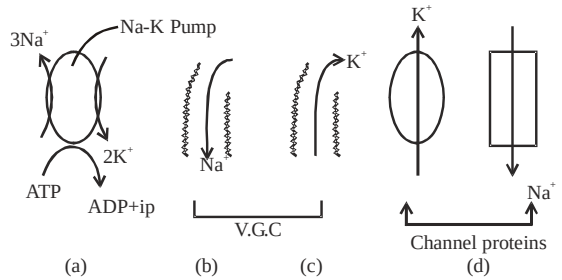


(1) (a) (2) (b)
(3) (d) (4) (c)

BNC270

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6. Find out the answers of given statements according to diagrams :-



Which of above diagram/pathways are closed during depolarisation of neuron ?

(1) Only (a) & (d) (2) Only (a) & (c)
(3) Only (a) & (b) (4) Only (b) and (c)

BNC271

7. Which one of the following is correct :-

(1)	Efferent fibres	Transmit regulatory impulses from peripheral tissues to CNS
(2)	Somatic neural	Transmit impulses from system CNS to skeletal muscles
(3)	Autonomous	Transmit impulses from nervous system voluntary muscles to CNS
(4)	Afferent nerve	Transmit impulses from fibres sympathetic nervous system to parasympathetic nervous system.

BNC272

8. Four healthy people in their twenties got involved in injuries resulting in damage and death of few cells of the following. Which of the cells are least likely to be replaced by new cells :-

(1) Osteocytes
(2) Malpighian layer of the skin
(3) Liver cells
(4) Neurons

BNC273

9. GABA acts by :-

(1) Increasing the opening of $Na^{\oplus}$ gated channels
(2) Increasing the opening of $Ca^{\oplus}$ gated channels
(3) Decreasing the opening of $K^{\oplus}$ gated channels
(4) Increasing the opening of Cl^{-} gated channels

BNC274

10. Which of the following parts of brain associated with regulation of circadian rhythms and human behaviour ?

(1) Thalamus
(2) Hypothalamus
(3) Cerebral cortex
(4) Medulla oblongata

BNC275

11. **Assertion (A):** Inner part of cerebral hemisphere is of greyish appearance.

Reason (R): Fibre of tract are covered with concentrated phospholipid sheath which constitute the outer part of cerebral hemisphere.

(1) Both A and R are correct but R is not the correct explain of A.
(2) Both A and R are correct and R is the correct explanation of A.
(3) A is correct but R is incorrect.
(4) Both A and R are incorrect.

BNC276

12. **Statement I:** The inner parts of cerebral hemispheres and a group of associated deep structures like amygdala, hippocampus, etc, form a complex structure called the limbic lobe.

Statement II: Hypothalamus along with limbic system is involved in regulation of sexual behaviour, expression of emotional reactions and motivation.

(1) Statement I is correct but statement II is incorrect.
(2) Statement I is incorrect but statement II is correct.
(3) Both statement I and II are correct.
(4) Both statement I and II are incorrect

BNC277

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13. The thermoregulatory centre in the body of human females is found in :-
(1) Diencephalon (2) Hypothalamus
(3) Pituitary (4) Skin

BNC278

14. Crura cerebri is present in _____ and is concerned with controlling the muscles of ___.
(1) Forebrain, limbs (2) Midbrain, limbs
(3) Hindbrain, limbs (4) Heart, limbs

BNC279

15. Match the following columns.

	Column – I		Column – II
i.	Layer in contact with brain tissue	a.	Pons
ii.	Integrates visual, tactile, and auditory inputs	b.	Mid brain
iii.	Convoluted surface provides additional space for neurons	c.	Pia mater
iv.	Interconnect different regions of brain	d.	Cerebellum

(1) (i)-(d), (ii)-(b), (iii)-(a), (iv)-(c)
(2) (i)-(c), (ii)-(b), (iii)-(d), (iv)-(a)
(3) (i)-(a), (ii)-(b), (iii)-(c), (iv)-(d)
(4) (i)-(b), (ii)-(c), (iii)-(a), (iv)-(d)

BNC280

16. A man is admitted in hospital. He is suffering from an abnomally low body temprature, loss of appetite and extreme thirst. His brain scan would probably show a tumor in :-
(1) Medulla oblongata
(2) Pons varoli
(3) Cerebellum
(4) Hypothalamus

BNC281

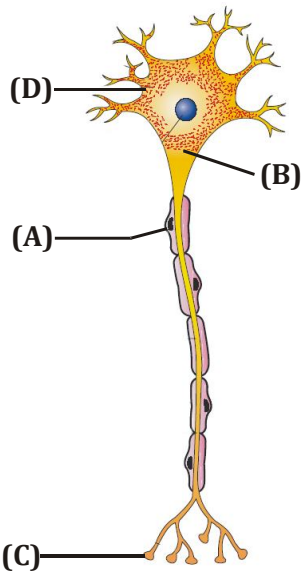
17. Which of the following is incorrectly matched?
(1) Crura cerebrai – Limb-muscle control
(2) Hypothalamus – Body temprature
(3) Cerebrum – vomiting centre
(4) Medulla oblongata – Respiratory centre

BNC282

18. Which of the following is not a function of basal part of diencephelon?
(1) It stimulates reabsorption of H_2O from renal tubules by it's secretion.
(2) Participates in regulation of biological clock
(3) Pathway of sensory signaling sent to crebrum
(4) Autonomic and emotional responses

BNC283

19. Choose the correct option for label (A), (B), (C) and (D) respectively in given diagram.



(1) Node of Ranvier, Trigger zone, myelin sheath Nissl's grannule
(2) Schwann cell, Trigger zone, synaptic knob Nissl's grannules.
(3) Nissl's grannule, synaptic knob, cell body Schwann cell
(4) Schwann cell, Synaptic knob, Trigger zone, Nissl's grannules

BNC284

20. Mark the incorrect one regarding the structures of a nerve cell ?
(1) *Dendrites* — short and branched fibres which contains receptor proteins on their surface
(2) *Nissl's grannules* — found in cyton, dendrites and axon part of neurons
(3) *Axon* — Functional part of neuron that transmit impulses away from the cyton
(4) *Schwann cell* — Synthesises myelin sheath around axons in spinal and cranial nerves.

BNC285

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21. Why neurilemma is absent in central nervous system ?
(1) Myelin sheath formation absent in CNS.
(2) Oligodendrocytes present in CNS in place of Schwann cells
(3) No exchange of ions in CNS
(4) Always non-medullated nerve fibres are found in CNS.

BNC286

22. Discontinuous covering of myelin sheath of nerve fibres helps in :-
(1) Continuous conduction of nerve impulse
(2) Nerve impulse become slow due to discontinuous sheath
(3) Nerve impulse become fast due to saltatory conduction
(4) There is no conduction of nerve impulse

BNC287

23. Find out the correct statement ?
(A) Somatic neutral system relay impulses from CNS to skeletal muscle.
(B) ANS transmits impulses from CNS to voluntary muscle
(C) Neural organisation is very simple in *Hydra* while better organised in insects.
(D) Neuron can detect, receive and transmit stimulus toward both side of axon.
(1) A & B are correct
(2) B & C are correct
(3) C & D are correct
(4) A & C are correct

BNC288

24. What are the causes of hyperpolarization?
(1) Due to excess K^+ outflux
(2) Due to influx of Cl^- ions
(3) After releasing of GABA
(4) All the above

BNC289

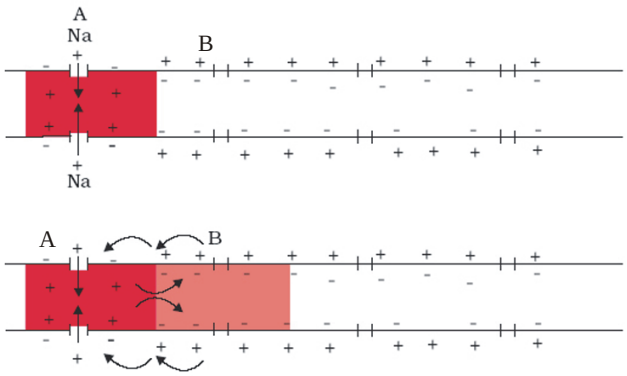
25. Choose the correct option :
Statement I: Neurons are excitable cells due to a differential concentration gradient of ions across the membrane.
Statement II: The nerve impulse is conducted along the axon membrane in the form of a wave of depolarisation and repolarisation.
(1) Statement I is correct but statement II is incorrect
(2) Statement I is incorrect but statement II is correct
(3) Both statement I and II are correct
(4) Both statement I and II are incorrect

BNC290

26. Mark the incorrectly matched pair w.r.t. conduction of nerve impulse through a nerve fibre.
(1) Depolarisation — Opening of $Na^{\oplus}$ channels and rapid influx of $Na^{\oplus}$ ions
(2) Polarisation — Ionic gradient is maintained by $Na^{\oplus}$ – $K^{\oplus}$ pump which transport $3Na^{\oplus}$ outwards for $2K^{\oplus}$ into the cell
(3) Depolarisation — High permeability of the axon membrane for $K^{\oplus}$
(4) Repolarisation—High permeability of the membrane for $K^{\oplus}$ and rapid outflux of $K^{\oplus}$ ions

BNC291

27. Which of the following helps in current flow on the inner surface of an axon from site-A to site-B as shown in the figure given below ?



(1) Negatively charged lipids
(2) Nissl's granules
(3) Neurofibrils
(4) Neurilemma

BNC292

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28. Read the following statements (A - D)

(A) The medulla contains centres, which control respiration, cardiovascular reflexes and gastric secretions.

(B) The mid brain receives and integrates visual tactile and auditory inputs.

(C) The neural system coordinates and integrates functions metabolic and homeostatic activities of all the organs.

(D) A very important part of the forebrain called medulla oblongata controls eating and drinking.

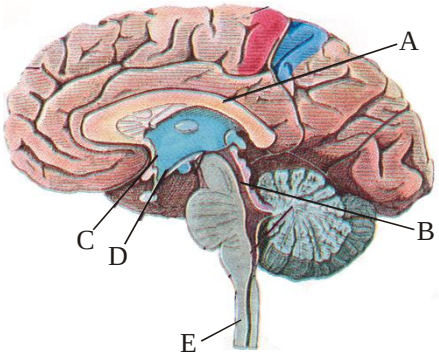
How many of the above statements are false:

(1) Three (2) Two

(3) One (4) Four

BNC293

29. Given below is a sagital section of human brain, identify the structures labelled as A, B, C, D and E with their important functions.



(A) Corpus callosum — Communication between two cerebral hemispheres

(B) Cerebral aqueduct — Connects IInd ventricle with IIIrd ventricle of brain

(C) Corpora quadrigemina — Controls Cardio-vascular reflexes

(D) Hypothalamus — Controls body temperature, urge for eating and drinking

(E) Limbic system — Regulation of sexual behaviour, emotions and motivation

(1) Only A is correct

(2) A and D are correct

(3) B, C and D are correct

(4) A, B, D and E are correct

BNC294

30. Which part of human brain controls the urge for eating and drinking ?

(1) Cerebellum (2) Cerebrum

(3) Pons (4) Hypothalamus

BNC295

31. Common neurotransmitter of peripheral nervous system is :-

(1) Colchicine (2) Epinephrine

(3) GABA (4) Acetylcholine

BNC296

32. All are true about midbrain except one: -

(1) Located between thalamus/ hypothalamus of forebrain and pons of hind brain.

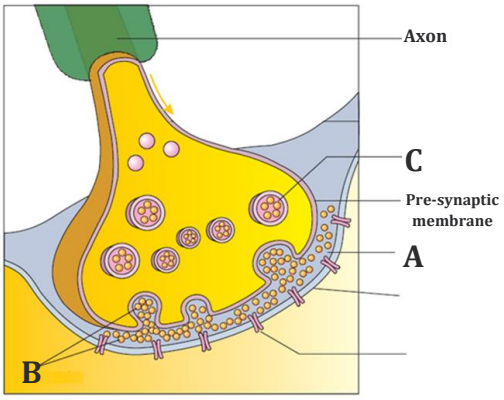
(2) It receiver and integrates visual, tactile and auditory inputs.

(3) It integrates information received from semi-circular canals of the ear and the auditory system

(4) The dorsal portion of midbrain consists mainly of four round swelling (lobes) called corpora quadrigemina.

BNC297

33. Mark the following in given figure-



A = Gap separating the pre-synaptic and post synaptic membrane.

B = Chemicals involved in the transmission of impulses at synapse.

C = They release their neurotransmitters in the synaptic cleft

(1) A = Synapse; B = Neurotransmitters. C = Synaptic vesicles

(2) A = Synaptic cleft; B=Neurotransmitters; C = Synaptic vesicles.

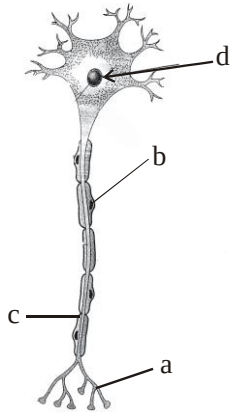
(3) A = Axon terminals; B=Neurons. C = Synaptic vesicles

(4) A = Synaptic cleft; B = Neurotransmitters. C = pre-synaptic membrane

BNC298

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34. Given below is a diagram of neuron, labelled by a,b,c and d points. Choose the correct match for these points :



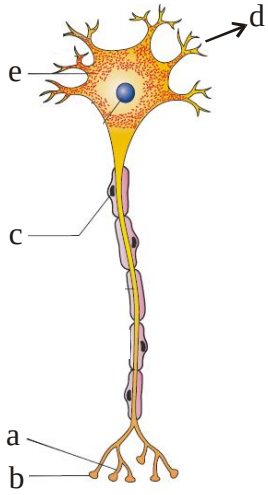
(1)	a = Dendrites	Efferent fibre
(2)	b = Schwan cell	Myelination in CNS
(3)	c = Node of Ranvier	Saltatory conduction
(4)	d = Nissl granules	Internal conduction of impulse

BNC299

35. Neural system composed of specialised cells called neuron, which has property of :-
(1) Excilability and flexiblity
(2) Conductivity and elasticity
(3) Excitability and conductivity
(4) Flexibility and contractibility

BNC300

36. Recognise the given figure and find the correct labelling?
(i) Schwann cell (ii) Synaptic knob
(iii) dendrites (iv) Axon terminal
(v) Nissl's granule (vi) Node of Ranvier



(1) a-iv, b-ii, c-i, d-iii, e-v
(2) a-ii, b-iv, c-iii, d-i, e-v
(3) a-ii, b-iv, c-vi, d-i, e-v
(4) a-iv, b-ii, c-iii, d-vi, e-v

BNC301

37. Find out correct statement :
(A) Somatic neural system relay impulses from CNS to skeletal muscles.
(B) ANS transmits impulses from CNS to voluntary muscles.
(C) Neural organisation is very simple in vertebrates where better organised in insects.
(D) Neuron can detect, receive and transmit stimuli.
(1) A and B correct (2) B and C correct
(3) C and D correct (4) A and D correct

BNC302

38. Scavanger cells of nervous tissue are:
(1) Oligodendrocytes (2) Astrocytes
(3) Cell body (4) Microglial cells

BNC303

39. Opaque white appearance of white matter is due to:-
(1) Fibers of tract covered by axolemma only.
(2) Fibers of tract not covered by myelin sheath
(3) Fibers of tract covered by myelin sheath.
(4) White matter absent in CNS.

BNC304

40. Read the following statements carefully :-
(A) Unipolar neurons has cell body with one axon only and are found in retina of eye.
(B) Unmyelinated nerve fibres don't have myelin sheath because they are never enclosed by schwann cells in PNS.
(C) The new potential developed on post-synaptic membrane after binding of neurotransmitter with receptors, is always excitatory
(D) Na^+–K^+ pump is responsible for resting stage
(E) Na^+ concentration is more in axoplasm.
Which of the above statements are **not correct**?
(1) A and B (2) A, B and C
(3) B, C and D (4) A, B, C and E

BNC305

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41. Which of the following statements is **false** about the electrical synapse ?

I. At electrical synapses, the membranes of pre and post synaptic neurons are in very close proximity.

II. Electrical current can flow directly from one neuron into the other across the synapses.

III. Transmission of an impulse across electrical synapses is very similar to impulse conduction along single axon.

IV. Electrical synapses pass electrical signal between cells with the use of Ach

V. Electrical synapses are fast.

VI. Electrical synapses are rare in our system.

(1) I and II (2) Only II

(3) Only IV (4) Only V

BNC306

42. When a stimulus is applied at a site on the polarised membrane. Then :-

(1) membrane becomes freely permeable to Na^+.

(2) membrane becomes more permeable to K^+.

(3) ionic gradient across the membrane are maintained by Na^+– K^+–pump.

(4) membrane becomes nearly impermeable to Na^+

BNC307

43. Some neurotransmitter excite the post synaptic neuron where others inhibit postsynaptic neuron. It depends on :-

(1) The permeability changes are induced in postsynaptic neuron by binding of neuro-tranismitter

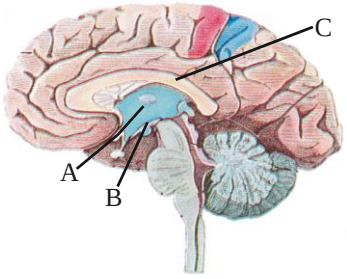
(2) Closing of specific ion channels in the post synaptic membrane by binding of Neurotransmitter

(3) It depends on electrical transmission of synapse by K^+

(4) This is not applicable for chemical synapse

BNC308

44. Given below is diagrammatic sketch of a sagittal section of the human brain. Identify A, B and C with their **correct** function



	A	B	C
(1)	Thalamus (Coordinating centre for Sensory and Motor Neuron)	Corpus callosum (Connects cerebral hemispheres)	Hypothalamus (Body temperature)
(2)	Hypothalamus (Coordinating centre for Sensory and Motor Neuron)	Corpus callosum (Connects cerebral hemispheres)	Thalamus (Body temperature)
(3)	Thalamus (Coordinating centre for Sensory and Motor Neuron)	Hypothalamus (Body temperature)	Corpus callosum (Connects cerebral hemispheres)
(4)	Corpus callosum (Connects cerebral hemispheres)	Hypothalamus (Body temperature)	Thalamus (Coordinating centre for Sensory and Motor Neuron)

BNC309

45. Velocity of nerve impulse is more in :-

(1) Unmyelinated axons

(2) Continuous myelinated axon

(3) Discontinuous myelinated axons

(4) In dendron and cyton

BNC310

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46. Due to presence of which structure cerebral cortex is related to storage of memory. Communication of information towards different parts and also integrates two sensory stimuli :-
(1) Sensory area (2) Motor area
(3) Association area (4) All of the above

BNC311

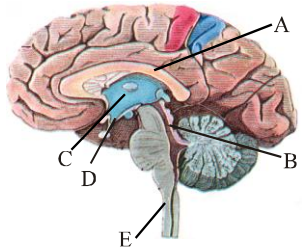
47. Association area are :-
(1) Sensory area found in deep part of cerebral cortex.
(2) Motor area found in surface layer of grey matter of cerebrum.
(3) Neither clearly sensory nor motor found in grey matter of cerebrum.
(4) Hypothalamic neurons which secrete hormones.

BNC312

48. Which option is correct regarding corpora quadrigemina?
(1) It is composed of four hollow lobes
(2) It is characteristic of amphibians
(3) It is located dorsally in midbrain
(4) It is not the part of brain stem

BNC313

49. Given below is a sagittal section of human brain, identify the structure labelled as A, B, C, D and E with their important function :-



(A)	Corpus callosum	Communication between two cerebellar hemisphere
(B)	Cerebral aqueduct	Connects III ventricle with IV ventricle
(C)	Pons	regulation of emotional behaviour
(D)	Hypothalaomus	control body temperature, urge for eating and drinking
(E)	Medulla oblongata	controls cardiovascular reflexes

Options :-
(1) Only A is correct
(2) A & D are correct
(3) C & D and E are correct
(4) Only B, D & E are correct

BNC314

50. **Assertion (A):** The chemical stored in the synaptic vesicles are termed as neurotransmitters.
Reason (R): Synaptic vesicles release these chemicals in the synaptic cleft.
(1) Both A and R are correct but R is not the correct explain of A.
(2) Both A and R are correct and R is the correct explanation of A.
(3) A is correct but R is incorrect.
(4) Both A and R are incorrect.

BNC315

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	4	2	2	3	2	2	4	4	2	4	3	2	2	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	4	3	3	2	2	2	3	4	4	3	3	3	3	2	4
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	4	3	2	3	3	1	4	4	3	2	3	1	1	3	3
Question	46	47	48	49	50										
Answer	3	3	3	4	1										

Chemical Coordination and Integration

1. Identify the correct match from column-I, II and III.

	Column-I	Column-II	Column-III
(1)	Interstitial cells	Progesterone	sexual behaviour
(2)	Graffian follicle	Estrogen	female secondary sexual characters
(3)	Graffian follicle	Androgen	suppressed by a neuro-transmitter
(4)	Corpus luteum	Mammotropic	Support pregnancy

BCI332

2. Complete the sentence :-
Hormones are non-nutrients chemicals which act as ___ messengers and are produced in ___ amounts ?
(1) Intracellular, large
(2) Intercellular, large
(3) Intracellular, small
(4) Intercellular, Trace

BCI333

3. A hormone that increases deposition of Ca^{+2} in bones by decreasing blood calcium level. It is antagonistic to a hormone secreted from a gland located at back side of most famous gland of body identify it :-
(1) Parathyroid hormone
(2) Thyrocalcitonin hormone
(3) Aldosterone hormone
(4) Calcitriol hormone

BCI334

4. Read the following statements (A-D):
(A) Hormones of posterior pituitary are synthesized in hypothalamic nuclei and transported through axon.
(B) Melatonin plays a major role in development of immune system.
(C) Melanocyte stimulating hormone is secreted by pars nervosa.
(D) In females, progesterone induces ovulation of Graafian follicle.
How many above statements are incorrect?
(1) One (2) Two
(3) Three (4) Four

BCI335

5. Alveolar growth in mammary gland is function of :-
(1) Thyroxine (2) Progestrogen
(3) ANF (4) Oxytocin

BCI336

6. Which of the following statements are true/false ?
(A) Calcitonin regulates the metabolism of calcium.
(B) Oxytocin stimulates contraction of uterine muscles during birth.
(C) Grave's disease is caused by malfunctioning of adrenal gland
(D) ADH stimulates absorption of water and *increase* the urine production
(1) A and C are true, B and D are false.
(2) A and B are true, C and D are false
(3) A and D are false, B and C are true
(4) A only true, B, C and D are false

BCI337

7. Vasopressin and cortisol are related with following activities performed inside the body :-
(a) Vasopressin is secreted from posterior adrenal gland where cortisol from anterior pituitary gland
(b) Vasopressin stimulates reabsorption of electrolytes by distal tubules and cortisol is related with maintenance of cardio vascular system.
(c) Cortisol is related to RBC formation where vasopressin reduces loss of water.
(d) Vasopressin is anti diuretic where cortisol is life saving substance of endocrine glands.
(1) a and b are true where c and d are false.
(2) a, b, c are true where only d is false.
(3) a is true where b, c, d are false
(4) a is false where b, c, d are true

BCI338

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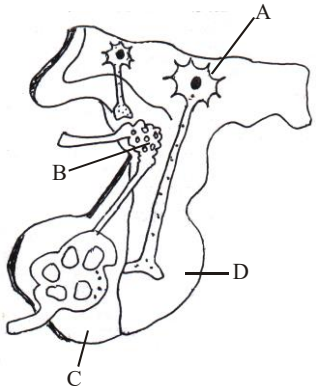
8. Which one of the following suitable hormones are related to stimulation of gluconeogenesis, lipolysis and proteolysis :-
(1) Mineralocorticoids (2) Gluococorticoids
(3) Progesterone (4) Aldosterone

BCI339

9. What will happened when any hormone molecule is bind with receptor ?
(1) It is non species specific interaction between hormone and receptor leads to degeneration of bio-membrane to form second messenger
(2) It is hormone specific interaction between hormone and receptor leads to target tissue metabolism and physiological functions.
(3) It is species specific interaction leads to complete formation of specific steroid hormone in all cell.
(4) It is specific interaction of hormone and receptor leads to cellular abnormality.

BCI340

10. Identify A, B, C and D in the given diagram



	A	B	C	D
(1)	Thalamic neurons	Portal circulation	Anterior Pituitary	Posterior Pituitary
(2)	Hypothalamic neurons	Portal circulation	Anterior Pituitary	Posterior Pituitary
(3)	Pituitary neurons	Portal circulation	Anterior Pituitary	Posterior Pituitary
(4)	Pineal neurons	Portal circulation	Anterior Pituitary	Posterior Pituitary

BCI341

11. Which of the following hormone stimulates a vigorous contraction of uterus at the time of child birth and milk ejection from the mammary gland?
(1) Progesterone
(2) Prolactin
(3) Oestrogen
(4) Oxytocin

BCI342

12. Which of the following gland is degenerated in old individuals resulting in a decreased production of hormone related with weak immune system ?
(1) Thyroid
(2) Thymus
(3) Parathyroid
(4) Pancreas

BCI343

13. Which of the following is not a function of thyroid gland ?
(1) Regulation of BMR
(2) Increases Ca^{+2} absorption from the digested food
(3) Support the process of RBC formation
(4) Control the metabolism of carbohydrate

BCI344

14. Match list I with list II and select the correct answer using the codes given below the list :-

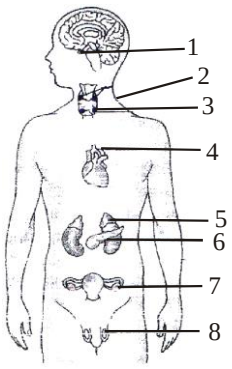
List I	List II
(a) LH	(i) Supression of GH
(b) Somatostatin	(ii) Dispersion of melanine
(c) Melatonin	(iii) Formation of corpus luteum
(d) MSH	(iv) Antigonadal action

	a	b	c	d
(1)	iii	i	ii	iv
(2)	iii	i	iv	ii
(3)	i	iii	ii	iv
(4)	i	iii	iv	ii

BCI345

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15. Study the given figure and choose the correct option -



(1) 2. → Parathormone
6. → islets of Langerhans
8. → ICSH production
3. → Thyroxine

(2) 5. → Adrenaline
1. → FSH production
8. → Testosterone
4. → Immunity

(3) 1. → Master gland
8. → ICSH production
2. → Thyroxine
5. → Adrenaline

(4) 7. → LH production
8. → Testosterone
6. → Insulin
4. → Thyroxine

BCI346

16. Match the hormone in column I with their function in column II.

	Column-I		Column-II
(i)	Oxytocin	(a)	Prepares alveoli of mammary gland for milk synthesis
(ii)	Progesterone	(b)	Increase uptake of glucose by cells
(iii)	Insulin	(c)	Contraction of uterine muscles
(iv)	Thyroxine	(d)	Support RBC formation

Choose the correct answer from the options given below :

(1) (i)-(c), (ii)-(a), (iii)-(d), (iv)-(b)
(2) (i)-(b), (ii)-(a), (iii)-(c), (iv)-(d)
(3) (i)-(a), (ii)-(d), (iii)-(b), (iv)-(c)
(4) (i)-(c), (ii)-(a), (iii)-(b), (iv)-(d)

BCI347

17. Select the correct matching of a hormone, its source and function: -

	Column-I	Column-II	Column-III
(1)	Prolactin Pituitary	Posterior	Regulates growth of mammary glands and milk formation in females
(2)	Glucagon	β–cells of islets of Langerhans	Stimulates glycogenolysis
(3)	Vasopressin	Prolactin Pituitary	Increases loss of water through urine
(4)	Nor epinephrine	Adrenal Medulla	Increases heart beat and rate of respiration

BCI348

18. Which one is not a group of antagonistic hormones?

(1) PTH & TCT
(2) Melatonin & MSH
(3) Insulin & glucagon
(4) Progesterone & oestrogen

BCI349

19. Which one is not a group of diabetogenic hormones?

(1) Adrenaline and nor adrenaline
(2) Cortisol and growth hormone
(3) Glucagon and insulin
(4) Thyroxine and glucagon

BCI350

20. Which one is correct match

(1) ANF = vasoconstrictor;
ADH = vasodilator
(2) Nor-adrenaline = Vasoconstrictor;
vasopressin = vasodilator
(3) ANF = vasodilator;
Nor-adrenaline = vasoconstrictor
(4) Angiotensin-II = vasodilator

BCI351

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21. Match the following columns and find out the answer?

	Column-I	Column-II
(a)	Estrogen	Formation of mammary gland alveoli
(b)	Aldosterone	Action on adipose tissue
(c)	Progesterone	Maintenance of body fluid volume
(d)	Insulin	Female sexual behaviour

(1) a-(i), b-(iii), c-(ii), d-(iv)
(2) a-(iv), b-(ii), c-(iii), d-(i)
(3) a-(iv), b-(iii), c-(i), d-(ii)
(4) a-(i), b-(ii), c-(iii), d-(iv)

BCI352

22. Consider the following statements –
(a) Catecholamines stimulate the breakdown of glycogen in glucose.
(b) Glucocorticoids produces anti-inflammatory reactions.
(c) LH produce wide ranging actions such as development of growing ovarian follicles, stimulation of growth and activities of female secondary sex organs.
(d) Cortisol is related with mixing of FSH and LH during embryo stage.
Choose the correct statement from above –
(1) a, b are true where c, d are false
(2) b, c are true where a, d are false
(3) a, d are true where b, c are false
(4) a is true where b, c, d are false.

BCI353

23. Hormones which interact with intracellular receptors are generally :–
(a) Iodothyronine (b) MSH
(c) Estradiol (d) Epinephrine
(e) Progesterone (f) Glucagon and Insulin
(1) a, b, d (2) b, e, f
(3) a, c, e (4) c, d, e, f

BCI354

24. Insulin hormone –
(1) Stimulating oxidation of glycogen and conversion of glycogen into glucose.
(2) Enhances release of glucose into blood.
(3) Provide general resistance to long-term stress by inflammatory and allergic responses.
(4) Stimulating transport of glucose from blood to muscle and adipose tissue.

BCI355

25. Thyroid hormone of thyroid gland is related with ?
(1) It's inactive form store in follicular cavity for some time.
(2) It support the red blood cell formation.
(3) Related with BMR
(4) All of these.

BCI356

26. Catechalomines of adrenal gland stimulate: -
(1) Decrease the rate and force of heart beat and conversion of liver glucose into glycogen
(2) Increase dilation of arterioles of heart
(3) Contraction in pupil part of eye
(4) Stimulate anti-inflammatory reactions

BCI357

27. Loss of glucose through urine and formation of harmful compounds like ketone bodies leads to a complex disorder like :-
(1) Diabetes insipidus showing polyurea and glycosurea
(2) Diabetes mallitus showing nodule formation and diuresis
(3) Hyperthyroidism showing adverse effect on body
(4) Diabetes mallitus showing glucosurea and ketoacidiosis

BCI358

28. Blood pressure is controlled by :-
(1) Thymus (2) Pineal gland
(3) Corpus luteum (4) Adrenal gland

BCI359

AIM : FREE EDUCATION TO ALL APUL

29. Tetany is condition in man caused by :-
(1) Hyposecretion of PTH
(2) Hyposecretion of thyroxine hormone
(3) Hypersecretion of PTH
(4) More ADH and Aldosterone

BCI360

30. An over dose of intravenous insulin may lead to the death of an individual due to :-
(1) An excessive increase of blood
(2) An excessive fall in ionic concentration of blood
(3) An excessive decrease in blood glucose
(4) An overproduction of histamine

BCI361

31. Match list-I with II and select the correct option :-

	List-I	List-II
(a)	Adrenalin	Myxodema
(b)	Hyper-parathyroidism	heat beat
(c)	Oxytocin	Salt-water balance
(d)	Hypothyroidism	Child birth
(e)	Aldosterone	Demineralization

(1) a-1 b-5 c-4 d-2 e-3
(2) a-2 b-5 c-4 d-1 e-3
(3) a-2 b-3 c-4 d-1 e-5
(4) a-4 b-3 c-2 d-1 e-5

BCI362

32. Adenohypophysis in humans consists of :
(1) Pars distalis and Pars nervosa
(2) Pars intermedia and Pars distalis
(3) Pars nervosa and Pars intermedia
(4) Anterior and posterior pituitary

BCI363

33. Which hormone dilates blood vessels during high B.P.
(1) ADH (2) ANF
(3) Aldosterone (4) MSH

BCI364

34. Which one is a group for salt & H_2O balancing in body ?
(1) Insulin and glucagon
(2) MSH and melatonine
(3) PTH and TCT
(4) ADH and Aldosterone

BCI365

35. Find out the incorrect statements :-
(1) Hormonal effect always produces when they bind with specific proteins located on cell membrane and inside the cell.
(2) Receptors are specific for a particular hormone molecule.
(3) Hormones act with membrane bound receptors and then enters into cell to bind with secondary messenger.
(4) Protein hormone produce secondary messengers during their action.

BCI366

36. Which one of the following hormones can change the chromosomal functions by binding with their receptors?
1. Cortisol 2. Iodothyronine
3. Testosterone 4. Estradiol
(1) Only 3 & 4 (2) Only 1, 2 and 3
(3) All 1, 2, 3 & 4 (4) Only 1 & 2

BCI367

37. In hormone action, if receptor molecules are removed from target organ, the target organ will-
(1) Continue to respond to hormone
(2) Not respond to hormone
(3) Continue to respond but require higher concentration
(4) Continue to respond but in the opposite way

BCI368

38. Which of these hormones uses cAMP as a second messenger?
(1) Estrogen
(2) Cortisol
(3) Epinephrine
(4) Testosterone

BCI369

39. Hormones produce their effect on target tissue by binding to specific proteins called as:
(1) Target protein
(2) Activator protein
(3) Inhibitor proteins
(4) Hormone receptors

BCI370

AIM : FREE EDUCATION TO ALL APUL

40. Hormone which interact with membrane bound recepters normally: -
(1) Enters into cell
(2) Do not enter in to the cell
(3) Generate secondary messenger
(4) Both (2) and (3)

BCI371

41. Steroid hormone typically alters the activity of target cells by :-
(1) activating primary messenger
(2) activating secondary messenger
(3) interacting with intracellular recepters
(4) none of above

BCI372

42. In a hormonal control system, hormone X, promote the secretion of substance Y. If this system works by negative feedback, which of the following be correct?
(1) Substance Y stimulates the production of hormone X
(2) Secretion of Y substance should stop completely.
(3) Level of hormone X should not be affected by level of substance Y.
(4) If level of substance Y highly increases secretion of hormone X should decrease

BCI373

43. Production of ADH, monitoring of body temperature and blood pressure are mainly controlled by -
(1) Cerebellum (2) Cerebrum
(3) hypothalamus (4) Medulla

BCI374

44. Which hormone promotes cell division, protein synthesis and bones growth?
(1) PTH (2) ADH (3) GH (4) ACTH

BCI375

45. FSH plays an important role in :-
(1) Ovulation
(2) Development of secondary sexual characters
(3) Spermatogenesis
(4) Control of blood sugar

BCI376

46. Which of the following endocrine glands doesn't work under regulation of pituitary gland :-
(a) Parathyroid
(b) Thyroid
(c) Adrenal cortex
(d) Adrenal medulla
(e) Testis/Ovary
(1) a & b (2) a, d & e
(3) b, c & e (4) a & d

BCI377

47. An adenohypophysis hormone which is regulated by feedback mechanism is :-
(1) TSH (2) Oxytocin
(3) Calcitonin (4) cortisone

BCI378

48. Find out the correct combinations:-
(a) LH - Development of ovarian follicle.
(b) ACTH - Synthesis of steroid hormone.
(c) Oxytocin - Ejection of milk
(d) FSH - maintenance of corpus luteum.
(1) a, b (2) b, d
(3) b, c (4) c, d

BCI379

49. Oxytocin responsible for vigorous contraction of uterus at the time of child birth is synthesize by:-
(1) posterior pituitary gland
(2) Hypothalamus
(3) Both (1) and (2)
(4) Anterior pituitary gland

BCI380

50. Which of the following statements are correct?
(a) Cortisol suppresses the immune response
(b) Thyroxine supports the process of red blood cell formation
(c) Glucagon stimulates glycogenesis process
(d) Androgens stimulate muscular growth, aggressiveness, high pitch of voice.
(1) a, b, c are ture, but d is false
(2) a, b, d are true, but c is false
(3) d is true, but a, b, c are false
(4) a, b are true, but c, d are false

BCI381

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51. Select the incorrect option :-
(1) Thyroid gland is the largest endocrine gland in human.
(2) Thyroid secrete T_3 and T_4.
(3) Thyroid gland is composed of follicle and stroma
(4) Thyroid consist of four lobes

BCI382

52. By which part of thyroid gland T_3 and T_4 hormones are synthesised ?
(1) Follicle (2) Stromal tissue
(3) Isthmus (4) Larynx

BCI383

53. Hormone that increases the blood calcium and decrease the excretion of calcium by reabsorption is :-
(1) Calcitonin (2) Parathormone
(3) Insulin (4) ACTH

BCI384

54. Match the column-I with column-II.

	Column-I		Column-II
(i)	ADH	(a)	Regulate pigmentation
(ii)	Parathormone	(b)	Maturation of Graafian follicle
(iii)	FSH	(c)	Maintain Ca^{+2} level
(iv)	MSH	(d)	Reabsorption of water

Choose the correct answer from the options given below :
(1) (i)-(a); (ii)-(d); (iii)-(b); (iv)-(c)
(2) (i)-(c); (ii)-(b); (iii)-(d); (iv)-(a)
(3) (i)-(d); (ii)-(b); (iii)-(c); (iv)-(a)
(4) (i)-(d); (ii)-(c); (iii)-(b); (iv)-(a)

BCI385

55. Which one of the following pairs is correctly matched :-
(1) Thyroxine — Tetany
(2) PTH — Growth impaired
(3) Insulin — Diabetes mellitus
(4) FSH — Failure of ovulation

BCI386

56. Glucagon produced by alpha cells of islet of Langerhans which :-
(1) Converts glucose to glycogen
(2) Converts glycogen to glucose
(3) Decreases concentration of glucose in blood
(4) None of the above

BCI387

57. Match the function of hormone in column I with their hormone in column II

	Column-I		Column-II
(A)	Produces anti-inflammatory reactions	(i)	Thyroxine
(B)	Regulation of Basal metabolic rate	(ii)	Aldosterone
(C)	Stimulates glycogenesis	(iii)	Cortisol
(D)	Stimulates reabsorption	(iv)	Insulin

(1) A-(iii), B-(i), C-(iv), D-(ii)
(2) A-(iii), B-(ii), C-(iv), D-(i)
(3) A-(i), B-(iii), C-(iv), D-(ii)
(4) A-(ii), B-(iii), C-(i), D-(iv)

BCI388

58. Which one is incorrect statement?
(1) Neurohypophysis store and release two hormone called oxytocin and vasopressin
(2) GH shows indirect effect on bones of body by acting with IGF.
(3) Melatonin plays a very important role in the regulation of a diurnal/circadian rhythm.
(4) Thymosin of thymus influences metabolism, pigmentation and defence capability.

BCI389

59. Choose a correct pair of peptide hormone from the following options -
(1) Adrenaline, non-adrenaline
(2) Atrial natriuretic factor, Parathyroid hormone
(3) Thyroxine, estradiol
(4) Aldosterone, Cortisol

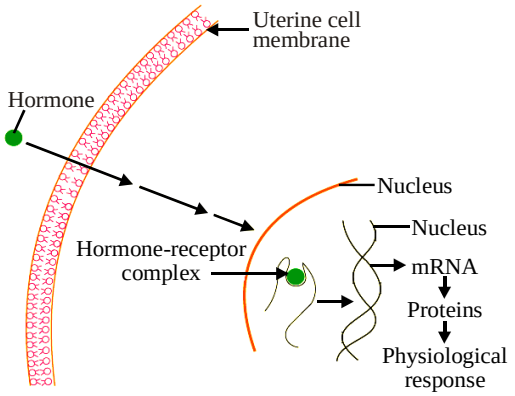
BCI390

AIM : FREE EDUCATION TO ALL APUL

60. The function of insulin hormone is :-
(1) To release glucose from liver cells and glycogenolysis promotion
(2) To increase cellular glucose uptake and consumption
(3) To increase blood sugar level
(4) To increase lipolysis

BCI391

61. Following is a diagrammatic representation of the mechanism of hormone action.



Which of the following statement does **not** support the hormone action depicted.
(1) It explains the action of a hormone which interacts with membrane bound receptors.
(2) It explains the action of a hormone which interacts with intracellular receptors in nucleus.
(3) The hormone regulates chromosome function by the interaction of hormone receptor complex with genome.
(4) This hormone does not generate a second messenger

BCI392

62. Secondary messengers: -
(1) Are substance that interact with first messengers inside cells
(2) Are substances that bind to first messengers in the cell membrane
(3) Are hormone secreted by cell in response to stimulation by another hormone
(4) Mediate the intracellular response to many different hormones and neuro-transmitters

BCI393

63. Hormones are :-
(1) Non-nutritive, non-antigenic & non species specific
(2) Nutritive, non antigenic & non species specific
(3) Non-nutritive, non-antigenic & species specific
(4) Non-nutritive, antigenic & non species specific

BCI394

64. Which of the following is correct about endocrine glands ?
(a) These are ductless glands
(b) Their secretions are called hormones
(c) Endocrine glands release hormones into the blood and transported to a distantly located target organ
(d) Invertebrates possess very simple endocrine systems with many hormones
(1) a, b, c (2) b, c, d (3) c, d (4) a, b, c, d

BCI395

65. Thyroxine receptor are :-
(1) Membrane bound receptor
(2) Intracellular receptor
(3) Nuclear receptor
(4) Both (2) and (3)

BCI396

66. Maximum protein hormones receptors are :-
(1) Membrane bound receptors
(2) Nuclear receptor
(3) Intracellular receptors
(4) Both (2) and (3)

BCI397

67. In a female labour pain is of low intensity during parturition, then what step should be taken by doctor ?
(1) Prolactin will given
(2) Oxytocin injection will given
(3) Progesterone will given
(4) ADH will given

BCI398

68. Releasing hormones and inhibiting hormones are produced by :-
(1) Hypothalamus
(2) Pituitary gland
(3) Upper lateral part of diencephalon
(4) Thyroid gland

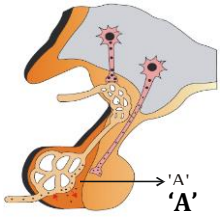
BCI399

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69. Which of the following is correct about human?
(a) Pars distalis produces GH, PRL, TSH, ACTH, LH, FSH
(b) Pars intermedia secretes only one hormone called melatonin
(c) Posterior lobe of pituitary is also called as pars nervosa
(d) Posterior pituitary stores and releases two hormones called oxytocin and vasopressin
(1) a, b and c (2) b, c and d
(3) a, c and d (4) a, b, c and d

BCI400

70.



'A'

Which of the following diseases are related to the deficiency of hormones secreted from the structure marked 'A' in the above given figure:-
(A) Diabetes insipidus (B) Dwarfism
(C) Acromegaly (D) Gigantism
(E) Grave disease
(1) A, B, C, D and E (2) C, D and E
(3) B only (4) 'C' only

BCI401

71. Which of the following is not a correct match of hormone and function :-
(1) LH⇒Production of testosterone by the testes
(2) Prolactin ⇒ milk production in the mammary glands
(3) FSH ⇒ Development of Graafian follicle
(4) Oxytocin ⇒ Relax the uterine muscles

BCI402

72. Oxytocin and vasopressin, which are actually synthesized by the __A__ and are transported __B__ to __C__.
Choose A, B, C from the given options :-
(1) A = Hypothalamus, B = by blood, C = Neurohypophysis
(2) A = Hypothalamus, B = axonally, C = Adenohypophysis
(3) A = Hypothalamus, B = by blood, C = Adenohypophysis
(2) A = Hypothalamus, B = axonally, C = Neurohypophysis

BCI403

73. Hypothalamus contains several group of neurosecretory cells called :-
(1) Parietal cells (2) Oxyntic cells
(3) Nuclei (4) G-cells

BCI404

74. Which of the following is **correct** ?
(a) Hypercalcaemic hormone is thyro-calcitonin (TCT) hormone
(b) Parathyroid hormone (PTH) increases the Ca^{+2} levels in the blood
(c) Thymosin play a major role in the differentiation of T-lymphocytes
(d) Parathyroid glands are present on back side of the thyroid gland
(1) b, c and d (2) a, b and c
(3) a, c and d (4) a, b, c and d

BCI405

75. Which of the following pair is not matched correctly: -
(1) Cretinism = Thyroid gland
(2) Acromegaly = Pituitary gland
(3) Addison's disease = Adrenal gland
(4) Cushing's syndrome = Pineal gland

BCI406

76. Which of the following option does not include the antagonistic hormones :-
(1) Insulin and glucagon
(2) PTH and calcitonin
(3) Epinephrine and oxytocin
(4) MSH and melatonin

BCI407

77. Which one of the following condition is not linked to deficiency of thyroid hormones :-
(1) Cretinism
(2) Goitre
(3) Myxodema
(4) Exophthalmic goitre

BCI408

78. Which of the following are not present in cretinism :-
(1) Low intelligence
(2) Dwarfism
(3) Deaf-mutism
(4) Normal healthy skin

BCI409

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79. Which of the following are the functions of thyroxine :-
(A) Development of CNS
(B) Stimulates erythropoiesis
(C) Influence maintenance of water and electrolyte balance.
(D) Regulates menstrual cycle
(1) A, B, C and D (2) C and D
(3) Only D (4) A and B only

BCI410

80. Identify the **correct** match from the columns-I, II and III

Column-I		Column-II		Column-III	
(1)	Zona glomerulosa	a.	Cortisol	i.	Osmotic pressure maintenance
(2)	Zona glomerulosa	b.	Androgenic steroids	ii.	Emergency hormone
(3)	Adrenal Medulla	c.	Aldosterone	iii.	Growth of axial hair at puberty
(4)	Zona glomerulosa	d.	Adrenaline	iv.	Anti-inflammatory response

(1) 1-a-iv, 2-c-i, 3-d-ii, 4-b-iii
(2) 1-c-i, 2-a-iv, 3-d-ii, 4-b-iii
(3) 1-a-i, 2-c-iv, 3-d-ii, 4-b-iii
(4) 1-b-iv, 2-a-i, 3-d-iii, 4-c-ii

BCI411

81. (A) induces ovulation of fully mature follicles and (B) acts mainly at the kidney and stimulates resorption of water and electrolytes.

	A	B
(1)	FSH	ADH
(2)	LH	Oxytocin
(3)	LH	Vasopressin
(4)	LH	Prolactin

BCI412

82. Which of the following is correct about glucagon?
(a) Stimulates glycogenolysis
(b) Stimulates gluconeogenesis
(c) Hypoglycaemic hormone
(1) a, c only (2) a, b only
(3) b, c only (4) a, b, c all

BCI413

83. Given below is an incomplete table about certain hormones, their source glands and one major effect of each on the body in human. Identify the correct option for the three blanks A, B and C :

Gland	Secretion	Effect on Body
A		Maintenance of secondary sexual characters
Alpha cells of Islets of Langerhans	**B**	Raises blood sugar level
Anterior pituitary	**C**	Over secretion leads to gigantism

Options :

	A	B	C
(1)	Ovary	Glucagon	Growth hormone
(2)	Placenta	Insulin	Vasopressin
(3)	Ovary	Insulin	Calcitonin
(4)	Placenta	Glucagon	Calcitonin

BCI414

84. Which of the following statement is false :-
(1) Glucagon reduces cellular glucose uptake and utilisation.
(2) Glucagon is a hyperglycemic hormone.
(3) Insulin stimulates gluconeogenesis
(4) Insulin stimulates glycogenesis

BCI415

85. Which of the given statement is false :-
(1) Androgen play a role in the process of spermatogenesis.
(2) Androgens act on the central neural system and influence the male sexual behaviour.
(3) Androgens stimulate high pitch of voice in males
(4) Androgens shows anabolic effect

BCI416

AIM : FREE EDUCATION TO ALL APUL

86. Identify the correct match from column I, II and III.

Column-I		Column-II		Column-III	
(1)	ADH	(a)	Parathyroid	(i)	Tetany
(2)	PTH	(b)	Hypothalamus	(ii)	Diabetes mellitus
(3)	Insulin	(c)	Anterior pituitary	(iii)	Diabetes insipidus
(4)	Growth hormone	(d)	Pancreas	(iv)	Dwarfism

(1) I-b-ii, II-a-i, III-d-iii, IV-c-iv
(2) I-b-iii, II-a-i, III-d-ii, IV-c-iv
(3) I-c-iii, II-a-i, III-d-ii, IV-b-iv
(4) I-c-ii, II-a-i, III-d-iii, IV-b-iv

BCI417

87. Read the following statements (A–D)
(A) Gastrin acts on gastric glands and stimulates secretion of HCl and pepsinogen.
(B) Deficiency of Insulin leads to Diabetes insipidus.
(C) Glucagon is antagonistic to insulin
(D) Thymus gland originates from embryonic endoderm
How many of the above statements are **correct**?
(1) Four (2) One (3) Two (4) Three

BCI418

88. **Assertion** : Neurohypophysis (Pars nervosa) is also known as posterior pituitary.
Reason : Posterior pituitary stores and releases two hormones called oxytocin and vasopressin.
(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) The assertion is true but the reason is false.
(4) Both the assertion and reason are false.

BCI419

89. **Assertion** : Immune response of old person becomes weak.
Reason: Thymus is degenerated in old individuals
(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both the assertion and reason are true but the reasn is not a correct explanation of the assetion.
(3) The assertion is true but the reason is false.
(4) Both the assertion and reason are false.

BCI420

90. **Assertion :** Hormones are nutrient substances which act as intercellular messenger.
Reason: Hormones are produced in large amount.
(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) The assertion is true but the reason is false.
(4) Both the assertion and reason are false.

BCI421

91. **Assertion:** Over secretion of GH leads to gigantism.
Reason: It leads to abnormal growth of the body.
(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) The assertion is true but the reason is false.
(4) Both the assertion and reason are false.

BCI422

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92. **Assertion:** CCK stimulates secretion of bile juice.
Reason: CCK act on liver.
(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) The assertion is true but the reason is false.
(4) Both the assertion and reason are false.
BCI423

93. **Assertion :** Cortisol is useful in organ transplantation.
Reason : Cortisol suppresses immune response.
(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) The assertion is true but the reason is false.
(4) Both the assertion and reason are false.
BCI424

94. **Assertion** : Vasopressin is also known as antidiuretic hormone.
Reason : Vasopressin stimulates the reabsorption of water and electrolyte by distal tubules and thereby reduces diuresis.
(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) The assertion is true but the reason is false.
(4) Both the assertion and reason are false.
BCI425

95. **Assertion :** Prolactin is also called the 'Milk ejection hormone'
Reason : Prolactin stimulates the smooth muscle contractions of the mammary glands.
(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) The assertion is true but the reason is false.
(4) Both the assertion and reason are false.
BCI426

96. **Assertion :** MSH regulates pigmentation of skin.
Reason : MSH acts on melanin containing cells.
(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) The assertion is true but the reason is false.
(4) Both the assertion and reason are false.
BCI427

97. **Assertion**: Secretin acts on endocrine part of pancreas.
Reason: Secretin stimulates secretion of insulin.
(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) The assertion is true but the reason is false.
(4) Both the assertion and reason are false.
BCI428

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	4	2	3	2	2	4	2	2	2	4	2	2	2	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	4	4	4	3	3	3	1	3	4	4	2	4	4	1	3
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	2	2	2	4	3	3	2	3	4	4	3	4	3	3	3
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	4	1	3	2	4	4	1	2	4	3	2	1	4	2	2
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	1	4	1	1	4	1	2	1	3	3	4	4	3	1	4
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	3	4	4	1	2	3	2	1	3	3	2	4	2	1	4
Question	91	92	93	94	95	96	97								
Answer	1	3	1	1	4	1	4								

Biomolecules-I & II

1. Which element is most abundant in protoplasm?
(1) Oxygen (2) Carbon
(3) Hydrogen (4) Nitrogen
BBM235

2. All monosaccharides have two types of ring structure (i.e. – Furanose & Pyranose) but it is not true for which monosaccharide?
(1) Glyceraldehyde
(2) Dihydroxy acetone
(3) Erythrose
(4) All
BBM236

3. Which of the following is ketose sugar ?
(1) Fructose (2) Sedoheptulose
(3) Dihydroxyacetone (4) All
BBM237

4. Priority order of organic compounds present in protoplasm –
(1) Water > Protein > Lipid > Carbohydrate
(2) Lipid > Protein > Carbohydrate > water
(3) Protein > Carbohydrate > Lipid
(4) Protein > Water > Lipid > Carbohydrate
BBM238

5. Which of the following disaccharide is reducing?
(1) Glucose (2) Glyceraldehyde
(3) Lactose (4) All
BBM239

6. Which of the following is correctly matched ?
(1) Mannose = 6 carbons sugar
(2) Erythrose = 4 carbons sugar
(3) Arabinose = 5 carbons sugar
(4) All
BBM240

7. Which sugar is not found in free form ?
(1) Galactose (2) Mannose
(3) Glucose (4) Both (1) & (2)
BBM241

8. Milk sugar has
(1) 1 – 2" Glycosidic linkage
(2) 1 – 4" Glycosidic linkage
(3) 1 – 1" glycosidic linkage
(4) None
BBM242

9. Commercial sugar is another name of –
(1) Maltose (2) Lactose
(3) Sucrose (4) Trehalose
BBM243

10. Which fatty acid is metabolically least active ?
(1) Palmitic Acid (2) Oleic Acid
(3) Linoleic Acid (4) Arachidonic Acid
BBM244

11. In phospholipid, which component is hydrophobic?
(1) Fatty acid (2) N-Compound
(3) Glycol (4) Monoglyceride
BBM245

12. Which chemical compound gives high amount of energy in metabolism ?
(1) Carbohydrate (2) Lipid
(3) Protein (4) Glycerol
BBM246

13. How many total double bonds are present in oleic acid between C–C atoms?
(1) 4 (2) 3 (3) 2 (4) 1
BBM247

14. Which element is compulsory in proteins ?
(1) N (2) I (3) P (4) Fe
BBM248

15. Which of the following amino acid is essential?
(1) Leucine (2) Tryptophan
(3) Arginine (4) All
BBM249

16. Which of the following amino acid is non essential ?
(1) Serine (2) Cysteine
(3) Valine (4) Both (1) & (2)
BBM250

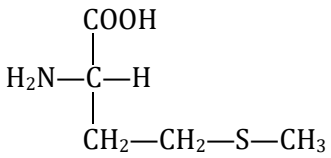
17. Which amino acid is aromatic ?
(1) Tyrosine (2) Tryptophan
(3) Phenyl Alanine (4) All
BBM251

18. Histidine amino acid is –
(1) Acidic (2) Alkaline
(3) Neutral (4) None
BBM252

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19. This structural formulae is related to which Amino acid



(1) Proline (2) Hydroxy proline
(3) Serine (4) Methionine

BBM253

20. Which types of bond are not broken in denaturation of protein –
(1) Peptide bonds
(2) Hydrophobic bond
(3) Hydrogen bonds
(4) Both (1) & (2)

BBM254

21. A monomeric protein having globular appearance represents which level of protein structure ?
(1) Primary (2) Secondary
(3) Tertiary (4) Quaternary

BBM255

22. Which is correct information about myosin protein ?
(1) Secondary configuration
(2) Intra molecular H-bonds
(3) Fibrous in appearance
(4) All

BBM256

23. In a peptide chain 6 amino acids are present and molecular weight of one amino acid is 60. Then calculate the molecular Weight of this peptide chain.
(1) 360 (2) 252
(3) 270 (4) Can't be calculate

BBM257

24. Which of the following is not simple globular protein ?
(1) Rubisco (2) Albumin
(3) Elastin (4) Histone

BBM258

25. 5-Methyl uracil is present in
(1) RNA (2) DNA
(3) RNA & DNA (4) None

BBM259

26. Cytidine & Guanylic Acids are respectively
(1) Nucleotide, Nucleoside
(2) Nucleoside, Nucleotide
(3) Nucleotide, Nucleotide
(4) Nucleoside, N-Base

BBM260

27. Wilkins & Franklin studied DNA by –
(1) Autoradiography
(2) Chromatography
(3) X-Ray Crystallography
(4) None

BBM261

28. In a double stranded DNA A = 35 % then the amount of Guanine will be ?
(1) 35 % (2) 65 %
(3) 30 % (4) 15 %

BBM262

29. In a segment of DNA 6.4 kilo base pairs are present. If DNA segment has 1640 Adenine molecules, then what will be the number of cytosine ?
(1) 1560 (2) 1640
(3) 4760 (4) 1500

BBM263

30. Which type of DNA is present in $\phi \times 174$ bacteriophase
(1) Double stranded circular
(2) Single Stranded circular
(3) Double stranded linear
(4) Single stranded linear

BBM264

31. Helix length of one turn in B-DNA is
(1) 34 Å (2) 3.4 Å
(3) 45.6Å (4) 28 Å

BBM265

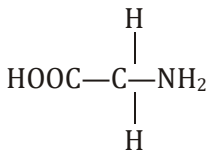
32. Sequence of N-base in one strand of DNA is A G T C T T G C. Then the sequence of N-base in it's complementary strand will be ?
(1) T C A G A A C G (2) U C A G A A C G
(3) C G T T C T G A (4) None

BBM266

33. Deoxythymidylic acid is :-
(1) Ribonucleotide
(2) Deoxyribonucleotide
(3) Ribonucleoside
(4) Deoxyribonucleoside

BBM267

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34.  is –

(1) Lysine (2) Glycine
(3) Methionine (4) None of these

BBM268

35. Amino acids may be –

(1) α-AA (2) β-AA
(3) γ-AA (4) All of these

BBM269

36. Which type of bonds are not present in DNA?

(1) Ester bond (2) Glycosidic bond
(3) H-bond (4) Peptide bond

BBM270

37. Which of the protein is simple fibrous ?

(1) Elastin (2) Histone
(3) Rubisco (4) All

BBM271

38. How many H-bonds are present in between Guanine and cytosine N-base in formation of DNA ?

(1) 1 (2) 2 (3) 3 (4) 4

BBM272

39. Which of the following is monosaccharide ?

(1) Sedoheptulose (2) Glycogen
(3) Trehalose (4) All

BBM273

40. Pectin is –

(1) Monosaccharide (2) Oligosaccharide
(3) Polysaccharide (4) Protein

BBM274

41. Which of the following is nitrogen containg homopolysaccharide –

(1) Glycogen (2) Inulin
(3) Cellulose (4) Chitin

BBM275

42. Which is correct statement about tertiary configuration of protein ?

(1) Soluble in water
(2) Most abundant configuration in maximum enzymes and proteins.
(3) Di sulphide bond stabilizes this configuration.
(4) All

BBM276

43. If percentage of Adenine in double stranded DNA is 17 %. Then calculate the percentage of cytosine.

(1) 33 % (2) 66 %
(3) 83 % (4) 17 %

BBM277

44. Which of the following type of DNA has 20 Å diameter ?

(1) A-DNA (2) B-DNA
(3) D-DNA (4) Z-DNA

BBM278

45. Enzymes are different from inorganic catalysts in :-

(1) Being used up in biochemical reactions
(2) Being functional at high temperatures
(3) Being proteinaceous in nature
(4) Being denatured at low temperature

BBM279

46. Enzyme carbonic anhydrase accelerate the rate of reaction $CO_2 + H_2O \rightarrow H_2CO_3$ by about:-

(1) 10 million times
(2) 10 thousand times
(3) 100 times
(4) 10 times

BBM280

47. Enzymes, accelerate the rate of biochemical reactions by :-

(1) Increasing the thermal stability of substrate
(2) lowering down the energy difference between substrate and product
(3) Lowering down the energy difference between substrate and a specific altered structural state
(4) Increasing energy difference between product and transition state.

BBM281

48. Non protein organic substance which can not be easily separated from apoenzyme is :-

(1) Prosthetic group
(2) Coenzyme
(3) Activator
(4) Holoenzyme

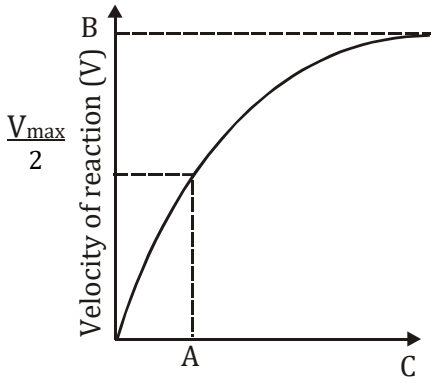
BBM282

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49. Which of the following statements is correct?
(1) Thermal stability is an important quality of enzymes of thermophilic organisms
(2) An enzyme can not has a primary structure of protein
(3) Enzymes generally function in a wide range of pH
(4) Coenzyme nicotinamide adenine dinucleotide contains the vitamin riboflavin

BBM283

50. Identify A, B and C in given graph of any enzymatic reaction :-



S = substrate

	A	B	C
(1)	S	V_{max}	K_m
(2)	K_m	V_{max}	S
(3)	S	K_m	V_{max}
(4)	V_{max}	S	K_m

BBM284

51. A substance, unrelated to the substrate, that reversibly changes the activity of enzyme by binding at a site other than the active site, is known as :-
(1) Catalytic activator
(2) Non competitive inhibitor
(3) Competitive inhibitor
(4) Allosteric modulator

BBM285

52. Which of the following statement is correct regarding enzyme ?
(1) Inhibition of succinate dehydrogenase enzyme by malonic acid is an reversible inhibition
(2) All non competitive inhibitions are not reversible inhibitions
(3) Allosteric inhibitions can not be checked by increasing substrate concentration
(4) All of the above

BBM286

53. Which of the following combinations denotes positive allosteric modulation?
(1) Succinate dehydrogenase - Malonate
(2) Folic acid synthesis - Sulpha drugs
(3) Cytochrome 'C' oxidase - Cyanide
(4) Phosphofructokinase - AMP

BBM287

54. One molecule of enzyme converts the 10 molecules of substrate into product in 10 minutes. 30 molecules of substrate and 2 molecules of enzyme are mixed in a test tube. After 10 minutes the test tube will have :
(1) 10 unreacted molecules of substrate and 20 molecules of products
(2) 20 molecules of products and 2 molecules of enzyme
(3) 10 unreacted molecules of substrate, 2 molecules of enzyme and 20 molecules of product
(4) only 20 molecules of product

BBM288

55. Find out the incorrect match from option given below?

(1)	Primary metabolites	Amino acids, chlorophylls
(2)	Secondary metabolites	Flavonoids, antibiotics
(3)	Biomacromolecules	Protein, Nucleic acid, Polysaccharides
(4)	Biomicromolecules	Nucleic acid, Amino acids

BBM289

56. Choose the correct option w.r.t. secondary metabolites?
Concanavalin A : _____ Codeine : _____
(1) Alkaloids, Lectins
(2) Carotenoids, Cellulose
(3) Lectins, Alkaloids
(4) Drugs, Terpenoides

BBM290

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57. Match List-I with List-II.

	List-I		List-II
(A)	Carotenoids	(i)	Drugs
(B)	Ricin	(ii)	Alkaloids
(C)	Diterpenes	(iii)	Terpenoides
(D)	Morphine	(iv)	Toxins
		(v)	Pigments

Choose the correct answer from the options given below :

	A	D	C	B
(1)	v	i	iii	iv
(2)	v	ii	iii	iv
(3)	v	iv	iii	ii
(4)	v	iv	iii	i

BBM291

58. Which of the following is the correct set of secondary metabolites?

(1) Abrin, Lipid, Morphine, Gums

(2) Cholesterol, Lecithin, Palmitic acid

(3) Scents, Anthocyanins, Codeine, Lectins

(4) Scents, Lectins, Codeine, Uridine

BBM292

59. Identify the statement which is incorrect?

(1) Primary metabolites have identifiable functions.

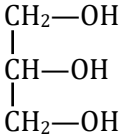
(2) Secondary metabolites have ecological importance.

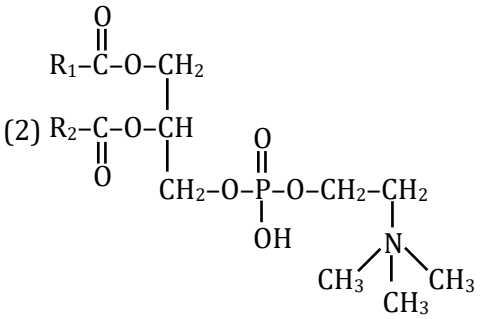
(3) Primary metabolites play known role in normal physiological processes.

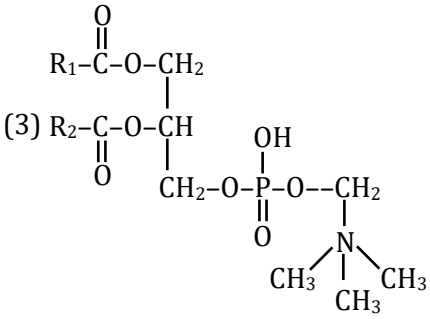
(4) Secondary metabolites are alkaloids, Ricin, Amino acids, Arachidic acids etc.

BBM293

60. Which of the following represents the structure of lecithin?

(1) 

(2) 

(3) 

(4) $CH_3-(CH_2)_{14}-COOH$

BBM294

61. Match the Column-I with Column-II.

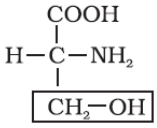
	Column-I		Column-II
(A)	Hydroxy methyl	(i)	Valine
(B)	Methyl group	(ii)	Glutamic
(C)	Aromatic	(iii)	Serine
(D)	Acidic	(iv)	Alanine
		(v)	Tyrosine

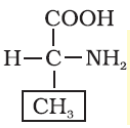
Choose the correct answer from the options given below :

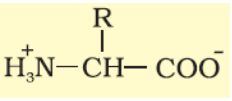
	A	B	C	D
(1)	iii	iv	v	ii
(2)	iv	iii	v	ii
(3)	iii	iv	v	i
(4)	i	ii	iii	iv

BBM295

62. Identify structure P, Q & R ?


(P)


(R)


(Q)

	P	R	Q
(1)	Serine	Zwitter ionic from	Glycine
(2)	Glycine	Serine	Alanine
(3)	Serine	Alanine	Zwitter ionic from
(4)	Amino acid	Valine	Lysine

BBM296

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63. **Assertion:** Enzymes isolated from thermophilic organisms are thermally stable.
Reason: They retain their catalytic power even at higher temperature..
(a) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(b) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(c) If the assertion is true but the reason is false.
(d) If both the assertion and reason are false

BBM297

64. Which one of the following statements is not correct regarding Amino acids.
(1) Amino acids are substituted methanes.
(2) The R group in proteinaceous amino acids could be a methyl group (Glycine).
(3) In solutions of different pH, the structure of amino acids changes.
(4) Ionizable nature of $-NH_2$ & –COOH groups.

BBM298

65. **Assertion :** Dietary proteins are the source of non-essential amino acids.
Reason : Each protein is a homopolymer, which can be essential or non-essential.
(1) Both (A) & (R) are correct but (R) is the correct explanation of (A).
(2) Both (A) & (R) are correct but (R) is not the correct explanation of (A).
(3) (A) is correct but (R) is incorrect.
(4) Both (A) & (R) are incorrect.

BBM299

66. **Statement I :** Collagen is the most abundant protein in animal world.
Statement II : Ribulose bisphosphate carboxylase – oxygenase is the most abundant protein in biosphere.
(1) Statement-I is true but statement-II is false
(2) Both statement-I & II are true
(3) Both statement-I & II are false
(4) None of these

BBM300

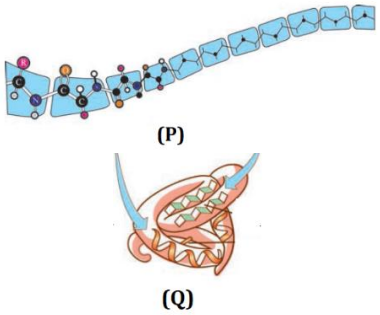
67. Match the column-I with column-II.

	Column-I		Column-II
(P)	Enzyme	(i)	Insulin
(Q)	Receptor	(ii)	Trypsin
(R)	GLUT-4	(iii)	Collagen
(S)	Hormone	(iv)	Sensory reception
		(v)	Enables glucose transport into cells

	P	Q	R	S
(1)	ii	iv	v	i
(2)	i	iv	v	ii
(3)	i	ii	iii	iv
(4)	iv	ii	v	i

BBM301

68. In below figure (P) & (Q) represents :



Choose the correct option :-
(1) P → Most stable structure
(2) Q → Most unstable structure
(3) P → First amino acid is C-terminal & last is N-terminal.
(4) Q → Necessary for many biological activities of proteins.

BBM302

69. **Assertion:** Tertiary structure is necessary for the many biological activities of proteins.
Reason: Tertiary structure gives 3-dimensional view of protein:
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.

BBM303

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70. Find out the correct match.
(1) Cellulose – Homopolymer
(2) Inulin – Polymer of Glucose
(3) Starch – Cannot hold I_2
(4) Glycogen – Reserved food of plants & monosaccharide

BBM304

71. **Statement I :** Exoskeletons of arthropods have a complex polysaccharide.
Statement II : These complex polysaccharide are mostly heteropolymers.
(1) Both statement-I & II are false.
(2) Statement-I is false & statement-II is true.
(3) Statement-I is true & statement-II is false.
(4) Both statement-I & II are true.

BBM305

72. Find out the correct set of statements :
(A) Nucleic acid is found in acid insoluble pool.
(B) Paper made from plant pulp & cotton fibre is a polymeric polysaccharide.
(C) Protein is a heteropolymer not a homopolymer.
(1) A, B are correct but C is incorrect
(2) A is correct but B & C is incorrect
(3) A, B and C are correct
(4) None of these

BBM306

73. Heterocyclic compound, monosaccharide & Phosphate or phosphoric acid are distinct components of :-
(1) Proteins
(2) Lipid
(3) Polysaccharides
(4) Building block of nucleic acid

BBM307

74. **Assertion :** 3-dimensional view of protein is indicated by a tertiary structure.
Reason : Hydrogen bond, Disulphide bond hold two or more protein molecule together.
(1) (A) is true but (R) is false
(2) Both (A) & (R) are false
(3) Both (A) & (R) are true but (R) is not the correct explanation of (A).
(4) Both (A) & (R) are true but (R) is the correct explanation of (A).

BBM308

75. Components of lecithin are :
(1) Glycerol + 1F.A. + H_3PO_4 + Choline
(2) Glycerol + 3F.A. + H_3PO_4 + Choline
(3) Trihydroxy propane + Diglycerides + H_3PO_4
(4) Trihydroxy propane + 2F.A. + H_3PO_4 + Choline

BBM309

76. Read the following statements :
(a) 20 carbon atoms are present including carboxyl carbon.
(b) Unsaturated fatty acids with one or more double bond.
(c) Polyunsaturated fatty acid.
Which of the following lipid is being described by above statements ?
(1) Arachidic acid (2) Arachidonic acid
(3) Palmitic acid (4) Oleic acid

BBM310

77. Which of the following set is incorrectly match to the category mentioned against them ?
(1) Polysaccharides → Inulin, glycogen, chitin
(2) Nucleotides → Adenosine, guanosine, thymidine
(3) Nitrogen bases → Cytosine, uracil, adenine
(4) Fatty acids → Palmitic acid, Arachidonic acid

BBM311

78. Match the column-I with column-II.

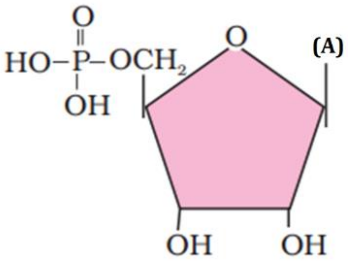
	Column-I		Column-II
(a)	Simple lipid	(i)	Hollow woolen ball
(b)	Phospholipid	(ii)	Nucleoside + H_3PO_4
(c)	Tertiary structure	(iii)	Glycerol
(d)	Nucleotide	(iv)	Lecithin

	a	b	c	d
(1)	iii	iv	i	ii
(2)	i	ii	iii	iv
(3)	iv	iii	ii	i
(4)	iv	iii	i	ii

BBM312

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79. The structure given below belongs to -



(1) Adenosine (2) Uracil
(3) Adenine (4) Adenylic acid

BBM313

80. Select the incorrect statement.
(1) Phospholipids are found in cell membrane.
(2) In macromolecule, mol. Weight > 1000 Daltons
(3) In lipid mol. weight exceed 800 Daltons
(4) In micromolecule, mol. weight < 1000 Daltons

BBM314

81. A nucleotide has three chemically distinct components.
One is a heterocyclic compound, the second is a _____ & third a _____.
(1) Nitrogenous bases, Phosphate
(2) Monosaccharide, Phosphate
(3) Sugar + Nitrogenous bases
(4) Nucleic Acid + Monosaccharide

BBM315

82. **Statement I :** Amino acids are amphoteric in nature.
Statement II : Fats & oils are glycerides in which fatty acids esterified to glycerol.
(1) Statement-I is true but statement-II is false.
(2) Both statement-I & II are false.
(3) Both statement-I & II are true.
(4) Statement-I is false but statement-II is true.

BBM316

83. Which of the following statement is correct?
(1) All nucleic acids behave like enzymes.
(2) All protein are enzyme.
(3) Organic catalysts work efficiently at high temperature.
(4) Almost all enzymes are protein.

BBM317

84. Which of the following statements about enzymes is incorrect?
(1) An active site of an enzyme is a crevice or pocket into which the substrate fits.
(2) Enzymes through their active site, catalyse reactions at a high rate.
(3) Enzymes are unstable & retain their catalytic power even at low temperatures.
(4) Crevices pockets are the active site.

BBM318

85. Ribozyme is a -
(1) Ribosome with enzyme
(2) Enzyme whose cofactor is RNA
(3) All nucleic acids that behave like enzymes
(4) RNA with enzyme activity

BBM319

86. Match the column-I with column-II.

	Column-I		Column-II
(A)	Lyases	(i)	Enzymes which catalyse joining of C–O, C–S etc bonds.
(B)	Hydrolases	(ii)	Catalyse the transfer H/e- between two substrates.
(C)	Dehydro-genases	(iii)	Catalyse removal of groups from substrates.
(D)	Ligases	(iv)	Catalysing hydrolysis of ether, ester.

	A	B	C	D
(1)	i	ii	iii	iv
(2)	iii	iv	ii	i
(3)	i	ii	iii	iv
(4)	iv	iii	ii	i

BBM320

87. Enzymes that catalyse a transfer of a group, G between a pair of substrate S & S' is ?

$$S - G + S' \rightarrow S + S' G$$

(1) Transferases (2) Hydrolases
(3) Isomerases (4) Ligases

BBM321

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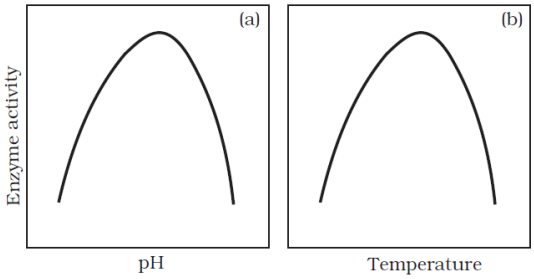
88. **A** → Non-protein constituents called apoenzyme.
B → Co-factors are bound to the enzyme to make the enzyme catalytically active.
(1) A & B are correct
(2) A & B are incorrect
(3) A is correct & B is incorrect
(4) A is incorrect & B is correct

BBM322

89. Choose the false statements?
(1) Ribozymes are nucleic acids with catalytic power.
(2) Apoenzyme = Holoenzyme + Coenzyme
(3) Holoenzyme = Apoenzyme + Cofactor
(4) Catalytic activity is lost when the co-factor is removed from the enzyme.

BBM323

90. Which one of the following conclusion cannot be made by this diagrams?



(X) **(Y)**

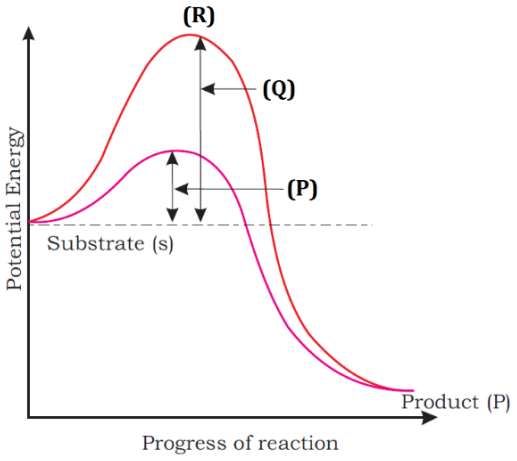
(1) Enzyme generally function in a narrow range of temperature & pH.
(2) X → Enzyme shows its highest activity at particular pH.
(3) Y → High temperature destroys enzymatic activity.
(4) Y → Low temperature preserve the enzyme in a temporarily active state.

BBM324

91. $CO_2 + H_2O \xrightarrow{(A)} B$
Choose the correct option from option given below :-
(1) A → Accelerate the reaction rate
(2) B → Transient state
(3) A → Rate of formation of product of this enzyme is 30 lakh molecules/sec.
(4) None of these

BBM325

92. What is denoted by P, Q & R in the given graph?



	P	Q	R
(A)	Activation energy with enzyme	Activation energy without enzyme	Intermediate substrate
(B)	Activation energy with enzyme	Activation energy without enzyme	Transition state
(C)	Activation energy without enzyme	Activation energy with enzyme	Transition state
(D)	Activation energy	Transition state	Product

BBM326

93. (A) Enzyme releases the products of reaction.
(B) Substrate binds to active site of the enzyme.
(C) New enzyme – product complex is formed
(D) Binding of the substrate induces the enzyme to alter its shape.
choose the correct sequence from option given below :-
(1) D → B → C → A
(2) B → D → C → A
(3) A → B → C → D
(4) B → C → D → A

BBM327

94. **Statement I :** When the binding of the chemical shuts off enzyme activity, the process is called inhibition.
Statement II : In inhibition process inhibitor is used.
(1) Both statement-I & II are false.
(2) Both statement-I & II are true.
(3) Statement-I is true but statement-II is false.
(4) Statement-I is false but statement-II is true.

BBM328

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95. **Assertion :** At high temperature enzymatic activity is destroyed.
Reason : Proteins are denatured by heat.
(1) Both (A) & (R) are true but (R) is the correct explanation of (A).
(2) Both (A) & (R) are false
(3) (A) is false but (R) is true
(4) Both (A) & (R) are true but (R) is not the correct explanation of (A).
BBM329

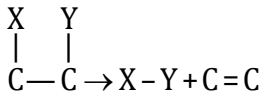
96. Why the velocity of enzymatic reaction is not exceeded by any further rise in the concentration of the substrate?
(1) Enzyme molecules are numerous than the substrate.
(2) The substrate molecules does not closely resemble the enzyme molecules.
(3) Enzyme molecules are fewer than the substrate molecules, and after saturation no free enzyme molecules is available to bind with substrate.
(4) Both (1) & (2)
BBM330

97. Which statement regarding enzyme inhibition is most appropriate?
(1) Inhibition of succinic dehydrogenase by malonate
(2) Competitive inhibition occurs when a substrate competes with enzyme for binding to inhibitor protein.
(3) In Competitive inhibition enzyme show close structural similarity with the inhibitors.
(4) Non- competitive inhibitors compete with substrate for binding the enzyme.
BBM331

98. Choose the enzyme which is responsible for inhibition of succinate by malonate.
(1) Hydrolases (2) Transferases
(3) Lyases (4) Dehydrogenase
BBM332

99. Select the incorrect statement from options given below :-
(1) Proteinaceous enzymes exhibit substrate specificity.
(2) The essential chemical components of many coenzymes are vitamins.
(3) Haem is the prosthetic group & part of the active site of the enzyme.
(4) Uncatlysed reactions proceed at rates vastly higher than that of catalysed ones.
BBM333

100. Select the type of enzyme involved in following reaction is :-



(1) Oxidoreductases (2) Transferases
(3) Lyases (4) Hydrolases
BBM334

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	4	4	3	3	4	4	2	3	1	1	2	4	1	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	4	4	2	4	1	3	4	3	3	2	2	3	4	1	2
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	1	1	2	2	4	4	1	3	1	3	4	4	1	2	3
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	1	3	1	1	2	4	4	4	3	4	3	2	3	4	2
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	1	3	1	2	4	2	1	4	2	1	3	3	4	1	4
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	2	2	1	4	3	2	3	4	3	4	2	1	4	2	4
Question	91	92	93	94	95	96	97	98	99	100					
Answer	1	2	2	2	1	3	1	4	4	3					

Structural Organisation in Animals (Animal Tissue)

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1. The simple ciliated columnar epithelium in human occurs in
(1) bronchioles and fallopian tubes
(2) bile duct and oesophagus
(3) bronchioles and bile duct
(4) Eustachian tube and stomach

BSA354

2. Which of the following type of cell junctions are found in animal tissues?
(a) adhering junctions (b) tight junctions
(c) gap junctions (d) plasmodesmata
(1) a , b and d (2) a , b and c
(3) b , c and d (4) d only

BSA355

3. Compound epithelium is found in
(1) stomach (2) oesophagus
(3) intestine (4) gall bladder

BSA356

4. Which of the following statements / is / are incorrect regarding compound epithelium ?
(a) It is made up of only one layer of cells.
(b) Their main function is to provide protection against chemical and mechanical stresses.
(c) They cover the dry surface of skin , moist surface of buccal cavity, pharynx, and inner lining of ducts of salivary glands.
(1) only a (2) a and b
(3) a and c (4) b and c

BSA357

5. Which of the following statement is incorrect regarding columnar epithelium?
(1) Cells are long column like
(2) Cells consist of basal rounded nucleus
(3) Found in stomach and intestine
(4) microvilli may be present on free surface

BSA358

6. Which of the following statement (s) regarding cell junctions is / are correct?
(1) **Tight junctions** help in stoppage of substances from leaking across a tissue
(2) **Adhering junctions** perform cementing to keep neighbouring cells together
(3) **Gap junctions** facilitate the cells to communicate with each other by connecting the cytoplasm of adjoining cells, far rapid transfer of ions, small molecules and sometimes big molecules
(4) All of the above

BSA359

7. Match the columns and choose the correct answer from the options given below:-

Column I (Terms)		**Column II (Features)**	
a.	Exocrine gland	i	Stop substances leaking across a tissue
b.	Endocrine gland	ii	Hormones are secreted directly into the fluid bathing the gland
c.	Tight junction	iii	They perform cementing to keep neighbouring cells together
d.	Adhering junction	iv	Secrets mucus, saliva, earwax, oil, milk, digestive enzymes and other cell products

(1) a-iv; b-ii; c-i; d-iii (2) a-ii; b-iv; c-i; d-iii
(3) a-iv; b-iii; c-i; d-ii (4) a-iii; b-iv; c-i; d-ii

BSA360

8. Lack of blood supply and presence of noncellular basement membrane are the characteristic features of the
(1) muscular tissue
(2) fluid connective tissue
(3) epithelial tissue
(4) nervous tissue

BSA361

9. Match the following columns

Column A		**Column B**	
a.	Loose connective tissue	i.	Bone
b.	Dense connective tissue	ii.	Areolar connective tissue
c.	Specialised connective tissue	iii.	Tendons
		iv.	Adipose connective tissue

(1) a-ii, iv; b-iii; c-i (2) a-ii, iii; b-iv; c-i
(3) a-i, ii; b-iii; c-iv (4) a-iv; b-iii; c-ii, i

BSA362

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10. Find out the incorrect match
(a) dense regular connective tissue ⇒ cartilage
(b) dense irregular connective tissue ⇒ tendons
(c) areolar connective tissue ⇒ store fats
(d) collagen fibers ⇒ yellow fibrous protein
(1) only a (2) only b
(3) a, b, c and d (4) only d

BSA363

11. Which statement is **incorrect** regarding areolar connective tissue?
(1) present beneath the skin.
(2) serves as a support framework for epithelium.
(3) cells of this tissue are specialized to store fats.
(4) it contains fibroblasts, mast cells and macrophages.

BSA364

12. Match the following tissues with their functions

Tissue		Function	
a	Areolar connective tissue	i	Connect bone to bone
b	Adipose connective tissue	ii	Connect bone to skeletal muscles
c	Tendons	iii	Store fats
d	Ligaments	iv	Serve as a support framework for epithelium

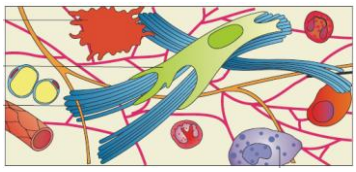
(1) a–iv , b–i , c–ii , d–iii
(2) a–iv , b–iii , c–i , d–ii
(3) a–iv , b–iii , c–ii , d–i
(4) a–iii ,b–iv , c–ii , d–i

BSA365

13. In the matrix of cartilage and bone, the cartilage and bone cells are situated in fluid filled spaces. These are known as
(1) vacuoles (2) endoderm
(3) lacunae (4) tubules

BSA366

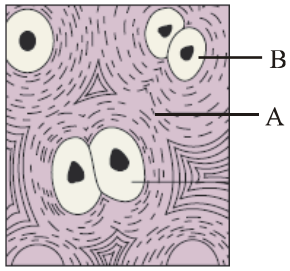
14. Refer the given figure and identify the correct characteristic feature



(a) It is a type of loose connective tissue.
(b) It contains fibroblasts, macrophages, collagen fibers and mast cells.
(c) The cells of this tissue are spaecialised to store fats.
(d) The outer wall of internal organs such as intestine and stomach contains this type of tissue.
(1) a and b
(2) a and c
(3) b and c
(4) c and d

BSA367

15. The intercellular material of the given figure is solid and resist compression. Identify the figure and the label marked as A and B.



(1) Cartilage : A-Collagen; B-Chondrocyte
(2) Cartilage : A-Collagen; B-Chondroclast
(3) Bone : A-Mitochondria; B-Chondroclast
(4) Bone : A-Collagen fibres, B-Osteoblast

BSA368

16. Which of the following type of tissue is being described by the given statements?
(a) They are named because of their special functions of linking and supporting.
(b) They include cartilage, bone, adipose and blood.
(c) They provide strength and elasticity to the tissues.
(1) Epithelial tissue
(2) Connective tissue
(3) Muscle tissue
(4) Nervous tissue

BSA369

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17. Read the following statements carefully
(a) They have a hard and non-pliable ground substance
(b) They support and protect softer tissues and organs
(c) Osteocytes are present in spaces called lacunae.
Which of the following type of tissue is being described by the above statements?
(1) Blood (2) Cartilage
(3) Bone (4) Neurons
BSA370

18. Largest cells of connective tissue proper are
(1) fibroblasts
(2) macrophages
(3) mast cells
(4) mesenchyme cells
BSA371

19. Ciliated epithelium in human body is present in
(1) Kidney & stomach
(2) Trachea & ureter
(3) Cavity of brain and fallopian tube
(4) Stomach & urinary bladder
BSA372

20. For the formation of a organ :-
(1) Only a single type of tissue is required
(2) Two types of tissues are required
(3) Any three types of tissues are required.
(4) All four types of tissues are required.
BSA373

21. A tissue which faces body fluid or the outside environment and provide a covering for some part of the body –
(1) Epithelium tissue (2) Connective tissue
(3) Muscular tissue (4) Neural tissue
BSA374

22. Which of the following statements is true ?
(1) Cells of epithelium are compactly packed with little intercellular matrix
(2) Simple epithelium lines our body cavity, ducts and tubes.
(3) Compound epithelium has protective function
(4) All of these
BSA375

23. The epithelium which is found in the wall of blood vessels and air sacs of lungs
(1) Cuboidal epithelium
(2) Squamous epithelium
(3) Columnar epithelium
(4) Ciliated epithelium
BSA376

24. An epithelium which is commonly found in ducts of glands and tubular parts of nephrons, and is specialised for secretion and absorption, can be identified as ?
(1) Simple squamous epithelium
(2) Simple cuboidal epithelium
(3) Simple columnar epithelium
(4) None of these
BSA377

25. Their function is to move particles or mucous in a specific direction over the epithelium, they are mainly present in the inner surface of hollow organs. Identify these structures and where they are found –
(1) Cilia : Stomach and Intestine
(2) Microvilli : Bronchioles
(3) Cilia : Fallopian tubes and bronchioles
(4) Steriocilia : Epididymis
BSA378

26. Which statement(s) is/are true ?
(A) Exocrine glands secrete mucus, Saliva, Earwax, Oil and milk
(B) Endocrine glands secrete hormones directly into the fluid
(C) Unicellular glands consisting of isolated glandular cells.
(1) A, B, C (2) Only C
(3) B, C (4) A, B
BSA379

27. What is true for compound epithelium ?
(1) It is made up of more than one layer of cells thus has a limited role in secretion and absorption.
(2) Its main function is to provide protection
(3) They cover the dry surface of the skin and moist surface of buccal cavity
(4) All statements are true.
BSA380

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28. Which cell junction help to stop substances from leaking across a tissue ?
(1) Gap junction (2) Tight junction
(3) Desmosomes (4) All of these
BSA381

29. Which tissue is most abundant and widely distributed in the body of complex animals ?
(1) Epithelium Tissue (2) Connective tissue
(3) Muscular tissue (4) Nervous tissue
BSA382

30. Which statements are true ?
(A) In loose connective tissue, cells and fibres are loosely arranged in a semi fluid ground substance.
(B) Areolar connective tissue provide support for epithelium beneath the skin.
(C) Adipose tissue located mainly beneath the skin
(1) A, B, C (2) A, B (3) B, C (4) A, C
BSA383

31. Which of the following is an example of dense connective tissue ?
(A) Ligament (B) Tendon
(C) Adipose tissue (D) Areolar tissue
(1) A, B (2) C, D
(3) A, B, C, D (4) Only B
BSA384

32. Which cell of connective tissue, produce and secrete fibres ?
(1) Mast cell (2) Fibroblast
(3) Macrophage (4) Adipocytes
BSA385

33. In which of the following preparation collagen fibres are present in the form of parallel bundles of fibres ?
(1) Tendon (2) Ligament
(3) Both (1) and (2) (4) None of these
BSA386

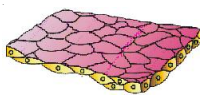
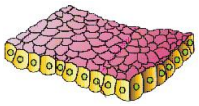
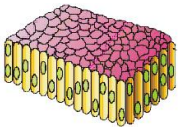
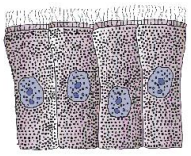
34. The tissue related with diffusion and filtration is :-
(1) Simple columnar
(2) Simple squamous
(3) Compound squamous
(4) Compound columnar
BSA387

35. Match the following columns :-

i	Unicellular gland	a	Pancreas
ii	Serous glands	b	Sebaceous gland
iii	Heterocrine	c	Goblet cell
iv	Holocrine glands	d	Watery secretion

(1) i (c), ii (a), iii (d), iv (b)
(2) i (c), ii (d), iii (a), iv (b)
(3) i (c), ii (d), iii (b), iv (a)
(4) i (a), ii (d), iii (c), iv (b)
BSA388

36. Select **incorrect** match with respect to simple epithelium shown and its location in the body?

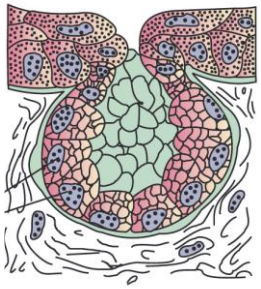
Simple epithelium		Location
(1)		Walls of blood vessels and air sacs of lungs
(2)		Ducts of glands and tubular parts of nephrons
(3)		PCT of nephron
(4)		Bronchioles and fallopian tubes

BSA389

37. Which of the following are **correct** for epithelial tissue?
(A) It has free surface which may face towards the body fluid.
(B) It provides covering for some part of body
(C) Intercellular matrix is abundant
(D) It lies on basement membrane.
Options :
(1) (A), (B), (C), (D)
(2) Only (C)
(3) (A), (C), (D)
(4) (A), (B), (D)
BSA390

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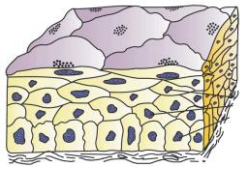
38. Choose the **incorrect** option for the secretion of given tissue in the given figure :-



(1) Saliva (2) Mucus
(3) Tear (4) Milk

BSA391

39. Choose the **correct** location of tissue for given diagram :-



(1) PCT of nephron
(2) Blood vessels
(3) Air sacs of lungs
(4) Covers the dry surface of skin

BSA392

40. Simple epithelium will be present in how many of the following structures?
Alveoli, Skin, Nephrons, Tendon, Cartilage, Blood vessel, Alimentary canal
(1) Three (2) Four
(3) Two (4) Five

BSA393

41. All the following structures are lined by transitional epithelium except :-
(1) Urinary bladder (2) Renal pelvis
(3) Ureter (4) Oesophagus

BSA394

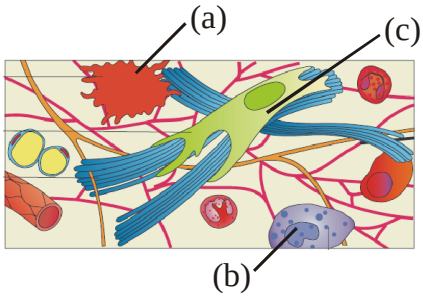
42. Goblet cells are the examples of :-
(1) Unicellular glands
(2) Multicellular glands
(3) Endocrine glands
(4) Both (2) and (3)

BSA395

43. Which modification of cell membrane increase maximum surface area for absorption?
(1) Villi (2) Microvilli
(3) Cilia (4) Stereocilia

BSA396

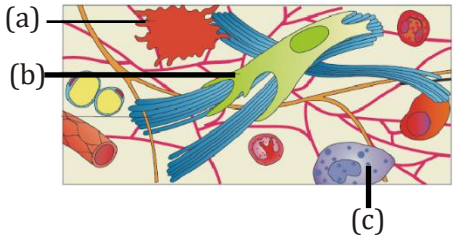
44. In the given figure, identify marked (a), (b) (c):-



(1) Fibroblast, macrophage, lymphocyte
(2) Macrophage, mast cell, fibroblast
(3) Plasma cell, mast cell, fibroblast
(4) Fibroblast, plasma cell, macrophage

BSA397

45. In the given diagram, what are the functions of (a), (b) and (c) ?



	a	b	c
(1)	Phagocytosis	Secrete fibres	Secrete heparin
(2)	Phagocytosis	Secrete heparin	Secrete fibres
(3)	Secrete heparin	Secrete fibres	Phagocytosis
(4)	Secrete fibre	Secrete heparin	Phagocytosis

BSA398

46. Tendons are :-
(A) Dense regular connective tissue with elastic fibres.
(B) Dense regular connective tissue with collagen fibres.
(C) Dense irregular connective tissue with collagen fibres.
(D) Dense irregular connective tissue with elastic fibres.
(1) Only (B) (2) (B) and (C)
(3) (A) and (D) (4) (A) and (C)

BSA399

AIM : FREE EDUCATION TO ALL APUL

47. Read the following statements (A–D)
(A) Muscular tissue is most abundant and widely distributed tissue.
(B) All connective tissue cells including blood secretes fibres of structural proteins called collagen or elastin.
(C) Epithelial cells rest on basement membrane.
(D) Blood often serves as support framework of epithelium.
How many of the above statements are incorrect?
(1) One (2) Three
(3) Two (4) Four

BSA400

48. Which one is an incorrect match ?

(1)	Compound epithelium	Main role in secretion and absorption
(2)	Areolar tissue	Support frame work for epithelium
(3)	Blood	Specialised C.T.
(4)	Dense regular connective	Tendon and ligaments

BSA401

49. Loose connective tissue is found :-
(1) Tendon and ligaments.
(2) Beneath skin.
(3) Blood.
(4) Cartilage and bone.

BSA402

50. Find out the incorrect match :-
(1) Widely distributed tissue – Connective tissue.
(2) Gap junction – Faciliate communication.
(3) Secretion of fibres – Blood cells.
(4) 40–50% part of body – Muscular tissue.

BSA403

51. ___fibres are oriented differently with ___ cells in dense irregular connective tissue.
(1) Elastic, Macrophages.
(2) Collagen, Mast cells.
(3) Reticular, Fibroblast.
(4) Collagen, Fibroblast.

BSA404

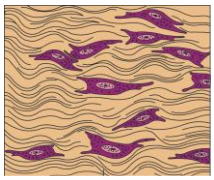
52. A tissue that is found in our body as connecting and supporting structures is :-
(1) Epithelial tissue (2) Muscular tissue
(3) Nervous tissue (4) Connective tissue

BSA405

53. Connective tissue represents different functions except :-
(1) Linking body structures.
(2) Support to body.
(3) Highest regeneration power.
(4) O_2 and CO_2 transportation.

BSA406

54. Observe the following diagram and find out correct option related to this :-



(1) Skin.
(2) Tendon and Ligaments.
(3) Pericardium.
(4) Glisson's capsule.

BSA407

55. Which of the given below structure belongs to areolar connective tissue?
(1) Ligament
(2) Tendon
(3) Tela subcutanea
(4) Panniculas adiposus

BSA408

56. **Assertion:** Ciliated epithelium mainly present in the inner surface of hollow organs like bronchioles and fallopian tube.
Reason: Cilia are microvilli.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.

BSA409

AIM : FREE EDUCATION TO ALL APUL

57. **Assertion:** Blood is connective tissue
Reason: In blood, cells use to secrete fibres of structural proteins called collagen or elastin.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.
BSA410

58. **Assertion:** Gap junctions facilitate the cells to communicate with each other.
Reason: Gap junction connects the cytoplasm of adjoining cells.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.
BSA411

59. **Assertion:** Non-striated muscles are said to be voluntary in nature.
Reason: Non-striated muscles can be moved according to will.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.
BSA412

60. **Assertion:** Intercalated discs are important regions of cardiac muscle cells.
Reason: Intercalated discs function as communication junction for muscle contraction waves.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.
BSA413

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	2	2	1	2	4	1	3	1	3	3	3	3	1	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	3	1	3	4	1	4	2	2	3	1	4	2	2	1
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	1	2	3	2	2	3	4	2	4	2	4	1	2	2	1
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	1	2	1	2	3	4	4	3	2	3	3	3	1	4	1

Structural Organisation in Animals (Cockroach)

AIM : FREE EDUCATION TO ALL APUL

1. The body cavity of arthropods is
(1) Pseudocoel (2) Coelom
(3) Haemocoel (4) Acoelom
BSA414

2. The head of the cockroach is formed by the fusion of how many segments–
(1) Five (2) Six
(3) Three (4) Ten
BSA415

3. Find out correct statements about cockroach–
(A) Anal cerci present both male and female
(B) Excretory organ are nephridia
(C) Nocturnal, Cursorial animal
(D) 3 Pair legs, 3 pair wings
(1) A, B, C (2) A, C
(3) C, D (4) B, C, D
BSA416

4. In cockroach, the position of heart is
(1) Central (2) Dorsal
(3) Ventral (4) Lateral
BSA417

5. Insects usually have ..... pairs of wings.
(1) One (2) Two
(3) Three (4) Four
BSA418

6. The heart is ..... chambered in *Periplaneta*
(1) 4 (2) 9 (3) 11 (4) 13
BSA419

7. The total number of ganglia in cockroach are
(1) Three thoracic and three abdominal
(2) Three thoracic and six abdominal
(3) Three thoracic and eight abdominal
(4) Two thoracic and six abdominal
BSA420

8. The type of circulatory system in *Periplaneta* is:
(1) Open (2) Fenestrated
(3) Peritrophic (4) Closed
BSA421

9. Fore wings of cockroach are articulated to–
(1) First abdominal segment
(2) Prothorax
(3) Metathorax
(4) Mesothorax
BSA422

10. Insects are
(1) Ammonotelic
(2) Uricotelic
(3) Ureotelic
(4) None above as they excrete only guanine
BSA423

11. In cockroach, wings are
(1) Prothoracic (2) Metathoracic
(3) Mesothoracic (4) Both (2) and (3)
BSA424

12. Cockroach has ..... kind of mouth parts
(1) Lapping and sponging
(2) Chewing and biting type
(3) Piercing and suking type
(4) Chewing and sucking
BSA425

13. Young one of the cockroach is called
(1) Naid (2) Grub
(3) Nymph (4) Maggot
BSA426

14. The functional units of compound eye of an insect are called
(1) Fenestrae (2) Ocelli
(3) Radula (4) Ommatidia
BSA427

15. How will you differentiate male from female cockroach ?
(1) Shape and body size
(2) Shape and size of wing
(3) Male has long antennae
(4) Males have anal styles
BSA428

16. Blood with corpuscles or plasma is absent in cockroach as
(1) Diffusion of gases takes place through integument
(2) Exchange of gases occur through tracheae
(3) Insects does not require oxygen
(4) Blood only carries respiratory gases
BSA429

17. In cockroach, fertilised eggs are encased in capsule called -
(1) Spermatophore (2) Ootheca
(3) Cocoon (4) Spermatheca
BSA430

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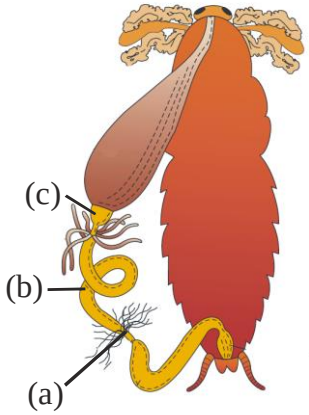
18. Male and female cockroach can be distinguished externally through–
(1) Anal styles in male
(2) Anal cerci in female
(3) Anal cerci and antennae in female
(4) Anal cerci in male

BSA431

19. Cockraoch is–
(1) Noctural – omnivorous
(2) Diurnal – omnivorous
(3) Noctural – carnivorous
(4) Diurnal – carnivorous

BSA432

20. Identify correctly matched (a), (b), (c)



(1) Colon, Mesenteron, Crop
(2) Ileum, Mesenteron, Gizzard
(3) Ileum, Mesenteron, Crop
(4) Colon, Mesenteron, Gizzard

BSA433

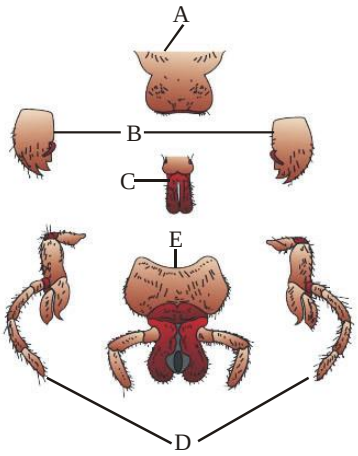
21. Which is incorrect for circulatory system of cockroach?
(1) Blood vessels are poorly developed.
(2) Plasma, haemocytes, erythrocytes are present.
(3) Heart chamber possess ostia on either side.
(4) Paired alary muscles are present.

BSA434

22. Which of the two parts in cockroach are fundamentally similar in function ?
(1) Anal style and labrum
(2) Mandibles and gizzard
(3) Maxillae & Legs
(4) Maxillae and anal cerci

BSA435

23. The figure is related with mouth part of cockroach. Identify A to E



(1) A = Maxillae B = Hypopharynx C = Labium D = Mandible E = Labrum
(2) A = Mandible B = Labium C = Maxilla D = Labrum E = Hypopharynx
(3) A = Labrum B = Mandible C = Hypopharynx D = Maxilla E = Labium
(4) A = Labium B = Hypopharynx C = Labrum D = Maxilla E = Mandible

BSA436

24. Ootheca is formed in cockroach by–
(1) Phallic gland
(2) Ejaculatory duct
(3) Collateral gland
(4) Utricular gland

BSA437

25. Which type of mouth parts present in lateral side of pre-oral cavity in cockroach ?
(1) Labrum and Mandibles
(2) Mandibles and Labium
(3) Mandibles and maxillae
(4) Mandibles and Hypopharynx

BSA438

26. The head of the cockroach is formed by the fusion of how many segments ?
(1) Five (2) Six
(3) Nine (4) Ten

BSA439

AIM : FREE EDUCATION TO ALL APUL

27. In cockroach a sac like structure present in alimentary canal and used for storing of food called :-

(1) Pharynx (2) Gizzard
(3) Crop (4) Stomach

BSA440

28. Mushroom gland is a part of :-

(1) Male reproductive system of earthworm.
(2) Female reproductive system of earthworm.
(3) Male reproductive system of cockroach.
(4) Female reproductive system of cockroach.

BSA441

29. In female cockroach ovary present in which abdominal segments ?

(1) 4th to 6th (2) 2nd to 6th
(3) 6th to 7th (4) 2nd to 4th

BSA442

30. In cockroach a pair of spermatheca is present in the 6th segment which open into the which type of structure :-

(1) Male gonopore
(2) Seminal vesicle
(3) Ejaculatory duct
(4) Genital chamber

BSA443

31. **Assertion:** Sperm glue together and form spermatophores in vas deferens in cockroach.

Reason: Sperm stores in vas deferens in cockroach.

(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.

BSA444

32. **Assertion:** Cockroach is pest.

Reason: Cockroach destroys food and contaminates it with their excreta.

(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.

BSA445

33. **Assertion:** Cockroach is monoecious.

Reason: Male and female sex organ is present in single organism.

(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.

BSA446

34. **Assertion:** Cockroaches are harmful to humans.

Reason: They are vector of several diseases.

(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.

BSA447

35. **Assertion:** If head of cockroach is cut o!, it will still live for as long as one week.

Reason: Head contain a bit of a nervous system.

(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.

(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.

(3) If the assertion is true but the reason is false.

(4) If both the assertion and reason are false.

BSA448

AIM : FREE EDUCATION TO ALL APUL

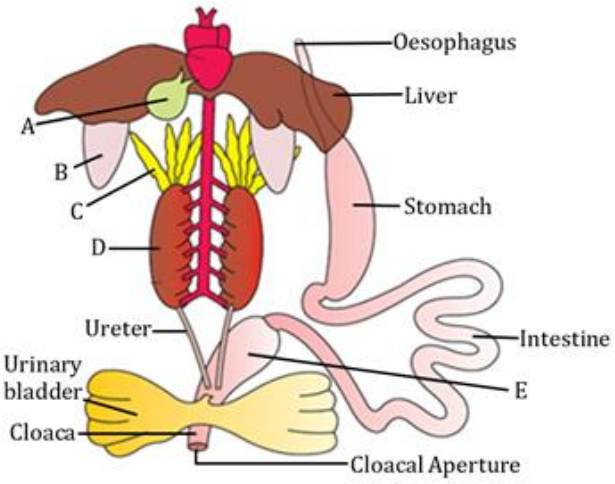
ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	2	2	2	2	4	2	1	4	2	4	2	3	4	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	2	1	1	2	2	2	3	3	3	2	3	3	2	4
Question	31	32	33	34	35										
Answer	4	1	4	1	1										

Structural Organisation in Animals (Frog)

AIM : FREE EDUCATION TO ALL APUL

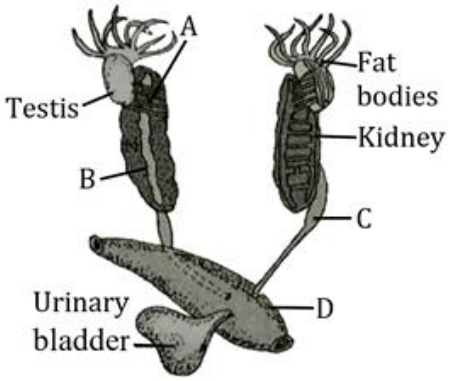
1. The above figure is associated with diagrammatic representation of internal organs of frog. Identify A to E.



(1) A-Gall bladder , B-Fat bodies , C-Lungs , D-Kidney, E-Rectum
(2) A-Gall bladder , B-Lungs , C-Rectum , D-Kidney, E-Fat bodies
(3) A-Gall bladder , B-Lungs , C-Rectum , D-Fat bodies, E-Rectum
(4) A-Gall bladder , B-Lungs , C-Fat bodies , D-Kidney, E-Rectum

BSA449

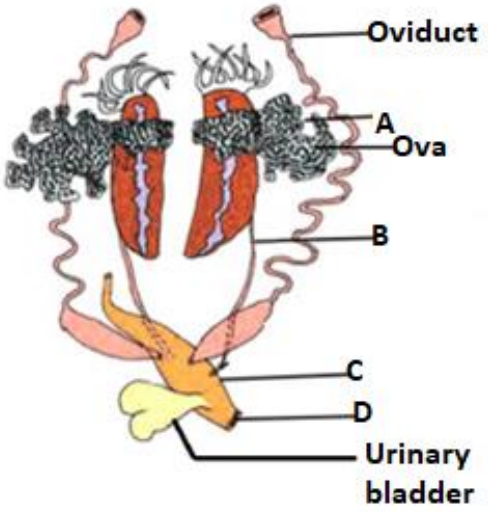
2. Go through the following figure indicating the male reproductive system of frog. Identify A to D.



(1) A-Bidder's canal; B-Adrenal gland; C-Urinogenital duct; D-Rectum
(2) A-Bidder's canal; B-Adrenal gland; C-Urinogenital duct; D-Cloaca
(3) A-Vasa efferentia; B-Adrenal gland; C-Urinogenital duct; D-Cloaca
(4) Vasa efferentia; B-Thyroid gland; C-Urinogenital duct; D-Cloaca

BSA450

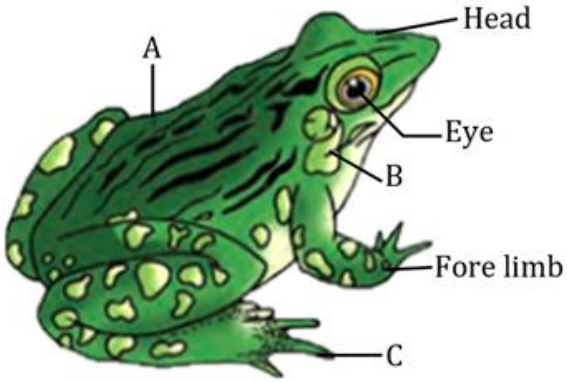
3. The above figure is related with female reproductive system of frog. Identify A to D.



(1) A-Ovary; B-Ureter; C-Oviduct ; D-Ovisac
(2) A-Ovary; B-Urinogenital duct; C-Bidder's canal; D-Ovisac
(3) A-Ovary; B-Ureter; C-Cloaca, D-Cloacal aperture
(4) A-Ovary; B-Urinogenital duct; C-Bidder's canal; D-Oviduct

BSA451

4. Identify A, B and C respectively-



(1) Trunk, Tympanum, Web
(2) Neck, Brown eye spot, Web
(3) Trunk, Tympanum, Hind limb
(4) Neck, Tympanum, Hind limb

BSA452

AIM : FREE EDUCATION TO ALL APUL

5. Most common species of frog found in India is-
(1) Rana tigrina
(2) Pheretima posthuma
(3) periplaneta americana
(4) Cliona

BSA453

6. Which of the following is incorrect about frog?
(1) They have constant body temperature
(2) They never drink water
(3) Color of dorsal side of body is generally olive green
(4) A pair of nostrils are found above the mouth

BSA454

7. Body of frog is divided into-
(1) 2 parts, head and tail
(2) 2 parts, tail and trunk
(3) 2 parts, head and trunk
(4) 3 parts, head, trunk and tail

BSA455

8. The alimentary canal of frog is short because-
(1) frog are ominovorous
(2) frog are carnivorous
(3) frog are herbivorous
(4) frog are never drink water throughout life

BSA456

9. During aestivation and hibernation gaseous exchange takes place by-
(1) Lungs
(2) Skin
(3) Gills
(4) Buccopharyngeal cavity

BSA457

10. Organ of touch in frog are called-
(1) nasal epithelium (2) tympanum
(3) sensory papillae (4) all of the above

BSA458

11. The partially digested food which is passed from stomach to intestine in frog is called–
(1) Chyle (2) Bolus
(3) Chyme (4) Bile

BSA459

12. Which is a correct statement w.r.t. frog–
(1) Sinus venosus is present on dorsal surface of heart
(2) Conus arteriosus is present on dorsal surface of heart
(3) 12 pairs of spinal nerves are present in frog
(4) Heart rate is 49 times/min.

BSA460

13. Which of the following structures is not present in digestive system of frog–
(1) Stomach (2) Rectum
(3) Esophagus (4) Caecum

BSA461

14. Which is incorrect w.r.t. frog–
(1) It can live on both land and marine water
(2) It never drinks water
(3) It is poikilothermous
(4) Hind limbs end in five digits

BSA462

15. During hibernation, frog performs–
(1) Pulmonary respiration only
(2) Cutaneous respiration only
(3) Both pulmonary and cutaneous respiration
(4) Buccal cavity respiration

BSA463

16. Each eye of frog has–
(1) Upper movable eyelid
(2) Lower immovable eyelid
(3) Thin, transparent freely movable third eyelid
(4) All of these

BSA464

17. The main excretory product in adult frog is–
(1) Ammonia (2) Uric acid
(3) Urea (4) Both (1) and (3)

BSA465

18. Select the incorrect statement w.r.t. Rana tigrina–
(1) The urinary bladder is present ventral to the rectum
(2) The medulla oblongata passes through the formen magnum and continues into spinal cord
(3) Hepatic portal system is absent but renal portal system is present
(4) Vasa efferentia are 10–12 in number

BSA466

19. A mature female frog can lay how many ova at a time–
(1) 4000 – 5000 (2) 2500 – 3000
(3) 1000 – 2000 (4) 1500 – 2500

BSA467

AIM : FREE EDUCATION TO ALL APUL

20. A fold of peritoneum called mesorchium attaches which structure to the kidneys in frog–

(1) Testes (2) Ovaries

(3) Cloaca (4) Oviduct

BSA468

21. **Assertion:** Frogs shows hibernation and aestivation

Reason: This protect frog from extreme heat and cold respectively.

(1) Both Assertion and Reason are true and Reason is a correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is not a correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

BSA469

22. **Assertion:** Frogs are said to be cold blooded.

Reason: Their body temperature varies with the temperature of environment.

(1) Both Assertion and Reason are true and Reason is a correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is not a correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

BSA470

23. **Assertion:** Frog shows sexual dimorphism

Reason: Male frog can be distinguished from female by presence of vocal sacs.

(1) Both Assertion and Reason are true and Reason is a correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is not a correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

BSA471

24. **Assertion:** Frog maintain ecological balance

Reason: Frog serves as important link in flood web and food chain in the ecosystem.

(1) Both Assertion and Reason are true and Reason is a correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is not a correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

BSA472

25. **Assertion:** Frogs are beneficial to humans

Reason: Frogs eat insect and protect our crops.

(1) Both Assertion and Reason are true and Reason is a correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is not a correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

BSA473

ANSWER KEY

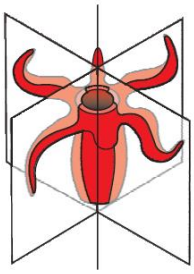
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Answer	4	3	3	3	1	1	3	2	2	3	3	1	4	1	2
Question	16	17	18	19	20	21	22	23	24	25					
Answer	3	3	3	2	1	1	1	1	1	1					

Animal Kingdom

AIM : FREE EDUCATION TO ALL APUL

1. Nervous system is found in
(1) Sponges (2) Coelenterate
(3) Annelida (4) Ctenophore
BAD452

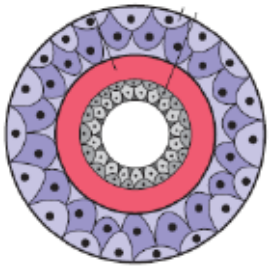
2. This type of symmetry is found in ?



(1) *Hydra*, *Pleurobrachia*, Tapeworm
(2) Tapeworm, Earthworm, Cockroach
(3) *Hydra*, Jelly fish, Starfish
(4) Sponges, Tapeworm, Mollusca
BAD453

3. How many statements are correct?
(a) In open circulatory system blood is circulated through series of vessels of varying diameter
(b) Sponges are bilaterally symmetrical
(c) When any plane passing through the central axis of the body divides the animal into two identical halves such symmetry is called bilateral symmetry.
(d) In Platyhelminthes digestive system is incomplete type.
(1) zero (2) one (3) two (4) three
BAD454

4. Choose the group which belongs to the given figure



(1) Coelenterates and ctenophores
(2) Platyhelminthes, aschelminthes and annelids
(3) Platyhelminthes, aschelminthes and chordates
(4) Sponges, coelenterates and chordates
BAD455

5. Match the columns and choose the correct answer from the options given below

COLUMN (A)		COLUMN (B)	
A	Triploblastic	i	Mesodermal in origin
B	Notochord	ii	Three germ layers
C	Metamerism	iii	true segmentation
		iv	Chordates

(1) A -ii, B-iii, C-iv (2) A-i, B-iii, C-iv
(3) A -ii, B-iv & i, C-iii (4) A-iii, B-ii, C-i
BAD456

6. Coelom is lined by
(1) Ectoderm (2) Mesoderm
(3) Endoderm (4) Mesogloea
BAD457

7. Organ level of organisation is found in
(1) Cnidaria
(2) Platyhelminthes
(3) Aschelminthes
(4) Annelida
BAD458

8. Segmentation with a serial repetition of atleast some organ is called
(1) Metagenesis (2) Metamerism
(3) Division (4) Repetition
BAD459

9. Find out the incorrect statement
(1) In pseudocoelom, mesoderm is found as scattered pouches
(2) Echinoderms, Arthropoda, Aschelminthes, Annelids are eucoelomates.
(3) In Tapeworm body cavity is absent.
(4) Young ones of echinodermata exhibit bilateral symmetry but their adults have radial symmetry.
BAD460

10. Triploblastic, complete digestive tract, eucoelomate, metameric segmentation are found in :-
(1) Tapeworm (2) Octopus
(3) Starfish (4) Cockroach
BAD461

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11. If mouth is present and anus is absent. This type of body plan is called
(1) Cell aggregate body plan
(2) Blind sac body plan
(3) Tube within tube body plan
(4) Tissue body plan

BAD462

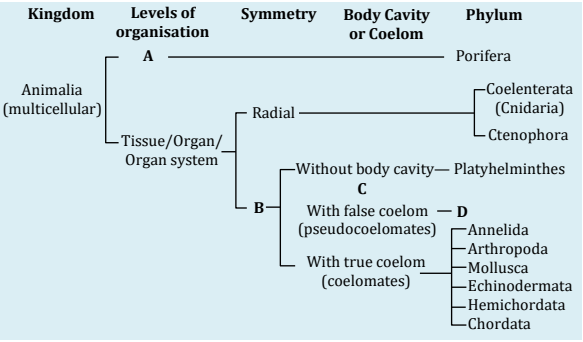
12. Which of the following statement is incorrect?
(1) Notochord is mesodermally derived structure formed on the dorsal side during development of all invertebrates
(2) In annelids closed type of blood vascular system is present
(3) Animals in which anus is formed earlier than mouth are called deuterostomiate
(4) Ctenophores are diploblastic animals

BAD463

13. Open type of circulatory system is the system in which blood is pumped out of the heart and the cells and tissues are directly bathed in it. This type of circulatory system is found in :-
(1) Arthropoda , mollusca , echinodermata
(2) Annelida , arthropoda , mollusca
(3) Arthropoda , mollusca , cephalopod mollusca
(4) Annelida , mollusca , cephalopod mollusca

BAD464

14. Complete the following chart by choosing correct option for A, B, C and D.



(1) A = Cellular level, B = Bilateral, C = Aschelminthes, D = Acoelomate
(2) A = Bilateral, B = Cellular level, C = Aschelminthes, D = Acoelomate
(3) A = Cellular level, B = Bilateral, C = Acoelomate, D = Aschelminthes
(4) A = Acoelomate, B = Aschelminthes, C = Bilateral, D = Cellular level

BAD465

15. Type of symmetry in *Sycon* and *Leucosolenia*
(1) Radial
(2) Bilateral
(3) These are assymetrical
(4) Adults radial, larva bilateral

BAD466

16. Which of the following is not the feature of animals of Porifera?
(1) Perforated body
(2) They have a water vascular system
(3) They are generally marine
(4) Asymmetrical animals except *Sycon* and *Leucosolenia*

BAD467

17. Exclusive feature of members of phylum Porifera
(a) Cellular level of organisation
(b) Ostia
(c) Spongocoel
(d) Choanocytes
(e) Hermaphrodite animals
(f) Internal fertilisation
(1) a and f (2) a, b and f
(3) a, b, c and d (4) a, b, c and e

BAD468

18. Type of digestion in Porifera and Coelenterata

	Porifera	Coelenterate
(1)	Extracellular	Intracellular
(2)	Intracellular	Extracellular Intracellular
(3)	Intracellular	Extracellular
(4)	Extracellular	Extracellular

BAD469

19. They are aquatic, mostly marine, sessile or free swimming, radially symmetrical animals. Cnidoblasts are present on the tentacles. These animals are belong to :-
(1) Porifera
(2) Coelenterata
(3) Ctenophora
(4) Platyhelminthes

BAD470

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20. Match the columns and choose the correct answer from the options given below

	Column A		Column B
a	Porifera	i	Sea walnuts
b	Coelenterate	ii	Diploblastic
c	Ctenophore	iii	Cnidoblasts
		iv	Spongocoel

(1) a-ii,iv; b-ii, iii; c-i,ii
(2) a-i,ii; b-ii,iii; c-ii, iv
(3) a-i,iii; b-ii,iv; c-i,iv
(4) a-iv; b-ii,iii; c-i,ii

BAD471

21. Which of the following is not a character of Ctenophora?
(1) Exclusively marine
(2) Body bears eight external rows of ciliated comb plates
(3) Digestion is only intracellular
(4) Fertilization is external with indirect development

BAD472

22. Pick out the odd one
(1) *Meandrina* ------- Brain coral
(2) *Pennatula* ------- Sea pen
(3) *Sycon* ------ Bath sponge
(4) *Gorgonia* ---- Sea fan

BAD473

23. Fill up with the suitable words
Cnidarians exhibit two basic body forms called __a__ and __b__. The former is ___c___ and ___d___ form whereas, the latter is ___e___ and ___f___.
(1) a-polyp; b-medusa; c-sessile; d-cylindrical; e-Umbrella shaped; f-Free swimming
(2) a-polyp; b-medusa; c-umbrella shaped; d-cylindrical; e-sessile; f-free swimming
(3) a-medusa; b-polyp; c-Umbrella shaped; d-cylindrical; e-Sessile; f-Free swimming
(4) a-medusa; b-polyp; c-sessile; d-cylindrical; e-Umbrella shaped; f-free swimming

BAD474

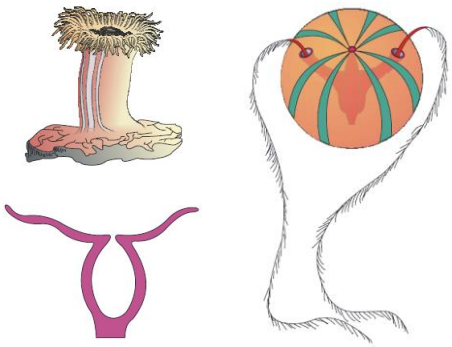
24. Identify the incorrect match
(1) Porifera ----ostia---choanocytes---indirect development----*Sycon*
(2) Porifera ----ostia---choanocytes---indirect development----*Euspongia*
(3) Coelenterata -----gastrovascular cavity---Cnidoblasts----direct development---*Obelia*
(4) Ctenophora ----comb plates----diploblastic ----indirect development---*Ctenoplana*

BAD475

25. Reproduction takes place by only sexual means in :-
(1) Ctenophora (2) Coelenterata
(3) Porifera (4) Protozoa

BAD476

26. Diagrams of some animals are given below. Identify the common characters in them



(1) Presence of cnidoblasts
(2) Presence of ciliary comb plates
(3) Extracellular and intracellular digestion
(4) Bioluminescence

BAD477

27. Choose the wrong statement with respect to Coelenterata
(1) Radial symmetry
(2) Coelom is absent
(3) Segementation is absent
(4) Cellular level of organisation

BAD478

28. Which of the following differ from others ?
(1) *Sycon* (2) *Spongilla*
(3) *Euspongia* (4) *Pleurobrachia*

BAD479

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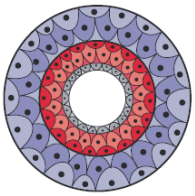
29. Many sponges are bigger than fishes, yet the sponge have no respiratory organs because they :-
(1) Do not need respiratory organs
(2) Are sedentary
(3) Respire anaerobically
(4) Have highly porous body

BAD480

30. Cnidoblasts are used for
(1) Anchorage
(2) Locomotion and defence
(3) Capture of prey
(4) All 1, 2 and 3

BAD481

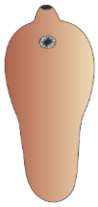
31. The cross section of the body of an invertebrate is given below. Identify the animal which has this body plan



(1) Cockroach (2) Roundworm
(3) *Taenia* (4) Earthworm

BAD482

32. Which of the following is a wrong statement about the given diagram ?



(1) Dorsoventrally flat body
(2) excretion by Malpighian body
(3) Bilaterally symmetrical
(4) Internal fertilization and indirect development

BAD483

33. Choose the option which describes the Aschelminthes with its exclusive features
(1) Rounded unsegmented body, muscular pharynx, pseudocoelomate, females are longer than males
(2) Rounded segmented body, muscular pharynx, pseudocoelomate, females are longer than males
(3) Rounded unsegmented body, muscular pharynx, acoelomate, females are longer than males
(4) Rounded unsegmented body, muscular pharynx, pseudocoelomate, females are smaller than males

BAD484

34. Read the following characters carefully
(a) Organ system level of organisation
(b) Sexes are separate
(c) Internal fertilization
(d) Bilateral symmetry
(e) Pseudocoelomate
How many animals given below consists of all the characters mentioned above a, b, c, d and e?
Ascaris, Ancylostoma, Nereis, Earthworm, *Taenia*, Liver fluke
(1) two (2) three
(3) four (4) six

BAD485

35. Find out the incorrect match about the characters of annelida
(1) Longitudinal and circular muscles------- help in locomotion
(2) Parapodia ----- help in swimming
(3) Flame cells ---- help in excretion
(4) Neural system ---- consists of paired ganglia

BAD486

36. From the following relationships between animal and character, mark the correct option :-
(a) Tapeworm ------ dorsoventrally flat
(b) Planaria -----low regeneration capacity
(c) *Ascaris* ----- Triploblastic, acoelomate
(d) *Nereis* ----- an aquatic form, dioecious
(1) a-correct, b ---incorrect, c ---incorrect, d ---correct
(2) a-correct, b ---incorrect, c ---incorrect, d ---incorrect
(3) a-correct, b ---incorrect, c ---correct, d ---incorrect
(4) a-incorrect, b ---incorrect, c ---incorrect, d ---correct

BAD487

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37. Which of the following statement / character is incorrect about the diagram given ?



(1) Possess lateral appendages, parapodia which help in reproduction
(2) Body surface distinctly marked out into segments
(3) Circulatory system is closed
(4) Nephridia help in excretion and osmoregulation

BAD488

38. Match the columns and choose the correct answer from the options given below

	Phylum	Common Name of Animal	Scientific Name of Animal
a	Platyhelminthes	(i) roundworm	(u) Taenia
b	Aschelminthes	(ii) earthworm	(v) Pheretima
c	Annelida	(iii) tapeworm	(w) Ascaris

(1) a-ii,v; b-i,w; c-iii,u
(2) a-iii,u; b-ii,w; c-i,v
(3) a-iii, u; b-ii,v; c-i,w
(4) a-iii,u; b-i,w; c-ii,v

BAD489

39. The body of these animals is circular in cross section. They may be free living, aquatic and terrestrial. They are bilaterally symmetrical and pseudocoelomate animals. These animals belong to which phylum of animal kingdom?

(1) Platyhelminthes (2) Aschelminthes
(3) Annelida (4) Arthropoda

BAD490

40. Pick out the odd one

(1) *Nereis* (2) *Pheretima*
(3) *Hirudinaria* (4) *Wuchereria*

BAD491

41. Identify the given diagram of the animal and the type of nervous system ?



(1) *Nereis*, neural system consists of paired ganglia connected by lateral nerves to a double ventral nerve cord
(2) *Nereis*, neural system consists of a paired ganglia connected by lateral nerves to a single ventral nerve cord
(3) Blood sucking leech, neural system consists of paired ganglia connected by lateral nerves to a double ventral nerve cord
(4) Blood sucking leech, neural system consists of paired ganglia connected by lateral nerves to a single ventral nerve cord

BAD492

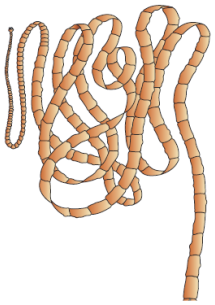
42. Open circulatory system is found in

(1) *Myxine* (2) *Pheretima*
(3) *Limulus* (4) *Nereis*

BAD493

43. (a) Hooks and suckers are present in parasitic forms
(b) Some of them absorb nutrients from the host directly through their body surface
(c) Specialised cells called flame cells help in excretion and osmoregulation
(d) Sexes are separate

Which of the above given character /s is /are not related with diagram given below?



(1) a (2) b (3) c (4) d

BAD494

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44. Roundworms differs from flatworms in having
(1) Circulatory system (2) Pseudocoel
(3) Dorsal nerve cord (4) Flame cells
BAD495

45. *Hirudinaria* is :-
(1) omnivorous (2) carnivorous
(3) herbivorous (4) sanguivorous
BAD496

46. In which of the following phyla, excretory organ is proboscis gland?
(1) Hemichordata (2) Mollusca
(3) Echinodermata (4) Arthropoda
BAD497

47. In phylum echinodermata, the adult echinoderms are _____a_____
Whereas larva are _____b_____
(1) a - radially symmetrical,
b - bilaterally symmetrical
(2) a - bilaterally symmetrical,
b - radially symmetrical
(3) a - bilaterally symmetrical,
b - asymmetrical
(4) a - radially symmetrical, b - asymmetrical
BAD498

48. A file like rasping organ for feeding called radula, present in the phylum
(1) Hemichordata (2) Mollusca
(3) Echinodermata (4) Arthropoda
BAD499

49. Which of the following phyla is being described by the given statements?
(a) They are bilaterally symmetrical, triploblastic, segmented and coelomate animals
(b) Circulatory system is open type
(c) Excretion takes place through malpighian tubules
(d) the body consist of paired jointed appendages
(1) Hemichordate (2) Mollusca
(3) Echinodermata (4) Arthropoda
BAD500

50. Which of the following is not a characteristic of phylum echinodermata?
(1) they have a water vascular system
(2) they have an internal skeleton
(3) they are pseudocoelomate
(4) they have bilateral symmetry at larval stage
BAD501

51. Which of the following statement is / are correct regarding phylum mollusca?
(a) They are bilaterally symmetrical, triploblastic and acoelomate animals
(b) Body is covered by calcareous shell and is unsegmented with a distinct head, visceral hump and muscular foot
(c) the mouth contain a file like rasping organ for feeding called radula
(d) Lithocysts are sense organs present over stalk for testing chemical and physical nature of water.
(1) a , b , d (2) a , b
(3) a , c , d (4) b , c
BAD502

52. Match the following columns and choose the correct answer from the options given below

	Scientific Name		Common Name
A	*Apis*	I.	King crab
B	*Bombyx*	II.	Lac insect
C	*Laccifer*	III	Silk worm
D	*Limulus*	IV	Honey bee

(1) A-IV; B-III; C-II; D-I
(2) A-IV; B-II; C-I; D-III
(3) A-IV; B-III; C-I; D-I
(4) A-III; B-IV; C-II; D-I
BAD503

53. Find the incorrect statement regarding hemichordate-
(1) True notochord is present
(2) Development is indirect
(3) Animals are fossorial
(4) Post anal tail is absent
BAD504

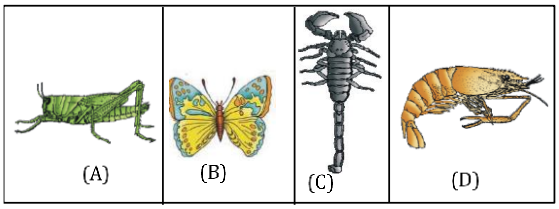
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54. In which of the following the animal name, its two characters and its phylum / class are correctly matched ?

	Animal name	Two Characters	Phylum
(1)	*Pila*	Body segmented Mouth with radula	Mollusca
(2)	*Asterias*	Spiny skin Water canal system	Echinodermata
(3)	*Periplanata*	Jointed appandages Chitinous exoskeleton	Arthropoda
(4)	*Saccoglossus*	Post anal tail present Respire by gills	Hemichordata

BAD505

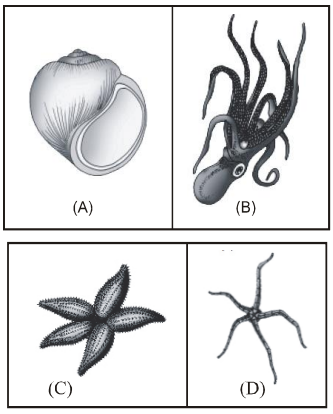
55. Refer the figures A, B, C and D given below. Which of the following options is correct for the figure ?



(1) A - *Locusta* , B - Butterfly , C - scorpion , D - Prawn
(2) A - *Locusta* , B - Prawn , C - Scorpion , D - Butterfly
(3) A - *Locusta* , B - Scorpion , C - Prawn , D - Butterfly
(4) A - Butterfly , B - Scorpion , C - Prawn , D - *Locusta*

BAD506

56. Identify the animals shown in the given figures A, B, C and D from the options given below.



	A	B	C	D
(1)	*Pila*	*Octopus*	*Asterias*	*Ophiura*
(2)	*Octopus*	*Pila*	*Asterias*	*Phhiura*
(3)	*Asterias*	*Ophiura*	*Pila*	*Octopus*
(4)	*Ophiura*	*Asterias*	*Pila*	*Octopus*

BAD507

57. Identify the correct point for the given figure?



(1) Diploblastic in nature
(2) Having bilaterally symmetrical body
(3) Dioecious with direct development
(4) Nervous system consist of two pairs of ganglia

BAD508

58. Which of the following features is common in silverfish , scorpion , dragonfly and prawn?
(1) Three pairs of legs and segmented body
(2) Chitinous cuticle and two pairs of antennae
(3) Jointed appandages and chitinous exoskeleton
(4) Cephalothorax and trachea

BAD509

59. Which of the following animals belong to the same phylum?
(1) *Pila* , *Octopus* , *Balanoglossus*
(2) *Pila* , *Pinctada* , *Sepia*
(3) *Pila* , *Asterias* , *Ophiura*
(4) *Saccoglossus* , *Loligo* , *Octopus*

BAD510

60. Pick out the odd character with respect to echinoderms :-
(1) Water vascular system
(2) Pedicellariae
(3) Autotomy
(4) Paired appendages

BAD511

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61. Animals belonging to phylum Chordata are fundamentally characterized by the presence of :_
(1) Notochord
(2) Hollow nerve cord
(3) Paired pharyngeal gill slits
(4) All of the above

BAD512

62. These are bilaterally symmetrical, triploblastic, coelomate with organ system level of organization. They possess a post anal tail and closed circulatory system.
The above mentioned lines are related to which phylum?
(1) Mollusca (2) Annelida
(3) Nematoda (4) Chordata

BAD513

63. Find out the wrong difference between chordates and nonchordates from the table given below :-

	Chordates	Non-chordates
(1)	Notochord present	Notochord absent
(2)	Heart is dorsal	Heart is ventral
(3)	Dorsal, hollow nerve cord is present	Ventral, solid nerve cord is present
(4)	Post anal tail is present	Post anal tail is absent

BAD514

64. Which of the following is not the sub phylum of chordata ?
(1) Tunicata (2) Cephalochordata
(3) Cyclostomata (4) Vertebrata

BAD515

65. In which of the following organism chordate character is present only in larval stage?
(1) *Ascidia* (2) Lancelet
(3) Hagfish (4) *Trygon*

BAD516

66. Besides the basic chordate characters, vertebrates have
(a) ventral muscular heart
(b) bony / cartilaginous vertebral column
(c) radula for feeding
(d) paired jointed appendages
Find out the characteristic which is wrong?
(1) a, b (2) c, d (3) d (4) a, b, d

BAD517

67. All living members of which class are ectoparasites on some fishes?
(1) Amphibia (2) Cyclostomata
(3) Chondrichthyes (4) Osteichthyes

BAD518

68. Match the following columns and choose the correct answer from the options given below:-

	Column A		Column B
i	*Petromyzon*	A	Sting ray
ii	*Myxine*	B	Saw fish
iii	*Pristis*	C	White shark
iv	*Carcharodon*	D	Lamprey
v	*Trygon*	E	Hagfish

(1) i-A; ii-B; iii-C; iv-D; v-E
(2) i-D; ii-E; iii-B; iv-C; v-A
(3) i-E; ii-D; iii-C; iv-A; v-B
(4) i-D; ii-E; iii-C; iv-B; v-A

BAD519

69. Which of the following statement is not correct related to cyclostomata ?
(1) Jawless circular mouth
(2) Body is devoid of scales but paired fins are present
(3) Cranium and vertebral column are cartilaginous
(4) Circulation is closed type

BAD520

70. **Assertion:** Cyclostomes are marine but migrate for spawning to fresh water.
Reason: Larvae of cyclostome is metamorphosed in marine water.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.

BAD521

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71. (a) Stream lined body
(b) Four pairs of gills
(c) Air bladder is present
(d) In males pelvic fins bear claspers
(e) Two chambered heart
(f) Fertilization is usually external
How many statements are correct for the members of class osteichthyes?
(1) one (2) two
(3) five (4) six

BAD522

72. Count the correct number of statements which are not related to the members of class chondrichthyes
(a) terminal mouth
(b) placoid scales on skin
(c) presence of air bladder
(d) poikilothermal
(e) internal fertilization
(f) vivipary
(1) one (2) two (3) three (4) Four

BAD523

73. Internal fertilisation takes place in :-
(1) Catla (2) Pterophyllum
(3) Pristis (4) Exocoetus

BAD524

74. ___ is/are marine but migrate for spawning to fresh water :-
(1) Lamprey
(2) Myxine and Torpedo
(3) Lamprey and Torpedo
(4) Torpedo and Trygon

BAD525

75. Read the following characters carefully :-
(a) Placoid scales are present
(b) Mouth is located ventrally
(c) Air bladder is absent
These characters are found in
(1) *Scoliodon* (2) *Catla*
(3) *Exocoetus* (4) *Labeo*

BAD526

76. Match the columns and choose the correct answer from the options given below

	Column A		Column B
i	*Bufo*	A	Tree frog
ii	*Rana*	B	Salamander
iii	*Hyla*	C	Limbless amphibian
iv	*Salamandra*	D	Common Indian frog
v	*Ichthyophis*	E	Toad

(1) i-c; ii-b; iii-a; iv-d; v-e
(2) i-e; ii-d; iii-a; iv-b; v-c
(3) i-e; ii-d; iii-c; iv-b; v-a
(4) i-e; ii-d; iii-b; iv-c; v-a

BAD527

77. Name of some animals are given below. Count the number of reptiles from the list given below
Chelone, Columba, Calotes, Canis
Testudo, Struthio, Alligator, Felis
Corvus, Chameleon, Bangarus, Ornithorynchus
(1) four (2) six
(3) seven (4) nine

BAD528

78. How many of the following show vivipary?
Pteropus, Camelus, Delphinus,Macropus, Ornithorhynchus, Neophron,Aptenodytes, Rattus, Equus
(1) 6 (2) 5 (3) 8 (4) 9

BAD529

79. How many of the following show pneumatic bones ?
Naja, Corvus, Psittacula, Pavo, Calotes, Bufo, Hyla, Felis, Balaenoptera, Neophron
(1) 5 (2) 6 (3) 4 (4) 9

BAD530

80. Catadromous migration is
(1) Migration from freshwater to marine water
(2) Migration from marine to fresh water
(3) Always found in fresh water
(4) Always found in marine water

BAD531

81. Universal feature of mammals without any known exception is
(1) seven cervical vertebrae
(2) body hair
(3) Corpus callosum
(4) mammary glands

BAD532

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82. Match the columns and choose the correct answer from the options given below

	Column A		Column B
i	*Chelone*	a	Garden lizard
ii	*Testudo*	b	Crocodile
iii	*Chameleon*	c	Tortoise
iv	*Calotes*	d	Turtle
v	*Crocodilus*	e	Tree lizard

(1) i-c; ii-b; iii-a; iv-e; v-d
(2) i-d; ii-c; iii-e; iv-b; v-a
(3) i-c; ii-d; iii-e; iv-b; v-a
(4) i-d; ii-c; iii-e; iv-a; v-b

BAD533

83. Match the columns and choose the correct answer from the options given below

	Column A		Column B		Column C
i	*Naja*	a	Ajgar	k	Poisonous
ii	*Bangarus*	b	Cobra	l	non-poisonous
iii	*Vipera*	c	Krait		
iv	*Python*	d	Viper		

(1) i/b/k; ii/c/k; iii/d/l; vi/a/l
(2) i/b/k; ii/c/k; iii/d/k; iv/a/l
(3) i/c/k; ii/b/k; iii/d/k; iv/a/l
(4) i/d/k; ii/b/k; iii/a/k; iv/c/l

BAD534

84. Which of the following is not the characteristic feature of amphibians?
(1) body is divisible into head and trunk
(2) moist skin
(3) oviparity and direct development
(4) eyes have eyelids and tympanum present

BAD535

85. Which of the following is not the feature of reptiles?
(1) external ear absent / tympanum present
(2) heart usually three chambered / four chambered in crocodile
(3) oviparity / indirect development
(4) poikilothermous

BAD536

86. Presence of feathers, beak, pneumatic bones, crop, gizzard, homeothermal,
Oviparity, external fertilization
How many characters are related to Aves?
(1) 4 (2) 5 (3) 7 (4) 6

BAD537

87. How many of the following characteristics are present in humans

Muscular diaphragm	Seven cervical vertebrae	Mammary glands
Body hair	Oviparity	Corpus callosum
Four chambered heart	Direct development	Thecodont dentition
One pair of forelimbs and one pair of hindlimbs		

(1) Seven (2) Eight
(3) Nine (4) Ten

BAD538

88. Which of the following is not a similarity between crocodile and humans?
(1) four chambered heart
(2) Thecodont dentition
(3) corpus callosum
(4) 2 pairs of limbs

BAD539

89. Match the columns and choose the correct answer.

	Column A		Column B
i	*Corvus*	a	Vulture
ii	*Columba*	b	Ostrich
iii	*Struthio*	c	Parrot
iv	*Struthio*	d	Pigeon
v	*Neophron*	e	Crow

(1) i-e; ii-b; iii-d; iv-c; v-a
(2) i-e; ii-d; iii-c; iv-b; v-a
(3) i-c; ii-a; iii-b; iv-d; v-e
(4) i-d; ii-e; iii-c; iv-b; v-a

BAD540

90. Find out the incorrect match :
(1) Felis ------ Cat
(2) Canis ------ Tiger
(3) Rattus ---- Rat
(4) Pteropus ---- Flying fox

BAD541

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91. **Assertion:** Acraniata is a group of organisms which do not have distinct cranium.
Reason: It includes small marine forms without head.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.
BAD542

92. **Assertion:** Digestive system of platyhelminthes is incomplete.
Reason: They have single opening to outside of the body, serve as both mouth as well as anus.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.
BAD543

93. **Assertion:** Coelenterates, Ctenophores and adult Echinoderms are said to be radial symmetrical.
Reason: Their body can be dived into two equal halves in any plane passing through central axis of the body.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.
BAD544

94. **Assertion:** Fertilisation in bony fishes is usually external.
Reason: In males in bony fishes pelvic fins bear claspers.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.
BAD545

95. **Assertion:** *Ascidia* belong to sub - phylum urochordata.
Reason: Notochord is present in tail of larva stage.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.
BAD546

96. **Assertion:** Aschelminthes are pseudocoelomate.
Reason: Body cavity in these organisms is not lined with mesoderm.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.
BAD547

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97. **Assertion:** Porifera to Echinoderms are non-chordates.
Reason: Notochord is not formed during embryonic development in these animals.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.
BAD548

98. **Assertion:** Most sponge's body are said to be asymmetrical.
Reason: Body of most sponge can't be divided into two equal half by any plane pass through the centre of body.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.
BAD549

99. **Assertion:** Platyhelminthes are known as flat worms.
Reason: Platyhelminthes have laterally compressed body.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.
BAD550

100. **Assertion:** In amphibian skin is generally moist.
Reason: They shows cutaneous respiration.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.
BAD551

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	3	2	1	3	2	2	2	2	4	2	1	1	3	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	3	2	2	4	3	3	1	3	1	3	4	4	4	4
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	3	2	1	1	3	1	1	4	2	4	3	3	4	2	4
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	1	1	2	4	3	4	1	1	3	1	1	2	3	2	4
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	4	4	2	3	1	2	2	2	2	3	3	2	3	1	1
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	2	2	1	3	1	4	4	2	3	3	3	3	3	2	2
Question	91	92	93	94	95	96	97	98	99	100					
Answer	2	1	1	3	1	1	1	1	3	1					

Human Reproduction # 01 (The Male Reproductive System)

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1. Read the given statements and choose correct option about them :-

(A) Each lobule of testis contains one to three highly coiled seminiferous tubules and it is lined inside by only columnar cells

(B) The vasa efferentia leave the testis and open into vas deferens

(C) Functional maturation of sperms take place in epididymis.

(D) Seminal plasma is rich in fructose, calcium and certain enzymes.

	A	B	C	D
(1)	True	False	True	True
(2)	False	False	True	True
(3)	False	False	False	False
(4)	True	True	False	False

BHR335

2. Fill in the blanks in the following statements : The human male ejaculates about ..... million sperms during a coitus of which, for normal fertility, at least ..... percent sperms must have normal shape and size and at least ..... percent of them must show vigorous motility.

(1) 400 - 500, 60, 40 (2) 100 - 120, 40, 60

(3) 200 - 300, 60, 40 (4) 200 - 300, 40, 60

BHR336

3. Which of the following statement is not correct about scrotum ?

(1) The temperature of scrotum is 2–2.5°C lower than normal body temperature.

(2) Scrotum is connected to the abdominal cavity through a passage termed as inguinal canal.

(3) During embryonic stage, testes develop in scrotum and remain in scrotum at the time of birth.

(4) Scrotum is a small bag like structure situated below abdominal cavity.

BHR337

4. Read the following four statements about the function of Sertoli cell and answer the question given below.

(A) To provide the nutrition of sperm.

(B) To phagocyte the dead or injured sperm.

(C) Secretions of testosterone.

(D) Act as gamete mother cell.

Which set of option is incorrect statements?

(1) A & B (2) C & D

(3) A & C (4) B & D

BHR338

5. Identify the correct pair :-

Hormones		Function	
I	Inhibin	a	Development of secondary sexual characters of male
II	Androgen binding protein (ABP)	b	Stimulates Leydig cells to secrete testosterone
III	Testosterone	c	Concentrates testosterone inside seminiferous tubule
IV	LH	d	Suppression of synthesis and reléase of FSH

(1) I-d, II-b, III-a, IV-c

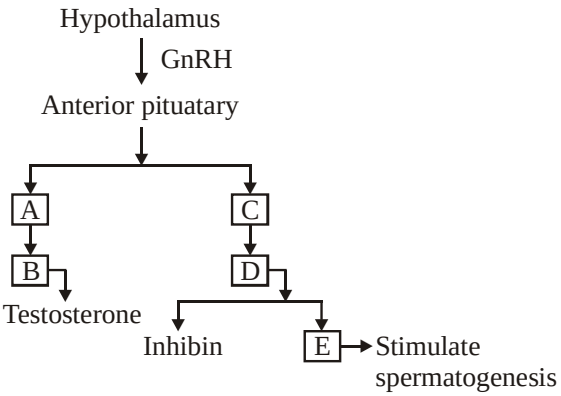
(2) I-d, II-c, III-a, IV-b

(3) I-c, II-d, III-a, IV-b

(4) I-c, II-b, III-a, IV-d

BHR339

6.



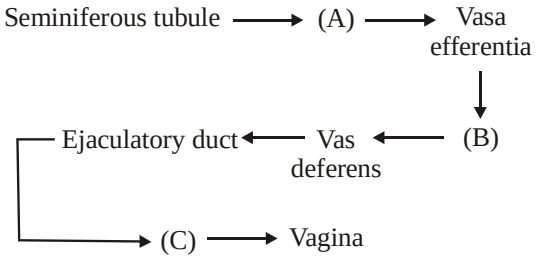
Find out A, B, C and E respectively :-

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	A	B	C	D	E
(1)	Testis	Leyding cells	Sertoli cells	AMH	ABP
(2)	ICSH	Testis	AMH	Sertoli cells	ABP
(3)	LH	Interstitial cells	FSH	Sertoli cells	AMH
(4)	ICSH	Leyding cells	FSH	Sertoli cells	ABP

BHR340

7. Here path of sperms in mammals is shown :-

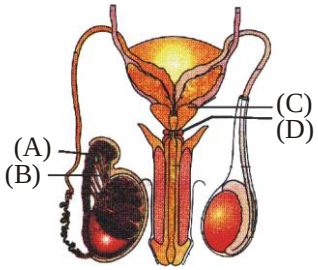


A, B & C respectively are -

	A	B	C
(1)	Epididymis	Testes	Urethra
(2)	Rete testis	Epididymis	Urethra
(3)	Ovary	Epididymis	Urethra
(4)	Testes	Rete testis	Epididymis

BHR341

8. Identify A, B, C and D in the given diagram. Which of them has been correctly identified along with its function :-



(1)	(A)	Epididymis	Formation of sperm
(2)	(B)	Rete testis	Functional maturation of sperms
(3)	(C)	Seminal vesicle	store and transport the sperms
(4)	(D)	Bulbourethral	destroy the acidity of urethra

BHR342

9. Which structures form the extra testicular pathways for sperms?

(1) Seminiferous tubules, Epididymis, Vas deferens
(2) Seminiferous tubules, Vasa efferentia, Epididymis
(3) Epididymis, Vas deferens, Ejaculatory duct
(4) Rete testis, Epididymis, Vas deferens

BHR343

10. After some time of ejaculation, semen liquefies due to presence of an enzyme which is found in secretion of :-

(1) Vagina (2) Seminal vesicle
(3) Prostate (4) Cowper's gland

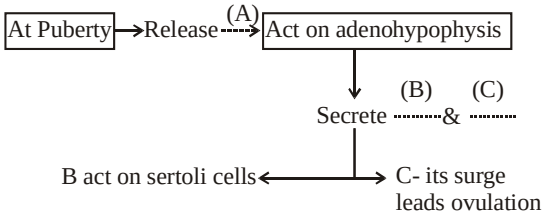
BHR344

11. Which of the following pair is correct ?

(1) Testosterone – Secreted by Sertoli cells
(2) FSH – Secreted by Leydig cells
(3) ABP – Concentrates the testosterone in seminiferous tubules
(4) ICSH – Stimulates the Leydig cells to secrete FSH

BHR345

12.



identify the A, B and C :-

	A	B	C
(1)	GnRH	LH	FSH
(2)	GnRH	FSH	LH
(3)	Somatotrophic	ICSH	STH
(4)	Somatotrophic	Androgen	STH

BHR346

13. Where sertoli cells are found and what function do they perform ?

(1) Pancreas and secrete cholecystokinin
(2) Ovaries and secrete progesterone
(3) Adrenal cortex and secrete adrenaline
(4) Seminiferous tubules and provide nutrition to germ cells

BHR347

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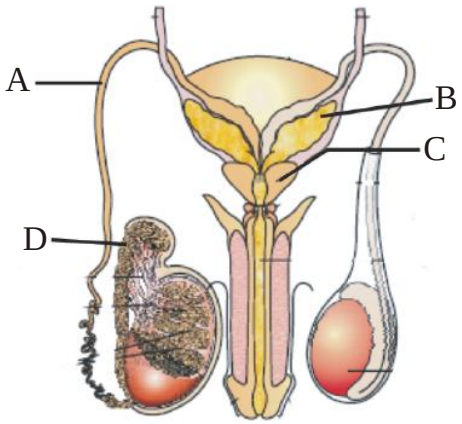
14. What is/are the function of bulbourethral gland:-
(1) Secretion of clear mucus that neutralizes acidic medium in the urethra
(2) Secretion also helps in the lubrication of the penis during coitus
(3) Release fructose for motility of sperm
(4) Both (1) and (2)

BHR348

15. Erection of penis occurs due to :-
(1) Sympathetic nervous system
(2) Parasympathetic nervous system
(3) Both (1) and (2)
(4) Cerebral cortex → ANS → Vasoconstriction

BHR349

16. Identify the diagram given below & choose correct A, B, C, D :-



	A	B	C	D
(1)	Vas deference	Seminal vesicle	Prostate gland	Epididymis
(2)	Vas deference	Prostate gland	Seminal vesicle	Tubuli recti
(3)	Ureter	Vas deference	Prostate gland	Epididymis
(4)	Epididymis	Prostate gland	Seminal vesicle	Vas deference

BHR350

17. Intratesticular genital duct system includes:-
(1) Tubuli recti, rete testis and vasa efferentia
(2) Tubuli recti, vas deferens and ejaculatory duct
(3) Urethra, epididymis and tubuli recti
(4) Seminal vesicle, ejaculatory duct and ampulla

BHR351

18. Optimal temperature for spermatogenesis is :
(1) Normal body temperature (37°C)
(2) 43° – 45°C
(3) 39° – 40°C
(4) 34.5 – 35°C

BHR352

19. Which is not a set of secondary sex organs ?
(1) Epididymis, seminal vesicle, penis
(2) Uterus, vagina, fallopian tube
(3) Prostate, bulbourethral gland, vas deferens
(4) Fallopian tube, ovary, penis, vagina

BHR353

20. Which is not a content of spermatic cord ?
(1) Vas deferens
(2) Gonadal artery
(3) Seminiferous tubules
(4) Gonadal vein

BHR354

21. **Assertion:** Testis is situated outside the abdominal cavity with in a pouch called scrotum.
Reason: Scrotum helps in maintaining the low temperature of the testes necessary for spermatogenesis.
(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) Assertion is true but the reason is false.
(4) Both the assertion and reason are false.

BHR355

22. **Assertion:** Male germ cells undergo meiotic division finally leading to sperm formation.
Reason: Sertoli cells provide nutrition to Leydig cells.
(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) Assertion is true but the reason is false.
(4) Both the assertion and reason are false.

BHR356

23. **Assertion:** Human are viviparous.
Reason: Human are sexually reproducing organism.
(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) Assertion is true but the reason is false.
(4) Both the assertion and reason are false.

BHR357

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	3	3	2	2	4	2	4	3	3	3	2	4	4	2
Question	16	17	18	19	20	21	22	23							
Answer	1	1	4	4	3	1	3	2							

Human Reproduction # 02 (The Female Reproductive System)

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1. Choose the correct option regarding mammary gland :-
(1) Areola is a non-pigmented and sensitive area around nipple
(2) Prolactin act on mammary ampulla and stimulate for milk synthesis
(3) The glandular tissue of each breast is divide into 2–5 mammary Lobes
(4) The cells of alveoli secrete milk

BHR358

2. Read the given statements and select the statements which are correct :-
(A) The uterus is single pair and is also called womb
(B) The shape of uterus is like an inverted pear
(C) The uterus open into vagina through a narrow cervix
(D) Birth canal is formed by cervical canal and vaginal canal
(E) Perimetrium of uterus wall is outermost and thick glandular layer
(1) A, B & C (2) B, C & D
(3) D & E (4) A & E

BHR359

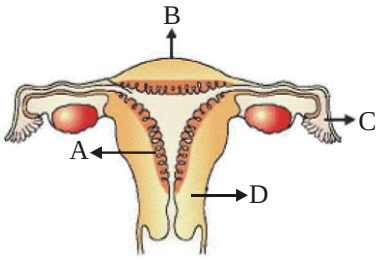
3. Which of the following statement is incorrect regarding female external genitalia ?
(1) Mons pubis is a cushion of fatty tissue covered by skin and pubic hairs
(2) Clitoris lies at the upper junction of labia minora below the urethral opening
(3) Hymen is not a reliable indicator of virginity
(4) Labia majora are fleshy folds of tissue

BHR360

4. A functional mammary gland is the characteristic feature of :-
(1) All mammals
(2) All primates
(3) All male mammals
(4) All female mammals

BHR361

5. Given diagram is of female reproductive system Identify the structures which are marked as A, B, C and D : -



	A	B	C	D
(1)	Endometrium	Uterine cavity	Infundibulum	Vagina
(2)	Myometrium	Uterine cavity	Infundibulum	Cervix
(3)	Endometrium	Uterine fündus	Infundibulum	Cervix
(4)	Perimetrium	Uterine fündus	Infundibulum	Cervix

BHR362

6. If ampulla of mammary gland is blocked, the milk will not go from :-
(1) Alveoli to mammary duct
(2) Mammary tubule to mammary duct
(3) Mammary duct to lactiferous duct
(4) Mammary duct to mammary tubule

BHR363

7. There are three layers present in uterus, one of them is thick muscular, second is thin visceral peritoneum and third glandular layer. Identify layers respectively :-
(1) Perimetrium, Myometrium, Endometrium
(2) Endometrium, Myometrium, Perimetrium
(3) Myometrium, perimetrium, Endometrium
(4) Myometrium, endometrium, Perimetrium

BHR364

8. Read the following
(A) It is a thick layer of smooth muscles
(B) It is highly glandular layer
(C) It exhibits strong contraction during delivery of the baby
(D) It is a thin membranous layer

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Which of the above statements are correct with respect to myometrium?

(1) A and B (2) B, C and D

(3) A and C (4) A, B and C

BHR365

9. The ovaries are located one on each side of the lower abdomen and is connected to pelvic wall and uterus by :-

(1) Ligament

(2) Tendon

(3) Loose connective tissue

(4) Dense irregular connective tissue

BHR366

10. Which is incorrectly matched pair among these?

(1) Labia majora – 1 pair

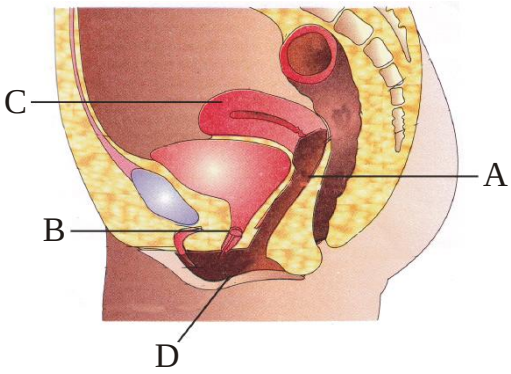
(2) Labia minora – 1 pair

(3) Clitoris – 1 pair

(4) Bartholin's gland – 1 pair

BHR367

11. Given diagram is of female reproductive system. Identify the structures which are marked as A, B, C and D



	(A)	(B)	(C)	(D)
(1)	Urethra	Vaginal orifice	Vagina	Uterus
(2)	Vagina	Urethra	Uterus	Vaginal orifice
(3)	Vaginal orifice	Vagina	Urethra	Uterus
(4)	Uterus	Vagina	Vaginal orifice	Urethra

BHR368

12. Why is the presence or absence of hymen not a reliable indicator of virginity or sexual experience?

(1) In some women, the hymen persists even after performing coitus

(2) Hymen can also be ruptured by sudden fall

(3) Hymen may be either present or absent in female

(4) All of the above

BHR369

13. Which of the following structure is incorrectly paired with its function ?

(1)	Seminal vesicle	Secrete sugar containing fluid.
(2)	Fimbriae	Helps in collection of the ovum after ovulation.
(3)	Ovary	Secrete steroid hormone.
(4)	Bartholin's gland	Destroy the basicity of the urethra.

BHR370

14. Which one of the following statements is wrong about mammary gland ?

(1) The lactiferous ducts of each lobe join to form mammary ampulla.

(2) Several mammary ducts join to form a mammary ampulla.

(3) Each breast is divided into 15-20 mammary lobes.

(4) Colostrum contains several antibodies.

BHR371

15. Read the given sentences carefully and answer accordingly.

(a)'A' is single

(b) The shape of 'A' is like an inverted pear

(c) 'A' opens into vagina

(d) The wall of the 'A' has three layers of tissue

Here 'A' is :-

(1) Ovary (2) Uterus

(3) Urethra (4) Fallopian tube

BHR372

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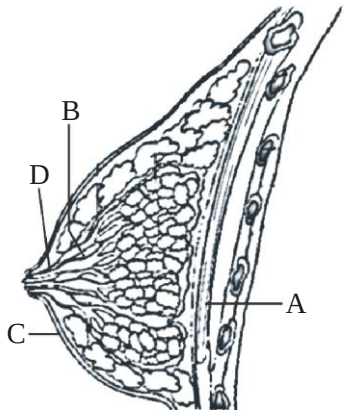
16. Which of the following structures are not included in external genitalia of female ?

(a) Mons pubis
(b) Labia majora
(c) Hymen
(d) Labia minora
(e) Cervix
(f) Fallopian tube

(1) c, e and f
(2) Only e and f
(3) Only d and f
(4) a, d, e and f

BHR373

17. The figure given below depicts a diagrammatic sectional view of mammary gland. Identify the structures which are marked as A, B, C, D.



	(A)	(B)	(C)	(D)
(1)	Muscles between ribs	Mammary lobe	Nipple	Mammary duct
(2)	Pectoralis major muscle	Ampulla	Areola	Lactiferous duct
(3)	Pectoralis major muscle	Mammary duct	Nipple	Lactiferous duct
(4)	Muscles between ribs	Ampulla	Areola	Mammary alveolus

BHR374

18. Choose the correct option to complete the given statement :-

The ..... of uterus undergoes cyclical changes during menstrual cycle while the ..... exhibits strong concentration during delivery of the baby.

(1) Perimetrium, myometrium
(2) Myometrium, endometrium
(3) Endometrium, myometrium
(4) Endometrium, perimetrium

BHR375

19. The clitoris is :-

(1) Represented by hymen.
(2) A male secondary sex organ.
(3) A tiny finger like structure that lies at the upper junction of the two labia minora.
(4) A tiny finger like structure present below the urethral opening in female.

BHR376

20. The milk synthesis in the mammary gland of a female after parturition occurs due to the action of :-

(1) Progesterone
(2) Prolactin
(3) Oxytocin
(4) Follicle stimulating hormone (FSH)

BHR377

21. **Assertion:** Myometrium undergoes cyclical changes during menstrual cycle.

Reason: Myometrium's lines the uterine cavity

(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) Assertion is true but the reason is false.
(4) Both the assertion and reason are false.

BHR378

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22. **Assertion:** Mammary glands differentiate during pregnancy.

Reason: Mammary glands secretes milk after child birth.

(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.

(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.

(3) Assertion is true but the reason is false.

(4) Both the assertion and reason are false.

BHR379

23. **Assertion:** Proximal end of fallopian tube (infundibulum) possess fimbriae (finger like projection).

Reason: Fimbriae help in catching of ovum after ovulation.

(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.

(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.

(3) Assertion is true but the reason is false.

(4) Both the assertion and reason are false.

BHR380

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	2	2	4	3	3	3	3	1	3	2	4	4	1	2
Question	16	17	18	19	20	21	22	23							
Answer	2	2	3	3	2	4	1	1							

Human Reproduction # 03 (Gametogenesis)

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1. Find out the correct match from the following table.

	Column I	Column II	Column III
(i)	Primary Spermatocyte	(n)	Maturation phase
(ii)	Secondary Spermatocyte	(n)	Maturation phase
(iii)	Spermatid	(n)	Multiplication phase

(1) (i) only (2) (i) and (ii)
(3) (ii) only (4) (ii) and (iii)

BHR381

2. Read the following statements carefully.
The .....(A)..... present on the inside wall of .....(B)..... multiply by .....(C)..... division. A .....(D)..... completes the .....(E)..... division leading to formation of two equal, haploid cells called .....(F)..... .
Which one of the following options gives the correct filling of A, B, C, D, E and F ?

	A	B	C	D	E	F
(1)	Spermato-gonia	Semini-ferous tubule	Mitotic	Primary Spermato-cyte	First meiotic	Secondary Spermato-cyte
(2)	Primary Spermato-cyte	Sertoli Cell	Mitotic	Secondary Spermato-cyte	First meiotic	Spermatid
(3)	Spermato-gonia	Sertoli Cell	First meiotic	Secondary Spermato-cyte	Second meiotic	Spermatid
(4)	Primary Spermato-cyte	Semini-ferous tubule	First meiotic	Spermatid	Second meiotic	Sperm

BHR382

3. During Oogenesis in human, first maturation division starts ___A___ and completes ____B___.
(1) A-Before birth, B-Just prior to ovulation
(2) A-After birth, B-Before menopause
(3) A-Before ovulation, B-After entry of sperm
(4) A-Before ovulation, B-After fertilisation

BHR383

4. How many structures in the list given below are haploid ?
Spermatid, secondary oocyte, primary spermatocyte, ovum, sperm, oogonia, spermatogonia, polar body
(1) Six (2) Four
(3) Two (4) Five

BHR384

5. Which one of the following structure of sperm is mismatched?
(1) Acrosome — Filled with enzyme that help in fertilisation of the ovum
(2) Nucleus — Contains an elongated and haploid nucleus
(3) Proximal centriole — Starts cleavage in zygote
(4) Tail — Contains numerous mitochondria, which produce energy for movement of tail

BHR385

6. If germ cells in female and male gonads begin undergoing meiosis simultaneously. What will be ratio of ova and sperms produced?
(1) 1 : 1 (2) 1 : 2
(3) 1 : 4 (4) 2 : 1

BHR386

7. Spermatogenesis starts at the age of puberty due to :-
(1) Significant increase in growth hormones in male body
(2) Significant increase in number of germ cells
(3) Significant increase in GnRH secreted from hypothalamus.
(4) Significant decrease in GnRH secreted from hypothalamus

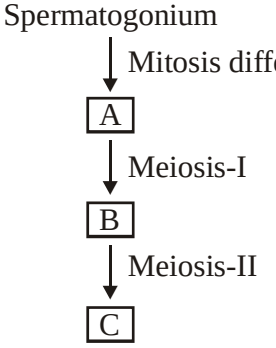
BHR387

8. Consider the following statements (a–d) and select the options, which includes all the correct ones only :
(a) Maturation phase is the longest phase in oogenesis process of human.
(b) Oogenesis is initiated at puberty.
(c) Primary oocyte temporarily arrested in prophase-I stage.
(d) During oogenesis meiotic-I division is an equal division leading to formation of secondary oocyte and polar body.
(1) Statement (a), (b), (c)
(2) Statement (a), (c)
(3) Statement (b), (d)
(4) Statement (a), (d)

BHR388

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9. Identify A, B and C in following figure :-



(1) A-Secondary spermatocyte, B-Primary spermatocyte, C-Spermatids
(2) A-spermatid, B-Primary spermatocyte, C-Secondary spermatocyte
(3) A-Spermatid, B-Secondary spermatocyte, C-Primary spermatocyte
(4) A-Primary spermatocyte, B-Secondary spermatocyte, C-spermatid

BHR389

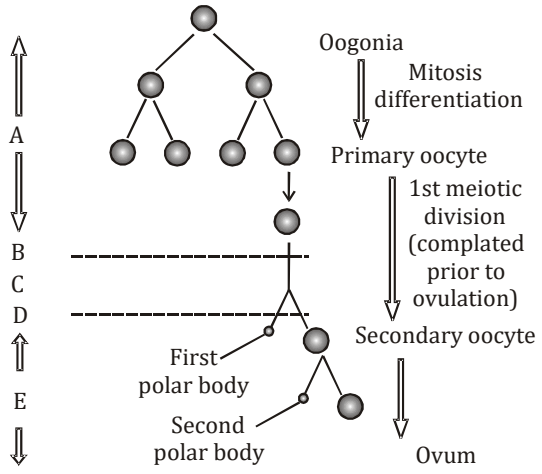
10. At the time of puberty total how many primary follicles are left in human female ?
(1) 60,000 – 80,000
(2) 40,000 – 60,000
(3) 1,20,000 – 1,60,000
(4) 80,000 – 1,20,000

BHR390

11. During oogenesis, the meiosis division in secondary oocyte is arrested at which stage ?
(1) Prophase I (2) Prophase II
(3) Metaphase I (4) Metaphase II

BHR391

12. The following refers to schematic representation of oogenesis.
Identify A to E correctly :



(1) A - Fetal life, B - Birth, C - Puberty
D - Adult reproductive life, E - Child body
(2) A - Fetal life, B - Birth, C - Child hood
D - Puberty, E - Adult reproductive life
(3) A - Adult reproductive life, B - Birth,
C - Puberty, D - Child hood, E - Fetal life
(4) A - Birth, B - Child hood, C - Fetal life
D - Puberty, E - Adult reproductive life

BHR392

13. Match list-I (Cell type) with list-II (Characteristic) and select the option with correct matchings.

	List-I		List-II
(A)	Spermatogonium	(i)	Does not divide
(B)	Spermatocyte	(ii)	Divides meiotically
(C)	Spermatid	(iii)	Divides mitotically
(D)	Sertoli cell	(iv)	Divides both mitotically and meiotically
		(v)	Nourishes other cell type

	A	B	C	D
(1)	(v)	(iii)	(i)	(iv)
(2)	(i)	(iii)	(iv)	(v)
(3)	(iii)	(ii)	(iv)	(i)
(4)	(iii)	(ii)	(i)	(v)

BHR393

14. How many sperm and ova will produce from 40 spermatogonium and 40 oogonium respectively.
(1) 160 sperm and 160 ova
(2) 40 sperm and 140 ova
(3) 160 sperm and 40 ova
(4) 40 sperm and 40 ova

BHR394

15. How many of the following are diploid structures ?
Primary oocyte, spermatozoa, spermatid, oogonium, ovum, second polar body, secondary spermatocyte, spermatogonium.
(1) 4 (2) 5
(3) 3 (4) 6

BHR395

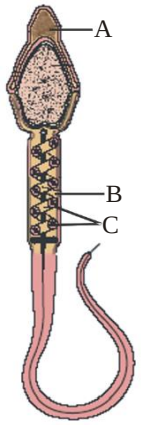
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16. What is true about polar bodies formed during gametogenesis in human?
(1) These are formed during the fetal life of female.
(2) These contain same number of chromosomes as in the secondary oocyte.
(3) These contain same number of chromosomes as in the primary oocyte.
(4) These are formed during both spermatogenesis and oogenesis.
BHR396

17. During oogenesis, first meiotic division and second meiotic division are respectively :-
(1) Unequal division, Unequal division
(2) Equal division, Unequal division
(3) Equal division, Equal division
(4) Unequal division, Equal division
BHR397

18. First maturation division in the process of oogenesis is completed :-
(1) Before birth (2) Before puberty
(3) At puberty (4) After menopause
BHR398

19. Identify A, B and C from above diagram and choose the correct option.



(1) A-Nucleus, B-Centriole, C-Mitochondria
(2) A-Acrosome, B-Middle piece, C-Mitochondria
(3) A-Plasma membrane, B-Mitochondria, C-Middle piece
(4) A-Plasma membrane, B-Mitochondria, C-Tail
BHR399

20. **Assertion:** Spermatogonia are haploid cells.
Reason: Spermatogonia multiply by binary fission.
(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) Assertion is true but the reason is false.
(4) Both the assertion and reason are false.
BHR400

21. **Assertion:** Androgens stimulate the process of spermatogenesis.
Reason: FSH acts on sertoli cells and stimulates the secretion of some factors which helps in the process of spermiogenesis.
(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) Assertion is true but the reason is false.
(4) Both the assertion and reason are false.
BHR401

22. **Assertion:** Oogenesis is initiated during the embryonic development stage when a couple of million gamete mother cell (oogonia) are formed with in each foetal ovary.
Reason: No more oogonia are formed and added after birth.
(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) Assertion is true but the reason is false.
(4) Both the assertion and reason are false.
BHR402

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	1	1	4	4	3	3	2	4	3	4	2	4	3	3
Question	16	17	18	19	20	21	22								
Answer	2	1	3	2	4	2	1								

Human Reproduction # 04

(Internal Structure of Ovary and Follicular development & Menstrual Cycle)

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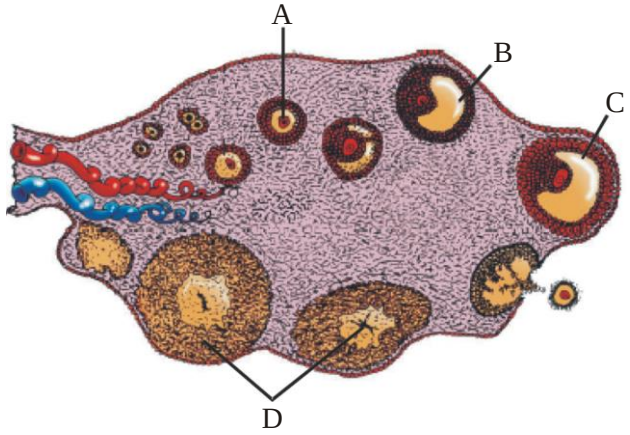
1. Level of which hormone reaches peak during follicular phase of menstrual cycle ?
(1) Estrogen
(2) Progesterone and estrogen
(3) Luteinizing hormone and hCG
(4) F.S.H. and progesterone

BHR403

2. How many statement are correct :-
(A) Corpus luteum secretes large amount of progesterone which is essential for the maintenance of endometrium.
(B) Maintenance of endometrium is necessary for implantation of fertilised ovum and other events of pregnancy.
(C) During pregnancy all events of menstrual cycle stop.
(D) In absense of fertilization, the corpus luteum degenerates this cause disintegration of the endometrium leading to menstruation.
(1) 1 (2) 2
(3) 3 (4) 4

BHR404

3. Following is diagrammatic sectional view of ovary with labelled parts A, B, C and D. Find out the incorrect identification of the labelled part and its description.



	Structure	Identification	Descritpion/ Function
(1)	A	Primary follicle	About 60,000-80,000 left in each ovary at puberty
(2)	B	Secondary follicle	Contains primary oocyte
(3)	C	Graafian follicle	Mature follicle containing secondary oocyte
(4)	D	Corpus luteum	Developed by the activity of LH, secrete mainly progesterone and some amount of estrogen

BHR405

4. Considered the following four statements (a-d) and select the option which include all correct ones only :
(a) Estrogen causes myometrial hypertrophy
(b) Progesterone decreases myometrial contraction
(c) Pre-ovulatory phase also known as progesteronic phase
(d) Oestrogen causes endometrial hyperplasia.
Options :-
(1) Statement (a), (c) and (d)
(2) Statement (a), (b) and (c)
(3) Statement (a), (b) and (d)
(4) Statement (b), (c) and (d)

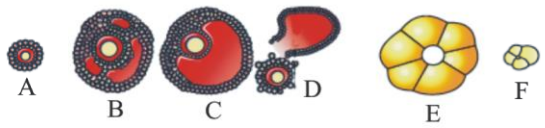
BHR406

5. In a female, undergoing hysterectomy, which one of the following will not occur ?
(1) Formation of Graafian follicle
(2) Ovulation
(3) Menstruation
(4) Completion of meiosis-I and formation of secondary oocyte

BHR407

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6. It is a diagrammatic representation of developing follicle events, identify the structure and choose correct option :-



(1) A = Primary follicle,
C = secondary follicle,
E = Corpus haemorragicum,
F = Corpus luteum

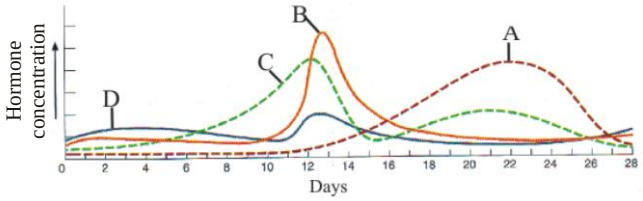
(2) B = Secondary follicle,
D = Mature follicle,
E = Corpus haemorragicum,
F = Corpus luteum

(3) A = Primary follicle,
C = mature follicle,
E = Corpus luteum,
F = Corpus albicans

(4) A = Primary follicle,
D = Mature follicle,
E = Corpus luteum,
F = Corpus albicans

BHR408

7. The figure given below show different phase of menstruation cycle's hormones. Select the option which gives correct identification together with its source of secretion.



(b) Changes in concentration of anterior pituaitary and ovartan hormones

(1) B–Estrogen ; Theca externa
(2) C–LH ; Graafian follicle
(3) A–Progesterone ; Corpus albicans
(4) D–FSH ; Anterior pituitary gland

BHR409

8. Which one of the following is the correct matching of the events occuring during the menstrual cycle ?

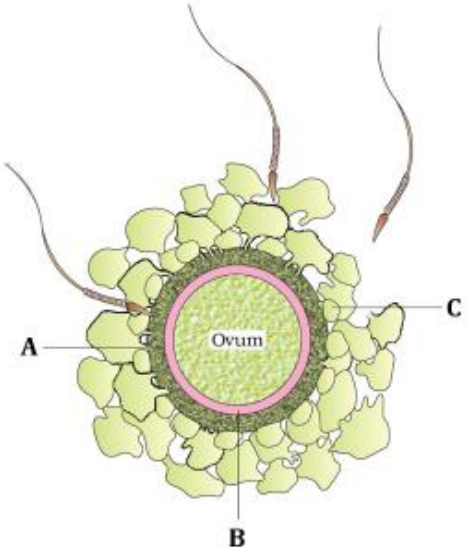
(1) Menstruation : Breakdown of myometrium and ovum not fertilised
(2) Ovulation : LH and FSH attain peak level and sharp fall in the secretion of progesterone.
(3) Proliferative phase : Rapid regeneration of myometrium and maturation of corpus luteum.
(4) Secretory phase : Development of corpus luteum and increased secretion of progesterone.

BHR410

9. Arrange following events of female reproductive cycle in (human) in sequence :-
(I) Secretion of FSH
(II) Growth of corpus luteum
(III) Growth of follicles
(IV) Ovulation
(V) Sudden increase in level of LH.
(VI) Formation of corpus albicans
(1) III → I → IV → II → VI → V
(2) I → III → V → IV → II → VI
(3) I → IV → III → V → II → VI
(4) II → I → III → IV → VI → V

BHR411

10. Find out correct match in given diagram-



(1) A – Corona Radiata
(2) B – Peri vitelline space
(3) C – Zona Pellucida
(4) All Above are correctly matched

BHR412

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11. When both ovaries of human are removed then which of the following hormone is decreased in blood ?
(1) Oxytocin
(2) Prolactin
(3) Estrogen
(4) Gonadotropin releasing factor

BHR413

12. Match the column I with Column II

Column-I		**Column-II**	
(a)	Relaxin	(i)	Anti abortion hormone
(b)	L.H.	(ii)	Testosterone secretion
(c)	Progesterone	(iii)	Widening of birth canal
(d)	F.S.H.	(iv)	Folliculogenesis

(1) a-(iii), b-(iv), c-(i), d-(ii)
(2) a-(iii), b-(ii), c-(i), d-(iv)
(3) a-(i), b-(ii), c-(iv), d-(iii)
(4) a-(ii), b-(i), c-(iii), d-(iv)

BHR414

13. Which of the following pair is correct ?
(1) Progesterone : steroid
Estrogen : Protein
(2) Progesterone and Estrogen : Protein
(3) Progesterone and Estrogen : Steroid
(4) Progesterone : Protein
Estrogen : Steroid

BHR415

14. Lack of mensturation (Amenorrhea) may be indicative of :-
(1) May be caused due to stress
(2) May be caused due to pregnancy
(3) Menarche
(4) Both (1) & (2)

BHR416

15. In case of pregnancy :-
(1) Progesterone level rises in blood
(2) Secretion of FSH and LH are increased
(3) Corpus albicans converts into corpus luteum
(4) All of the above

BHR417

16. Menstruation is initiated by :-
(1) A sudden release of FSH from the anterior pituitary
(2) A lack of progesterone due to degeneration of the corpus luteum
(3) An increased release of estrogen and progesteron from the corpus luteum
(4) A sudden drop in FSH level

BHR418

17. In the absence of fertilisation, the corpus luteum:
(1) immediately stop secretion of progesterone
(2) converts into Graafian follicle
(3) degenerates
(4) develops as embryo

BHR419

18. At the time of puberty, how many primary follicles are left in each ovary ?
(1) 60,000 – 80,000
(2) 7 million
(3) 10,000 – 20,000
(4) 1,20,000 – 1,60,000

BHR420

19. **Assertion:** Rupture of graffian follicle occur before the 14th day of onset of next menstrual cycle.
Reason: LH level is maximum during the mid of cycle, i.e., (14th day).
(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) Assertion is true but the reason is false.
(4) Both the assertion and reason are false.

BHR421

20. **Assertion:** In humans the menopause occurs at approximately 50 years of age.
Reason: Cyclic menstruation is an indicator of abnormal reproductive phase.
(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) Assertion is true but the reason is false.
(4) Both the assertion and reason are false.

BHR422

21. **Assertion:** Reproductive cycle in female primates is called menstrual cycle.
Reason: Menstruation begins at puberty and is known as menarche.
(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) Assertion is true but the reason is false.
(4) Both the assertion and reason are false.

BHR423

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	4	2	3	3	3	4	4	2	2	3	2	3	4	1
Question	16	17	18	19	20	21									
Answer	2	3	1	1	3	2									

Human Reproduction # 05

(Structure of Human egg, Fertilization, Cleavage, Embryonic Development and Parturition)

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1. Match the column-A and B about the embryonic development of human :-

	Column-A		Column-B
(i)	End of one month	(a)	Most of the major organ systems developed
(ii)	End of second month	(b)	Appearance of hair on the head
(iii)	End of 12 weeks	(c)	Heart formation
(iv)	During fifth month	(d)	Eye lid separated
(v)	End of about 24 weeks	(e)	Limbs and digits formation

	(i)	(ii)	(iii)	(iv)	(v)
(1)	c	e	a	b	d
(2)	c	a	e	d	b
(3)	c	e	b	a	d
(4)	d	e	b	a	c

BHR424

2. What is wrong about cleavage?
(1) Size of blastomeres continues to decrease
(2) Interphase is very long during cell cycle
(3) No significance of synthesis of cytoplasm takes place
(4) Karyoplasmic index of blastomeres increases after each cleavage

BHR425

3. Fertilisation has following processes :-
(a) Plasmogamy (b) Karyogamy
(c) Syngamy (d) Amphimixis
Arrange these in correct sequence :
(1) a → b → c → d (2) b → a → d → c
(3) c → a → b → d (4) c → a → d → b

BHR426

4. Perivitelline space is occurs between :-
(1) Zona pellucida and cells of corona radiata
(2) Sperms and zona pellucida
(3) Sperms and cells of corona radiata
(4) Membrane of ovum and zona pellucida

BHR427

5. The figure below show four stage (a, b, c, d) of human development. Select the option giving correct identification together with site of occurance ?

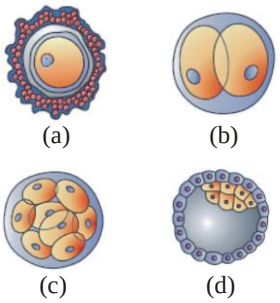
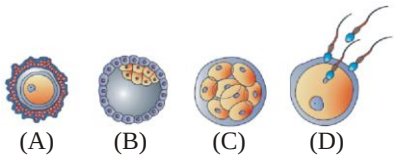


	Figure	Developmental stage	Siteof occurrence
(1)	(a)	Fertilised egg	Isthmus part of fallopian tube
(2)	(b)	Zygote	End part of fallopian tube
(3)	(c)	Mórula	Starting point of Fallopian tube
(4)	(d)	Blastocyst	Uterine cavity

BHR428

6. Choose the correct sequence for given diagrams (A, B, C and D) of human development events :-



(1) A → B → C → D (2) B → C → A → D
(3) C → B → A → D (4) A → D → C → B

BHR429

7. Identify incorrect match –
(1) Amnion – secrets amniotic fluid in which foetus remains hanging.
(2) Yolk sac – Ist haemopoietic organ of embryo
(3) Chorion – forms chorionic villi which get embedded into uterine endometrium.
(4) Allantois – function to store the excretory wastes.

BHR430

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8. Match the following and choose the correct answer :-

(a)	Inner cell mass	(i)	Embedding of blastocyst in the endometrium
(b)	Cleavage	(ii)	Group of cells that would differentiate as embryo
(c)	Trophoblast	(iii)	Outer layer of blastocyst attached to the endometrium
(d)	Implantation	(iv)	Mitotic division of zygote

(1) a-(ii), b-(i), c-(iii), d-(iv)
(2) a-(iii), b-(iv), c-(ii), d-(i)
(3) a-(ii), b-(iv), c-(iii), d-(i)
(4) a-(iii), b-(i), c-(ii), d-(iv)

BHR431

9. Consider the following statements and select the wrong one :-
(1) Parturition is a complex neuro endocrine mechanism
(2) The first movement of the foetus & appearance of hairs on the head are usually observed during fifth month of pregnancy
(3) During parturition, secretion of progesterone decreases but secretion of estrogen increases from placenta
(4) Progesterone and estrogen are produced in women only during pregnancy

BHR432

10. Which one of the following is not the function of placenta? It:-
(1) Facilitates supply of oxygen and nutrients to embryo
(2) Secretes estrogen and progesterone
(3) Facilitates removal of CO_2 and waste material from embryo
(4) Facilitates supply of blood to embryo from mother

BHR433

11. Which one of the following hormone helps in relaxation of pubic symphysis during child birth ?
(1) Estrogen (2) Progesterone
(3) Prolactin (4) Relaxin

BHR434

12. A structural and functional unit between developing embryo and mother is called___?
(1) Ovary (2) Uterus
(3) Placenta (4) Zona pellucida

BHR435

13. Which of following is an incorrect statement:-
(1) The embryo with 8 to 16 blastomere is called morula
(2) Placenta act as a temporary endocrine gland which secrete several hormone like hCG, hPL, estrogen, progesterone etc.
(3) Appearance of hairs on the head are usually observed during first trimester
(4) Menstrual cycles ceases around 50 years of age that is called menopause

BHR436

14. Match the column-I with column-II :-

	Column-I		**Column-II**
(A)	Acrosin	i	Blocks polyspermy
(B)	Fertilization membrane	ii	Dissolves cementing materials between the corona radiata cells
(C)	Hyaluronidase	iii	Digest zona pellucida

	A	**B**	**C**
(1)	iii	ii	i
(2)	i	ii	iii
(3)	ii	i	iii
(4)	iii	i	ii

BHR437

15. What is significance of amniotic fluid ?
(1) It prevents desiccation and helps in body movements of foetus
(2) It provides nutrition
(3) It helps in gaseous exchange
(4) It helps in excretion

BHR438

16. In human female, the blastocyst :-
(1) Gets implanted in uterus three days after ovulation
(2) has mulberry like appearance
(3) Gets implanted in endometrium by the trophoblast cells
(4) Forms placenta even before implantation

BHR439

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17. Which of following is a correct sequence followed in human embryology ?
(1) Syngamy → Completion of second meiotic division → Plasmogamy → Karyogamy
(2) Completion of second meiotic division → Syngamy → Plasmogamy → Karyogamy
(3) Syngamy → Plasmogamy → Completion of second meiotic division → Karyogamy
(4) Syngamy → Plasmogamy → Karyogamy → Completion of second meiotic division

BHR440

18. Select the hormone that is not secreted by placenta:-
(1) hCG (2) Estrogen
(3) Progesterone (4) LH

BHR441

19. During implantation, the blastocyst becomes embeded in the which layer of uterus :-
(1) Perimetrium (2) Endometrium
(3) Myometrium (4) Trophoblast

BHR442

20. Embryo is termed a foetus after about :-
(1) 8 weeks (2) 9 months
(3) 4 months (4) 3 weeks

BHR443

21. The pregnancy test is based on detection of :
(1) Progesterone (2) Estrogen
(3) hCG (4) LH

BHR444

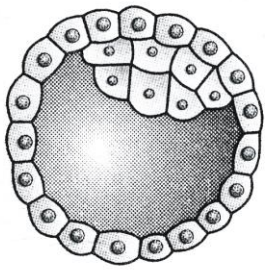
22. How many eggs and sperms are required to produce "non identical" (fraternal) twins ?
(1) One egg and one sperm
(2) One egg and two sperm
(3) Two eggs and one sperm
(4) Two eggs and two sperms

BHR445

23. How many meiotic divisions are essential for the formation of 128 zygotes in human ?
(1) 126 (2) 160
(3) 131 (4) 63

BHR446

24. Identify the human embryonic stage shown below and select the correct option with its matching with site of occurance in a normal pregnant female.



Developmental stage		Site of occurrence	
(1)	Blastocyst	(i)	Uterine cavity
(2)	8-celled morula	(ii)	Early part of fallopian tube
(3)	Late morula	(iii)	Middle part of fallopian tube
(4)	Blastocyst	(iv)	End part of fallopian tube

BHR447

25. Transport of sperm from vagina to ampulla occurs due to :-
(1) Sperm motility
(2) Appropriate ciliary activities in the fallopian tube
(3) Muscular movement of the uterus
(4) All of the above

BHR448

26. **Assertion:** Zygote of human contains 46 chromosomes.
Reason: Zygote is formed by fusion of haploid cells namely sperm and ovum.
(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) Assertion is true but the reason is false.
(4) Both the assertion and reason are false.

BHR449

27. **Assertion:** Sex of human baby is determined by father not by mother.

Reason: XY sex chromosome is present in human male.

(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.

(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.

(3) Assertion is true but the reason is false.

(4) Both the assertion and reason are false.

BHR450

28. **Assertion:** Doctors use to inject oxytocin to induce delivery.

Reason: Oxytocin causes uterine contraction which helps in delivery

(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.

(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.

(3) Assertion is true but the reason is false.

(4) Both the assertion and reason are false.

BHR451

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	2	3	4	4	4	4	3	4	4	4	3	3	4	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28		
Answer	3	1	4	2	1	3	4	2	1	4	1	1	1		

Reproductive Health

1. Which of the following is an indicative of improved reproductive health of the society ?
(a) Better awareness about sex related matters
(b) Increased number of medically assisted deliveries
(c) Better post-natal care
(d) Increased maternal and infant mortality rates
(e) Decreased number of couples with small families
(f) Better detection and cure of STDs
(g) Overall increased medical facilities for all sex-related problems
(1) a, c, d and g (2) a, e and f
(3) a, b, d, e and f (4) a, b, c, f and g

BRH115

2. Diaphragms, cervical caps and vaults prevent conception by :-
(1) Increasing phagocytosis of sperms within uterus
(2) Suppressing sperm motility
(3) Inhibiting ovulation and implantation
(4) Blocking the entry of sperms through the cervix

BRH116

3. Primary steps that can be taken towards attaining reproductive health are :-
(a) Counselling and creating awareness among people about reproductive organs, adolescence and associated changes.
(b) Safe and hygienic sexual practices
(c) Creating awareness about STDs including AIDS, etc
(d) Ban on MTP and amniocentesis
(1) a and b (2) b and c
(3) a, b and c (4) a, b, c and d

BRH117

4. Medical assistance and care to people is required in reproduction related problems like:
(a) Pregnancy (b) Delivery
(c) STDs (d) Abortions
(e) Contraception
(f) Menstrual problems
(g) Infertility
(1) a, c, e, and f (2) b, d and g
(3) a, b, c, d e and g (4) a, b, c, d, e, f and g

BRH118

5. Which of the following is not true about lactational amenorrhoea.
(1) Inhibition of menstrual cycle in lactating mother is due to high concentration of prolactin
(2) As a natural method of contraception, it has been reported to be effective only upto a maximum period of six months following parturition
(3) Side effects are almost nil, if we consider it as a natural method of contraception
(4) Chances of failure of this natural method of contraception are very low

BRH119

6. Match the column-I, II and III and choose the correct option.

Column-I Contraceptive method		Column-II Example		Column-III Mode of action	
A	Copper releasing IUD's	i	Nim-76	a	Suppress sperm motility
B	Weekly oral pills	ii	LNG-20	b	Act as spermicidal agent
C	Hormone releasing IUD's	iii	Multiload-375	c	Anti estrogenic activity in uterus
D	Chemical method	iv	Saheli	d	Makes uterus unsuitable for implantation and cervix hostile to sperms

(1) A-iii-a, B-iv-c, C-ii-d, D-i-b
(2) A-ii-b, B-iv-d, C-i-a, D-iii-c
(3) A-ii-c, B-i-a, C-iii-d, D-iv-b
(4) A-iii-a, B-iv-d, C-ii-b, D-i-c

BRH120

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7. Consider the following statements given below and choose the option. Which include correct statements only ?
(A) Contraception pills include administration of progesterone or progesterone–estrogen combinations.
(B) Medical termination of pregnancy (MTP) is safe during third trimester.
(C) First seven days and last seven days of menstrual cycle is called risk period.
(D) Intrauterine devices like Cu–T are effective contraceptives.
(1) B & D (2) A & D
(3) B & C (4) A, B & D
BRH121

8. In the given four statements (A–D) select the options which include all the correct ones only:-
(A) Tubectomy is the terminal method of contraception in males
(B) Oral pills prevents the ovulation from ovary.
(C) Copper releasing IUD's releases Cu ions which suppress sperm motility and the fertilizing capacity of sperm.
(D) Condoms are the barrier method of conception.
(1) Statement A, B, C (2) Statement B, C, D
(3) Statement A, C, D (4) Statement A, B, D
BRH122

9. Choose the incorrect for MTP :-
(1) Nearly 45 to 50 million MTPs are performed in a year all over the India
(2) Government of India legalised MTP in 1971
(3) MTPs are considered relatively safe during the first trimester
(4) It is also called induced abortion
BRH123

10. The technique called gamete intrafallopian transfer (GIFT) is recommended for those females :-
(1) Whose cervical canal is too narrow to allow passage for the sperms.
(2) Who can not provide suitable environment for fertilisation.
(3) Who fail to produce an ovum
(4) Who fail to retain the foetus inside uterus
BRH124

11. Given below are four statements (a-d) regarding assisted reproductive technologies:
(a) **ZIFT** - The zygote or early embryo (with upto 8 blastomere) transferred into the fallopian tube
(b) **ICSI** - A sperm is directly injected into the fallopian tube to form an embryo in the laboratory
(c) **IUI** - The semen collected either from the husband or healthy donor is artificially introduced into the uterus
(d) **GIFT** - Transfer of zygote collected from a donor into the fallopian tube of another female who cannot produce ovum but can provide suitable environment for fertilisation and development
Which two of the above statements are correct?
(1) a and d (2) b and c
(3) a and c (4) b and d
BRH125

12. Given below are some assisted reproductive technologies (ARTs), in which how many technologies involve(s) invitro fertilisation?
GIFT, ZIFT, ICSI, IUT, IUI
(1) Four (2) Three
(3) Two (4) Only one
BRH126

13. In _____ the egg is first fertilised outside the body and then it is inserted into the oviduct.
(1) GIFT (2) ZIFT (3) AI (4) IUI
BRH127

14. The mechanism of action of non medicated IUD's is -
(1) To suppress motility and fertilizing capacity of sperms
(2) To make the uterus unsuitable for implantation and cervix hostile to the uterus
(3) To prevent the ovulation from ovary
(4) To promote phagocytosis of sperms within the uterus
BRH128

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15. Here are some statements about recent amendment in MTP act, find the incorrect one:
(1) It's intention was to reduce illegal abortions, and consequent maternal mortality and morbidity.
(2) It was amended in 2017
(3) If pregnancy has lasted more than 24 weeks but less than 36 weeks, two registered medical practitioners must be of opinion that ground required for MTP exists
(4) MTP can be performed if there is risk for pregnant woman or the child if born will suffer from physical or mental abnormality as to be seriously handicapped.

BRH129

16. If a female is already having 3 children and all are absolutely normal. Now she wants to use sterilisation method. Which of the following method will be most suitable for this female?
(1) Oral contraceptive pill
(2) I.U.D.
(3) Tubectomy
(4) Hysterectomy

BRH130

17. How many statements are correct :-
(A) MTPs are considered relatively safe during first 12 weeks of pregnancy
(B) IUI- Transfer of an ovum collected from a donor into the fallopian tube
(C) Vaults are a barrier made of rubber that are inserted into the female reproductive tract
(D) IUDs are ideal contraceptives for the female and it is one of most widely accepted method of contraception in India
(1) 1 (2) 2 (3) 3 (4) 4

BRH131

18. Mechanism of action of oral combination hormonal contraceptive pills is :-
(1) to prevents ovulation
(2) to decrease thickening of cervical mucous
(3) to acts as a spermicidal agent
(4) to increases basal body temperature

BRH132

19. If male is sterile and female is fertile then which assisted reproductive technology may be used for treating the infertility in this particular couple ?
(1) GIFT
(2) Intra uterine insemination
(3) ZIFT
(4) Intra uterine transfer

BRH133

20. GIFT technique is used in :-
(1) In female who cannot produce ovum but can produce suitable environment for fertilisation.
(2) In female who cannot provide suitable environment for fertilisation.
(3) sterile male
(4) impotent male

BRH134

21. What are the characteristics of an ideal contraceptive ?
(a) User-friendly (b) Easily available
(c) Effective (d) Reversible
(e) No or least side effects
(f) No way to interfere with the sexual drive, desire and / or the sexual act of the user.
(1) a, b, c and e (2) b, d, e and f
(3) a, b, c, d, e and f (4) a, b, c and f

BRH135

22. The permissible use of the technique amniocentesis is for :-
(1) detecting any genetic abnormality.
(2) detecting sex of the unborn foetus.
(3) artificial insemination.
(4) transfer of embryo into the uterus of a surrogate mother.

BRH136

23. What is the mechanism of action of 'Saheli' oral pill?
(1) to show anti estrogenic effect on uterus
(2) to show anti progesteronic effect
(3) to prevent implantation by releasing progesterone
(4) inhibit motility of sperms by releasing copper ions

BRH137

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24. Which of the following contraceptive methods makes the uterus unsuitable for implantation?
(1) Condom (2) Today
(3) Diaphragm (4) Hormonal IUD
BRH138

25. Which of the following hormone stimulates milk synthesis and secretion and participates in contraceptive measure?
(1) Oxytocin (2) Prolactin
(3) hCG (4) Dopamine
BRH139

26. Choose the odd one out about contraceptive method :-
(1) Condoms (2) Cervical cap
(3) Sterilisation (4) Pills
BRH140

27. Pick up the correct match :-

(A)	I.U.T.	(1)	Transfer of embryo into the uterus after in vitro fertilization
(B)	GIFT	(2)	Transfer of semen into female genital tract
(C)	AI	(3)	Sperm is directly injected into ovum
(D)	ZIFT	(4)	Transfer of gamete into fallopian tube
(E)	I.C.S.I.	(5)	Transfer of zygote into fallopian tube

	A	B	C	D	E
(1)	1	2	5	3	4
(2)	1	3	5	4	2
(3)	1	4	3	5	2
(4)	1	4	2	5	3

BRH141

28. In which of the following contraceptive method side effects are almost nil ?
(1) Pills (2) IUD's
(3) Implants (4) Coitus interruptus
BRH142

29. Select the incorrect statement.
(1) MTP is safe during the first trimester.
(2) Removal of gonads is best and permanent method of contraceptive.
(3) 'Saheli' is a nonsteroidal oral contraceptive pill
(4) Intense lactation helps as a natural method of contraception
BRH143

30. Consider the following statements with two blanks A and B. Select the option which correctly fills up these blanks.
Government of India legalised MTP in ___A___ with some strict conditions to avoid its misuse. Such restrictions are all the more important to check indiscriminate and illegal ___B___ foeticides which are reported to be high in India.

	A	B
(1)	1951	Female
(2)	1971	Male
(3)	1971	Female
(4)	1951	Male

BRH144

31. **Assertion:** There is decrease in maternal and infant mortality rates nowadays.
Reason: Better awareness about sex related matter, increased number of medical assisted deliveries and better post-natal care leading to it.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.
BRH145

32. **Assertion:** Spermicidal creams, jellies and foams are usually used along with barrier methods.
Reason: All of the above products increases their contraceptive efficiency.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.
BRH146

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33. **Assertion:** Inability to conceive or produce children even after years of unprotected sexual cohabitation is called infertility.
Reason: ART is commonly used for such couples.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.
BRH147

34. **Assertion:** Saheli is the oral contraceptive.
Reason: They are non steroidal oral preparation inhibits implantation of fertilized ovum.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.
BRH148

35. **Assertion:** STDs are a major threat to a healthy society.
Reason: Complication of STDs are PID, abortions, still births, infertility, ectopic pregnancies and even cancer of the reproductive tract.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.
BRH149

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	4	3	4	4	1	2	2	1	3	3	2	2	4	3
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	3	1	2	1	3	1	1	4	2	3	4	4	2	3
Question	31	32	33	34	35										
Answer	1	1	2	1	1										

Biotechnology : Principles and Processes

1. Which of the following is not applicable to *Agrobacterium tumifaciens?*
(1) Pathogen of several dicot plants
(2) Has ability to transform mutant species
(3) Delivers gene of our interest
(4) Ti plasmid of it is always pathogenic to plants without any exception
BBP226

2. Which is not true about PCR (Polymerase Chain Reaction) ?
(1) PCR can amplify very small amount of DNA
(2) It can be used to detect HIV in suspects
(3) Only RNA can be used as primer
(4) Used to detect mutation in genes
BBP227

3. In agarose gel electrophoresis–
(1) DNA migrates towards negative electrode
(2) Larger molecules migrate faster than smaller molecules
(3) Ethidium bromide can be used to visualize the DNA fragments.
(4) Pink coloured bands of DNA can be visualized when gel exposed to UV light
BBP228

4. Which of the following vector is used extensively for genetic engineering in animals?
(1) *E.coli*
(2) Ti plasmid
(3) *Bacillus thuringiensis*
(4) Retrovirus (disarmed)
BBP229

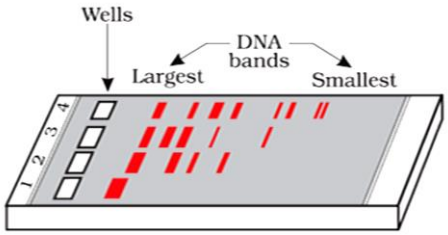
5. A good cloning vector should have all except:-
(1) Origin of replication to start replication
(2) Many recognition sites for single restriction enzyme
(3) High copy number
(4) Selectable marker
BBP230

6. If a recombinant DNA bearing gene for tetracycline resistance is transferred into E.coli cells and the host cells are spread on agar plates containing tetracycline then :-
(1) Both transformed and non-transformed recipient cells will die
(2) Both transformed and non transformed cells will grow
(3) Transformed recipient cells will die and non transformed recipient cells will grow
(4) Transformed recipient cells will grow and non-transformed cells will die
BBP231

7. Which one of the following is not required in PCR?
(1) Oligonucleotide primers
(2) Helicase enzyme
(3) DNA template
(4) Taq polymerase
BBP232

8. After the formation of products in the bioreactors, it undergoes through a series of processes before it is ready for marketing as a finished product, is called ?
(1) Upstream processing
(2) Dowsteram processing
(3) Elution
(4) Transformation
BBP233

9. Given below is a typical agarose gel in which electrophoresis had occurred.



In this agarose gel -
(1) Lane - 1 shows migration of digested DNA fragments
(2) Only Lane - 4 shows migration of digested DNA fragments
(3) Lane - 2 to 4 shows migration of digested set of DNA fragments
(4) Wells are always placed towards positive electrode during electrophoresis
BBP234

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10. A direct method of gene transfer which is suitable for plants, in this cells are bombarded with high velocity micro-particles of gold or tungsten coated with DNA is ?
(1) Micro-injection
(2) Biolistics or Gene gun
(3) Disarmed pathogen
(4) Electroporation

BBP235

11. **Assertion :** Continuous culture system produces a larger biomass.
Reason : It maintains the cells in their physiologically most active lag phase.
(1) Both assertion and reason are true and reason is the correct explanation of assertion.
(2) Both assertion and reason are true and reason is not the correct explanation of assertion.
(3) Assertion is true but reason is false.
(4) Both assertion and reason are false.

BBP236

12. Genes are located on long molecules of DNA intertwined with histones. On the process of DNA isolation, histones can be removed by :
(1) DNase (2) Ribonuclease
(3) Protease (4) Lipase

BBP237

13. Techniques of genetic engineering include all of the following except.
(1) Creation of recombinant DNA
(2) Use of gene cloning
(3) Gene transfer
(4) Gene recombination (Crossing over)

BBP238

14. Find the incorrect match :-
(1) Heat shock — Bacterial cells
(2) Microinjection — Animal cells
(3) Biolistics — Plant cells
(4) Gene gun — Animal cells

BBP239

15. Bioreactors are used in biosynthesis for all of the following except :
(1) To convert raw materials into specific products, enzymes etc.
(2) To provide optimal growth conditions.
(3) To yield products in large quantities.
(4) For separation and purification of products.

BBP240

16. Match column-I with column-II.

Column-I		Column-II	
(a)	Microinjection	(i)	Competent cells
(b)	Biolistics	(ii)	Plant cells
(c)	Heat shock	(iii)	Bacterial cells
(d)	Treatment with Ca^{2+} ions	(iv)	Animal cells

(1) a-(iv), b-(ii), c-(iii), d-(i)
(2) a-(ii), b-(iv), c-(i), d-(iii)
(3) a-(iv), b-(i), c-(ii), d-(iii)
(4) a-(i), b-(ii), c-(iii), d-(iv)

BBP241

17. Which of the following statements in incorrect?
(1) The first restriction endonuclease isolated was Hind II.
(2) Restriction enzymes cut at specific base sequence is known as recognition sequence.
(3) From over 230 strains of bacteria, 900 restriction enzymes have been isolated.
(4) Restriction enzymes belong to a larger class of enzymes called nuclease.

BBP242

18. In vitro synthesis of multiple copies of the gene of interest is done by using :
(1) Set of primers
(2) Taq polymerase & nucleotides
(3) DNA template
(4) All of the above

BBP243

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19. **Statement-I :** Bioreactors have volume of 100-1000 liters.
Statement-II : They provide optimal conditions for achieving the desired product by providing optimum growth conditions.
(1) Both statement I and statement II are correct.
(2) Both statement I and statement II are incorrect.
(3) Statement I is correct but statement II is incorrect.
(4) Statement I is incorrect but statement II is correct.

BBP244

20. If any protein encoding gene is expressed in a heterologous host, it is called as
(1) Nonhistone protein
(2) Histone protein
(3) Recombinant protein
(4) Selectable marker

BBP245

21. **Statement-I :** After completion of the biosynthetic stage, the product is directly marketed.
Statement-II : Process like separation and purification are involved in downstream processing.
(1) Both statement I and statement II are correct.
(2) Both statement I and statement II are incorrect.
(3) Statement I is correct but statement II is incorrect.
(4) Statement I is incorrect but statement II is correct.

BBP246

22. **Statement-I :** The normal *E. coli* cells do not carry resistance against any of the antibiotics.
Statement-II : Selectable marker helps in identifying and eliminating non transformants and selectively permitting the growth of the transformant.
(1) Both statement I and statement II are incorrect.
(2) Statement I is correct but statement II is incorrect.
(3) Statement I is incorrect but statement II is correct.
(4) Both statement I and statement II are correct.

BBP247

23. Which of the following is/are not the applications of Biotechnology.
(1) Test-tube baby
(2) DNA vaccine
(3) Correcting the defective gene
(4) Cross-breeding

BBP248

24. Techniques to alter the chemistry of genetic material (DNA and RNA) to introduce these into host organisms and thus change the phenotype of the host organism.
(1) Breeding
(2) Bioprocess engineering
(3) Genetic engineering
(4) Tissue culture

BBP249

25. Bioprocess engineering enables-
(1) Microbial contamination free environment.
(2) Growth of only the desired microbe/ eukaryotic cells.
(3) Production in large quantities.
(4) All of the above

BBP250

26. **Statement-I :** Exonucleases remove nucleotides from the ends of the DNA.
Statement-II : Endonucleases make cuts at specific positions within the DNA.
(1) Both statement I and II are incorrect
(2) Statement I is correct but statement II is incorrect
(3) Statement I is incorrect but statement II is correct
(4) Both statement I and II are correct

BBP251

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27. Techniques of genetic engineering include –
(a) Recombinant DNA
(b) Gene cloning I Gene transfer
(d) Traditional hybridization I Domestication
(1) a & b (2) a, b & c
(3) a, b, d (4) a, b, c, d

BBP252

28. Match column-I with column-II :

Column-I		Column-II	
(a)	Restriction enzymes	(i)	Recognition sites
(b)	Disarmed pathogen	(ii)	Plant cells
(c)	Biolistics	(iii)	Retroviruses as vectors
(d)	Palindromic sequences	(iv)	Nucleases

(1) a-(i), b-(ii), c-(iii), d-(iv)
(2) a-(iv), b-(iii), c-(ii), d-(i)
(3) a-(i), b-(iii), c-(ii), d-(iv)
(4) a-(iv), b-(ii), c-(iii), d-(i)

BBP253

29. **Statement-I:** In PCR, multiple copies of gene of interest is synthesized *in vivo*.
Statement-II: Thermostable DNA polymerase is isolated from a fungus *Thermus aquaticus*.
(1) Statement-I is correct but Statement-II is incorrect.
(2) Both Statement-I and Statement-II are correct.
(3) Both Statement-I and Statement-II are incorrect.
(4) Statement-I is incorrect but Statement-II is correct.

BBP254

30. **Assertion:** The development of bioreactors where large volumes of culture can be processed was required for biotechnology.
Reason: Small volume cultures can not yield appreciable quantities of products.
(1) Both assertion and reason are true and reason is the correct explanation of assertion.
(2) Both assertion and reason are true and reason is not the correct explanation of assertion.
(3) Assertion is true but reason is false.
(4) Both assertion and reason are false.

BBP255

31. The first recombinant DNA was constructed by using plasmid of :
(1) *E. coli*
(2) *Bacillus thuringiensis*
(3) *Salmonella typhimurium*
(4) *Thermus aquaticus*

BBP256

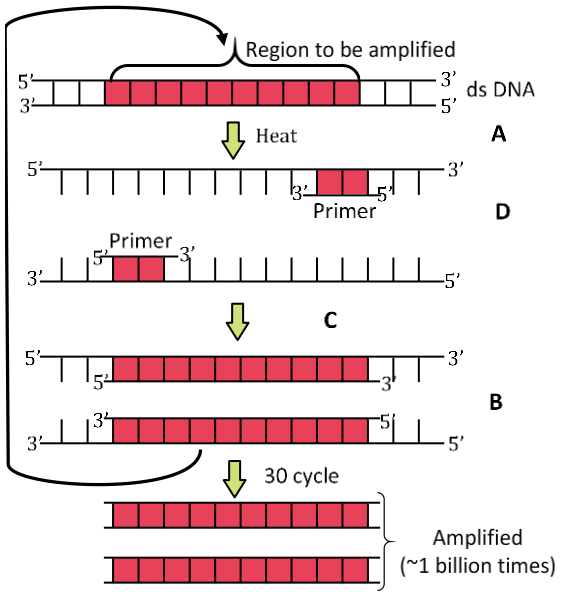
32. When a gene of interest is ligated at Pvu-II site, in pBR322 vector. The recombinant plasmids will lose the resistance against -
(1) Ampicillin
(2) Tetracycline
(3) Both (1) and (2)
(4) None of the above

BBP257

33. **Statement I :** When cut by the same restriction enzyme the resultant DNA fragment have the same kind of sticky ends.
Statement II : Restriction endonuclease are used in genetic engineering to form recombinant molecules of DNA, which are composed of DNA from same / sources / organism.
(1) Both statement I and statement II are correct
(2) Both statement I and statement II are incorrect
(3) Statement I is correct and statement II is incorrect
(4) Statement II is correct and statement I is correct

BBP258

34. Which of the following represents the correct labelling of the given figure?



	A	B	C	D
(1)	Denaturation	Annealing	DNA polymerase + deoxynucleotides	Extension
(2)	Denaturation	DNA polymerase + deoxynucleotides	Annealing	Extension
(3)	Denaturation	DNA polymerase + deoxynucleotides	Extension	Annealing
(4)	Denaturation	Extension	DNA polymerase + deoxynucleotides	Annealing

BBP259

35. Which of the following statements is incorrect?

(1) The separated DNA fragments can be visualised only after staining the DNA with ethidium bromide.

(2) Bright yellow coloured bands of DNA are seen in ethidium bromide stained gel exposed to UV light.

(3) The separated bands of DNA are cut out from agarose gel and extracted from the gel piece.

(4) Agarose is a natural polymer extracted from sea weeds.

BBP260

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	3	3	4	2	4	2	2	3	2	3	3	4	4	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	1	1	4	1	3	4	4	4	3	4	4	2	2	3	1
Question	31	32	33	34	35										
Answer	3	4	3	4	2										

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Biotechnology and its Applications

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1. Which one is incorrectly matched ?
(1) Golden rice – Rich in vitamin A
(2) 'Flavr savr' tomato – Polygalacturonase
(3) Brassica napus – Hirudin
(4) *Meloidogyne incognita* – Bt proteins
BBN182

2. GM plants, whose genes have been altered by manipulation, are useful in many ways. Match column-I (Genetic modification) to column-II (example) and choose the correct option.

	Column-I		Column-II
(I)	Reduce reliance on chemical pesticides	(i)	Salt tolerant transgenic tomato
(II)	Enhanced nutritional value of food	(ii)	Flavr-Savr tomato
(III)	Reduce post harvest loses	(iii)	Nematode
(IV)	Tolerent to abiotic stresses	(iv)	Golden rice

(1) I-iii, II-iv, III–i, IV–ii
(2) I–ii, II–iv, III–iii, IV–i
(3) I–ii, II–iv, III–i, IV–iii
(4) I–iii, II–iv, III–ii, IV–i
BBN183

3. Why insulin is not administered orally to diabetic patient ?
(1) It is bitter in taste
(2) It is peptide & can be digested
(3) It will lead to sudden decrease in blood sugar if given orally
(4) It leads to peptic ulcer if given orally
BBN184

4. In which of the following genetically modified organism polygalacturonase gene is silenced?
(1) Golden rice (2) Bt-cotton
(3) Flavr-savr tomato (4) Bt-corn
BBN185

5. Actually, the Bt toxin protein exist as inactive protoxins but once an insect ingest the inactive toxin, it is converted into an active form of toxin due to the :-
(1) Acidic pH of the gut
(2) Alkaline pH of the gut
(3) Neutral pH of the gut
(4) Either acidic or alkaline pH of the gut
BBN186

6. As most Bt toxins are insect group specific, so the choice of Bt genes depends upon :-
(1) crop (2) targeted pest
(3) only crop (4) Both (1) and (2)
BBN187

7. In human insulin is synthesised as a prohormone, which contains an extra stretch called :-
(1) A-peptide (2) C-peptide
(3) B-peptide (4) a-peptide
BBN188

8. The techniques that serve the purpose of early diagnosis :-
(1) Polymerase chain Reaction
(2) ELISA
(3) Recombinant DNA technology
(4) All of the above
BBN189

9. The Indian parliament has recently cleared the ___ amendment of the Indian patents Bill that takes such issues into consideration, including patent terms emergency provisions and research and development initiative :-
(1) First (2) Third
(3) Second (4) Fifth
BBN190

10. At present about How many recombinant therapeutics have been approved for human use the world over :-
(1) 12 (2) 30 (3) 27 (4) 40
BBN191

11. Which of the following is a permanent cure for ADA deficiency?
(1) Genetically engineered lymphocytes
(2) Enzyme replacement therapy
(3) ADA functional gene is introduced at early embryonic stage.
(4) Bone marrow transplantation
BBN192

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12. A single stranded DNA or RNA tagged with a radioactive molecule is known as :
(1) Primer (2) Probe
(3) Plasmid (4) Both (1) and (2)
BBN193

13. Transgenic models exist for human diseases except :
(a) Cancer
(b) emphysema
(c) Cystic fibrosis
(d) Rheumatoid arthritis
(e) Alzheimer's
(f) SCID
(1) (b), (c), (d) (2) (b) and (f)
(3) (c) and (d) (4) (d), (e), (f)
BBN194

14. Genetically identical plants are produced by :
(1) Somatic hybridization
(2) Meristem culture
(3) Micro- propagation
(4) Embryo culture
BBN195

15. The gene for adenosine deaminase is necessary because it helps in
(1) DNA replication
(2) ATP production
(3) Removal of amino group from amino acids
(4) Functioning of immune system
BBN196

16. The permanent cure for adenosine deaminase (ADA) deficiency.
(1) Bone marrow transplantation.
(2) Enzyme replacement therapy.
(3) Functional gene isolated from bone marrow cells producing ADA is introduced into cells at early embryonic stage.
(4) Functional ADA cDNA introduced into the lymphocytes which are subsequently returned to patient.
BBN197

17. About _____ recombinant therapeutics have been approved for human use the world over. In India _____ of these are presently marketed.
(1) 12, 30 (2) 30, 21
(3) 20, 12 (4) 30, 12
BBN198

18. Plants whose genes have been altered by manipulation are result of :
(1) Organic culture
(2) Agrochemical based agriculture
(3) Genetically engineered crop-based agriculture
(4) All of the above
BBN199

19. Which of the following molecular diagnosis method is not used for detection of mutation.
(1) Probe (2) PCR
(3) ELISA (4) Both (1) and (2)
BBN200

20. Early detection of disease is not possible by :
(1) ELISA (2) PCR
(3) Serum analysis (4) Probe
BBN201

21. Indian patients bill take issues into consideration such as :
(1) Patent terms emergency provisions
(2) Emergency provisions
(3) Research and Development initiative
(4) All of these
BBN202

22. **Assertion** : Genetically engineered lymphocytes are not a permanent cure for ADA deficiency.
Reason : Lymphocytes are immortal.
(1) Both Assertion and Reason are true and the Reason is a correct explanation of the Assertion.
(2) Both Assertion and Reason are true but Reason is not a correct explanation of the Assertion.
(3) Assertion is true and Reason is false
(4) Both assertion and Reason are false.
BBN203

23. The capacity to generate a whole plant from any cell explant is :
(1) Pleuripotency (2) Degeneration
(3) Totipotency (4) Plant tissue culture
BBN204

24. PCR is routinely used for the following except:
(1) To detect mutation in genes in suspected cancer patients.
(2) To identify many genetic disorder.
(3) To detect the presence of antigens.
(4) To detect HIV in suspected AIDS patients.
BBN205

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25. **Assertion :** The double stranded RNA initiates the RNAi.
Reason : The specific mRNA is silenced by preventing its transcription.
(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(2) Assertion is true but Reason is false.
(3) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
(4) Both Assertion and Reason are false.

BBN206

26. Match the column :

(a)	Vaccine safety	(i)	Phenylketonuria
(b)	Transgenic model	(ii)	Transgenic cow
(c)	Biological products	(iii)	Rheumatoid arthritis
(d)	Rosie	(iv)	Transgenic mice

(1) a-(iv), b-(iii), c-(i), d-(ii)
(2) a-(iv), b-(i), c-(iii), d-(ii)
(3) a-(i), b-(iv), c-(iii), d-(ii)
(4) a-(ii), b-(i), c-(iii), d-(iv)

BBN207

27. Find the incorrect match :-

(1)	Transgenic model	Phenylketonuria
(2)	Biological product	α-1-antitrypsin
(3)	Vaccine safety	Transgenic mice
(4)	Toxicity testing	Transgenic animals

BBN208

28. Match column-I with column-II :

Column-I		**Column-II**	
A	*Agrobacterium* vectors	i	Gene therapy
b	*Bacillus thuringiensis*	ii	Transgenic cow
c	*Retrovirus* vectors	iii	Insecticidal protein
d	Rosie	iv	RNAi

(1) a-(i), b-(ii), c-(iii), d-(iv)
(2) a-(iv), b-(iii), c-(i), d-(ii)
(3) a-(i), b-(iii), c-(iv), d-(ii)
(4) a-(ii), b-(iv), c-(iii), d-(i)

BBN209

29. Find the incorrect match.

(1)	Micropropagation	Produce large no. of plants in short duration
(2)	Meristem culture	To obtain virus free plants
(3)	Somatic hybridization	Protoplast fusion
(4)	Totipotency	Animal cells

BBN210

30. Which of the following genes are used to control bollworms.
(a) Cry IAc
(b) Cry IAb
(c) Cry IIAb
(1) Both (b) and (c) (2) Only (b)
(3) Both (a) and (b) (4) Both (a) and (c)

BBN211

31. **Assertion :** RNAi involves silencing of specific RNA.
Reason : It prevents transcription of the mRNA.
(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(2) Assertion is true but Reason is false.
(3) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
(4) Both Assertion and Reason are false.

BBN212

32. Arrange the following steps of RNAi process.
(a) Initiated RNAi.
(b) Introduction of DNA which produce both sense and antisense RNA in host cells.
(c) Double stranded RNA is formed.
(d) Silenced the specific mRNA of nematode.
(1) a → c → b → a (2) b→ c→ d→ a
(3) b→ c→ a→ d (4) c→ b→ d→ a

BBN213

33. Which of the following statements is/are incorrect regarding recombinant insulin?

(a) In mammals, excluding human insulin is synthesized as pro-hormone.

(b) The C-peptide is not present in mature insulin.

(c) In 1972, Eli Lilly an American company produced insulin.

(d) Chain A and B are linked by disulphide bridge.

(e) Chains A and B were produced together in plasmid of *E. coli.*

(1) a, c and d (2) a, c and e

(3) a, b and e (4) c, d and e

BBN214

34. Biological products are produced by transgenic animals to treat which of the following?

(a) Emphysema

(b) Phenylketonuria

(c) Cancer

(d) Cystic fibrosis

(e) Alzheimer's disease

(f) Rheumatoid arthritis

(1) a, b, d, e (2) a, b, c, e, f

(3) a, b and d (4) All of the above

BBN215

35. Match column I with column II

Column I		Column II	
(a)	Explant	(i)	Genetically identical plants
(b)	Somaclones	(ii)	Fusion of protoplast of different varieties of plants
(c)	Totipotency	(iii)	Any part of plant taken out and grown is a test tube
(d)	Somatic hybridisation	(iv)	Capacity to generate whole plant from any cell

(1) a-(i); b-(ii); c-(iii); d-(iv)

(2) a-(iii); b-(i); c-(iv); d-(ii)

(3) a-(iv); b-(ii); c-(iii); d-(i)

(4) a-(iv); b-(iii); c-(ii); d-(i)

BBN216

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	4	2	3	2	4	2	4	3	2	3	2	2	3	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	4	3	3	3	4	3	3	3	2	1	1	2	4	4
Question	31	32	33	34	35										
Answer	2	3	2	3	2										

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Human Health & Disease # 01

AIM : FREE EDUCATION TO ALL APUL

1. Which of the following is not a component of innate immunity ?
(1) Interferons (2) Phagocytes
(3) Skin (4) Antibodies
BHD258

2. Which of the following cells are not included in cellular barrier of innate immunity ?
(1) Neutrophils (2) Helper T cells
(3) NK cells (4) Macrophages
BHD259

3. Who disproved "Good humor hypothesis" ?
(1) Hippocrates (2) Bowditch
(3) William Harvey (4) Metchnikoff
BHD260

4. Virus infected cells secrete proteins called as interferons which protect non-infected cells from viral infection. They are a part of :-
(1) Physical barriers
(2) Physiological barriers
(3) Cellular barriers
(4) Cytokine barriers
BHD261

5. In which one of the following options examples are incorrectly matched with their particular type of immunity ?

	Examples	Type of immunity
(1)	Saliva in mouth and tears in eyes	Physiological barrier
(2)	Skin, HCl in stomach and mucus coating	Physical barrier
(3)	Polymorphonuclear leukocytes and monocytes	Cellular barrier
(4)	N.K. cell	Cellular barrier

BHD262

6. Fever forms ..... barrier of innate immunity:
(1) Physical (2) Physiological
(3) Cellular (4) Cytokine
BHD263

7. **Incorrect** about bone marrow is :-
(a) Primary lymphoid organ
(b) Main lymphoid organ where all blood cells excluding lymphocytes are produced
(c) Provide micro environment for the development of T-cells
(d) Produces B-cells
(1) b (2) c
(3) b + c (4) All statements are correct
BHD264

8. The exaggerated response of the immune system to certain antigens is called :-
(1) Primary immune response
(2) Secondary immune response
(3) Immune suppression response
(4) Allergy
BHD265

9. Mark the **incorrect** match :-
(1) Colostrum = Yellowish fluid, IgA present
(2) Antivenom = Passive immunisation
(3) Peyer's patches = PLO of small intestine
(4) Antibody = H_2L_2
BHD266

10. Primary and secondary immune response are carried out with the help of two special types of _____ present in our blood.
(1) Macrophages (2) Lymphocytes
(3) Neutrophils (4) Monocytes
BHD267

11. Health for a long time was considered as a state of body and mind where there was a balance of certain ___?
(1) Chemicals (2) Ions
(3) Humors (4) Hormons
BHD268

12. It was thought that person with ___ belonged to hot personality and would have fevers :-
(1) High temperature (2) Bile
(3) Black bile (4) Yellow bile
BHD269

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13. Which of the following statement is correct?
(A) Health is affected by genetic disorders, life style and infections.
(B) Balanced diet personal hygiene and regular exercise are very important to maintain good health.
(C) Health is only "absence of disease" or "physical fitness"
(D) "Good humor" hypothesis was disproved by William harvey.
(1) A, C (2) A, C, D
(3) A, B, D (4) All are correct
BHD270

14. Health is a state of complete _(A)_, _(B)_ and __(C)__ well being.

	(A)	(B)	(C)
(1)	Physiological	Mental	Social
(2)	Mental	Physical	Economical
(3)	Physical	Mental	Social
(4)	Both (2) and (3)		

BHD271

15. When people are healthy they are?
(A) More efficient at work
(B) Decreases productivity
(C) Brings economic prosperity
(D) Increases longevity of people
(E) Increases infant and maternal mortality
How many statements are correct about healthy people?
(1) Two (2) Three (3) Four (4) One
BHD272

16. Interferon is considered with :-
(1) First line of defence
(2) Second line of defence
(3) Third line of defence
(4) All the above
BHD273

17. Which antibodies activate the B–lymphocyte for further release of antibodies ?
(1) IgG (2) IgA (3) IgE (4) IgD
BHD274

18. All are the true statements about IgM except :
(1) Pentamer form
(2) Can cross placenta
(3) Synthesis after acute infection in blood
(4) Mostly intravascular in distribution
BHD275

19. All are the function of antibody except :-
(1) Antigen binds with antibody leading to cell lysis
(2) After binding of antigen with antibody facilitate phagocytosis
(3) Body cell directly binds with antigen and kill the antigen related cell by releasing lymphotoxic substances
(4) Antibodies also bind with bacterial products and inhibit the activity of bacterial product
BHD276

20. Inflammation is related to :-
(1) IgE
(2) Histamin
(3) Protective mechanism in our body
(4) All the above
BHD277

21. IFN is used in :-
(1) Cancer treatment
(2) Viral growth inhibition
(3) Bacterial growth inhibition
(4) Both (1) and (2)
BHD278

22. Maximum number of paratopes are present in
(1) IgA (2) IgD (3) IgG (4) IgM
BHD279

23. Which of the following is primary lymphoid organ ?
(1) Liver (2) Spleen
(3) Bone marrow (4) Yolk sac
BHD280

24. Which category of lymphocyte is produced by T-cell and is absent in B-cells ?
(1) Memory cell (2) Plasma cell
(3) Suppressor cell (4) Cartwheel cells
BHD281

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25. Humoral immune system is mediated by
(1) B-cells (2) T-cells
(3) NK cells (4) Mast cells
BHD282

26. Hepatitis-B vaccine produced from :-
(1) Bacteria
(2) Recombinant RNA Technology
(3) Fungus
(4) Both (2) and (3)
BHD283

27. Muscles degenerate in which disease ?
(1) Hashimoto disease
(2) Myasthenia gravis
(3) Diabetes mellitus
(4) Pernicious anaemia
BHD284

28. When a person is injected with antiserum, the type of immunity developed is
(1) Natural active (2) Artificial active
(3) Natural passive (4) Artificial passive
BHD285

29. Which vaccine is administred to protect the child from tuberculosis ?
(1) DPT (2) DT (3) BCG (4) MMR
BHD286

30. Toxoid is used in vaccination against
(1) Tetanus (2) Botulism
(3) Diphtheria (4) All of these
BHD287

31. For successful transplantation, matching of HLA allele and MHC protein of donor and recipient is done, In which category of grafting, this matching is required ?
(1) Allograft (2) Heterograft
(3) Xenograft (4) Isograft
BHD288

32. Pannus formation takes place in :-
(1) IDDM
(2) SCID
(3) Rheumatoid arthritis
(4) Hashimoto's disease
BHD289

33. In which disease maximum IgE is produced ?
(1) Asthma
(2) Anaphylactic shock
(3) Hay fever
(4) Skin rashes
BHD290

34. Agents which cause allergy are :-
(1) Immunogens (2) Allergens
(3) Antigens (4) Both (1) & (3)
BHD291

35. **Assertion:** Infectious diseases are very common.
Reason: These diseases are easily transmitted from one person to another.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.
BHD292

36. **Assertion:** Chill and high fever occurs due to malaria.
Reason: Release of the toxin, hemozoin from ruptured RBCs in malaria.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.
BHD293

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37. **Assertion:** Ringworms generally occur in skin fold. Such as groin or between toes.
Reason: Heat and moisture helps the fungi to grow.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.
BHD294

38. **Assertion:** Saliva in mouth is physiological barriers.
Reason: Interferon are cytokine barriers.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.
BHD295

39. **Assertion:** Yellowish fluid colostrum secreted by mother during the initiated days of lactation is protective for infant.
Reason: Colostrum contains abundant IgA antibodies.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.
BHD296

40. **Assertion:** Rheumatoid arthritis is an autoimmune disease.
Reason: Body attack self-cells in rheumatoid arthritis.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.
BHD297

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	2	3	4	2	2	1	4	3	2	3	3	3	3	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	4	2	3	4	4	4	3	3	1	3	2	4	3	4
Question	31	32	33	34	35	36	37	38	39	40					
Answer	1	3	2	2	1	1	1	2	1	1					

Human Health & Disease # 02

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1. Which one of the following group is not of infectious diseases ?
(1) Cancer, Allergy, Rheumatoid arthritis
(2) Dysentery, Plaque, Diphtheria
(3) Polio, Pneumonia, Tetanus
(4) Smallpox, Ringworm, Common Cold
BHD298

2. House flies are mechanical carriers of :-
(1) Amoebiasis (2) Malaria
(3) Common cold (4) Plague
BHD299

3. Haemophilis influenze bacteria are responsible for :-
(1) Influenza (2) Pneumonia
(3) Dengue (4) Filariasis
BHD300

4. *Salmonella typhi* is a bacteria and causes typhoid, this Pathogen causes lesions and ulcertion in the ____ .
(1) Stomach wall (2) Intestinal wall
(3) Gall bladder (4) Blood vessels
BHD301

5. Match the columns w.r.t. diseases and caustive agents :-

	Column-I		Column-II
(a)	Pneumonia	(p)	Rhino viruses
(b)	Common cold	(q)	*Haemophilus influenze*
(c)	Filariasis	(r)	*Microsporum*
(d)	Ringworms	(s)	Wuchereria bancrofti

(1) a – p, b – q, c – r, d – s
(2) a – r, b – p, c – s, d – q
(3) a – q, b – p, c – s, d – r
(4) a – p, b – q, c – s, d – r
BHD302

6. (a) Nasal congestion and discharge
(b) Sore throat, hoarseness, cough
(c) Headache, Tiredness etc, are characters of
(1) Malaria (2) Typhoid
(3) Common cold (4) Pneumonia
BHD303

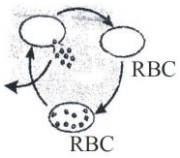
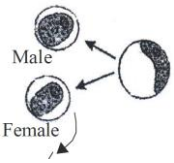
7. Many fungi belonging to genera *Microsporum, Trichophyton* and *Epidermophyton* are responsible for :-
(1) Filariasis (2) Ascariasis
(3) Ringworms (4) Amoebiasis
BHD304

8. Infectious form of *Plasmodium* for human is :-
(1) Sporozoite (2) Mosquito
(3) Haemozoin (4) Schizont
BHD305

9. Find odd one out w.r.t. disease causative agent
(1) Pneumonia (2) Common cold
(3) Plague (4) Diphtheria
BHD306

10. Widal test is done for -
(1) Diphtheria (2) Dengue fever
(3) Typhoid (4) Malaria
BHD307

11. Make a glance of a part of the life cycle of *Plasmodium* give below as events A, B, C and D along with their description :-

	Events	Description
(A)		Parasites reproduce asexually in the red blood cells, bursting the red blood cells and causing cycles of fever
(B)		Sexual stage (gametocytes) develop in RBCs of mosquito
(C)		Fertilization and development takes place in GUT of mosquito
(D)		Mature infective stages (sporozoites) escape from the GUT and migrate to the mosquito salivary glands

How many of the above events are correctly described ?
(1) One (2) Two (3) Three (4) Four
BHD308

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12. When a female *Anopheles* mosquito bites an infected person, the parasite enter the mosquitoes bodies in many forms. Which of these is the infective form for mosquito ?

(1) Sporozoite (2) Gamete form

(3) Gametocyte (4) Schizoint

BHD309

13. Select the incorrect match in the table given below :-

	Disease	Causative agent
(1)	Common cold	Rhino virus
(2)	Typhoid	*Salmonella typhi*
(3)	Pheumonia	*Haemophilus infuluenzae*
(4)	Plague	Xenopsylla

BHD310

14. Choose the odd one out w.r.t. viral disease :-

(1) Pneumonia (2) Influenzae

(3) Dengue (4) Small pox

BHD311

15. Which of the following is not correct about pneumonia ?

(A) It is caused by *Streptococcus pneumoniae*.

(B) Infects the alveoli of the lungs.

(C) Symptoms of pneumonia include fever, chills, cough and headache.

(D) In all cases the lips and finger nails turn blue to grayish in colour.

(1) A, B, D (2) B and D

(3) D only (4) A, B, C, D

BHD312

16. Most common cause of jaundice which is produced by DNA virus :–

(1) Hepatitis–B

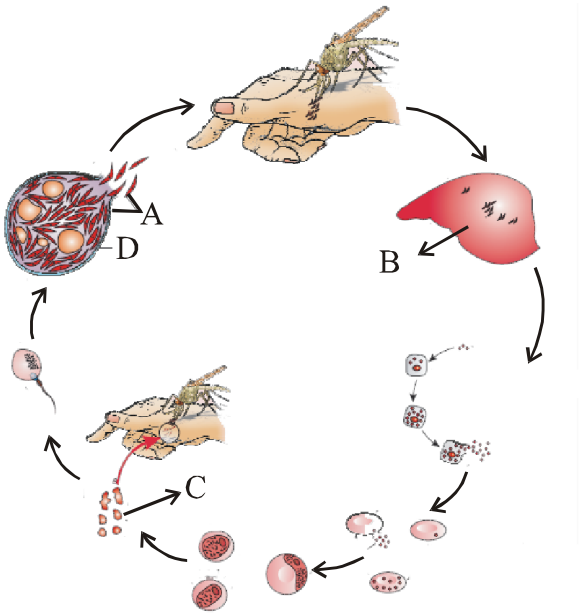
(2) Hepatitis–A

(3) Megaloblastic anemia

(4) Yellow fever

BHD313

17. Identify the likely A, B, C, D in the following life cycle : -



	(A)	(B)	(C)	(D)
(1)	Gametocytes	Salivary gland	Liver	Sporozoites
(2)	Sporozoites	Salivary gland	Liver	Gametocytes
(3)	Sporozoites	Liver	Salivary gland	Gametocytes
(4)	Sporozoites	Liver	Gametocytes	Salivary gland

BHD314

18. Which one of the following option gives the correct matching of a disease with its pathogen and mode of infection?

	(A)	(B)	(C)
(1)	Pneumonia	Haemophilus influenzae	Droplet infection
(2)	Typhoid	Salmonella Typhi	Airborne infection
(3)	Malaria	Plasmodium	Bite by male Anopheles
(4)	Filariasis	Wuchereria Bancrofti	Infected food and water

BHD315

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19. Read the following four statements (A-D)
(A) The lips and finger nails may turn gray to bluish in colour in severe cases of Typhoid
(B) Rhino viruses infect the nose and upper respiratory passage but not the lungs.
(C) Intestinal perforation and death may occur in severe cases of typhoid.
(D) *Streptococcus pneumoniae* infects the alveoli of the lungs.
How many of the above statements are right :-
(1) One (2) Two (3) Three (4) Four
BHD316

20. How many of the following diseases are vector borne Malaria, Dengue, Common cold, Filariasis, Chikungunya, Ascariasis, Typhoid, Pneumonia-
(1) Seven (2) Six (3) Five (4) Four
BHD317

21. Which of the following is a viral disease?
(1) Dysentery (2) Plague
(3) Diphtheria (4) Chicken pox
BHD318

22. How many statements are correct about common cold?
(A) It is caused by rhino virus
(B) Infection of alveoli of the lungs
(C) Symptoms are nasal congestion and discharge, sore throat, hoarseness, cough, headache and tiredness.
(D) Usually last for one or two days.
(1) One (2) Two (3) Three (4) Four
BHD319

23. Which of the following is correct about malaria?
(A) Malignant malaria is caused by *Plasmodium falciparum*.
(B) Plasmodium enters the human body as sporozoites.
(C) It is caused through the bite of infected female anopheles.
(D) The rupture of RBCs is associated with release of a toxic substance haemozoin.
(1) B, C (2) A, C, D
(3) A, B, C, D (4) C, D
BHD320

24. Which of the following is not a sign/symptom of amoebiasis?
(1) Constipation
(2) Abdominal pain and cramps
(3) Stool with excess mucous and blood clots
(4) Intestinal blockage
BHD321

25. _____ cause a slowly developing chronic inflammation of the organs :-
(1) Common round worm
(2) Filarial worm
(3) Guinea worm
(4) Ascaris
BHD322

26. **Assertion:** HIV enters in macrophages only in human body.
Reason: In macrophages, the RNA genome of virus get incorporated into host cell DNA as such.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.
BHD323

27. **Assertion:** Cancer cells shows uncontrolled growth.
Reason: Cancer cells losses property of contact inhibition.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.
BHD324

28. **Assertion:** Crack is a potent stimulant of CNS, producing sense of euphoria and increased energy.
Reason: It interferes with the transport of neurotransmitter dopamine.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.
BHD325

29. **Assertion:** Innate immunity is present by birth.
Reason: Innate immunity is non specific type of defence.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.
BHD326

30. **Assertion:** PMNL are cellular barriers.
Reason: Acid in stomach is physiological barrier.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.
BHD327

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	1	2	2	3	3	3	1	2	3	3	3	4	1	3
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	1	4	1	3	4	4	2	3	4	2	4	1	1	2	2

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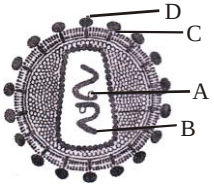
Human Health & Disease # 03

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1. Which biological response modifier activates immune system and helps in destroying the tumor ?
(1) α-interferons (2) Taxol drug
(3) Cytoleukins (4) Vincristin drug
BHD328

2. Burkitt' lymphoma is caused by :-
(1) EBV (2) HPV
(3) HIV (4) POX-Virus
BHD329

3. Following diagram of HIV shows which parts marked as A, B, C and D respectively :-



(1) ss-RNA, Reverse transcriptase, Lipoprotein, GP-120
(2) GP-120, ss-RNA, Reverse transcriptase, Lipoprotein
(3) Reverse transcriptase, ss-RNA, Lipoprotein, GP-120
(4) GP-120, Lipoprotein, ss-RNA, Reverse transcriptase
BHD330

4. How many statements are correct for AIDS ?
(A) Core of HIV has 2 identical molecule of dsDNA
(B) GP120 has complementary sequence for CD-8 receptors
(C) AIDS caused by HIV virus
(D) Macrophage become HIV factory after HIV infection
(1) One (2) Three (3) Two (4) Four
BHD331

5. Replication of HIV takes place in :-
(1) TH cells (2) TK cells
(3) Macrophage (4) Both (1) and (3)
BHD332

6. Which infection more often sprends due to concious behaviour pattern ?
(1) HIV (2) Cancer
(3) Pneumonia (4) Typhoid
BHD333

7. Which property is not shown by cancer cells?
(1) Metastasis
(2) Uncontrolled division
(3) Contact inhibition
(4) Invasion
BHD334

8. Breast cancer is type of :-
(1) Carcinoma (2) Sacroma
(3) Leukemia (4) Lymphoma
BHD335

9. The genetic material of HIV is :-
(1) dsDNA (2) dsRNA
(3) ssDNA (4) ssRNA
BHD336

10. HIV that causes AIDS, first starts destroying:
(1) Erythrocytes
(2) Helper T- Lymphocytes
(3) Thrombocytes
(4) B- Lymphocytes
BHD337

11. A certain patient is suspected to be suffering from Acquired Immuno Deficiency Syndrome. Which diagnostic technique will you recommend for its detection ?
(1) ELISA (2) MRI
(3) Ultra sound (4) WIDAL
BHD338

12. Which of the following cancer is not the type of carcinoma?
(1) Breast cancer (2) Lung cancer
(3) Cervical cancer (4) Bone cancer
BHD339

13. Which statement is false?
(1) Normal cells show a property called contact inhibition
(2) Metastasis is the most feared property of benign tumours
(3) Proto oncogenes when activated under certain conditions, could lead to oncogenic transformation
(4) Early detection of cancer is essential as it allows the disease to be treated successfully in many cases
BHD340

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14. Consider the following characteristics and answer the question.
(a) Metastasis
(b) Contact inhibition
(c) Uncontrolled meiosis
(d) Malignancy
(e) Invasion
(f) Oncogenic transformation
The properties shown by malignant tumor are?
(1) a, b, d, e, f (2) a, d, e, f
(3) a, b, d, e (4) a, b, c, d, e, f
BHD341

15. What is wrong statement about AIDS ?
(1) It is not detected during asymptomatic phase
(2) During AIDS, CD_4 : CD_8 count are 1 : 1
(3) HIV–I is common in India
(4) Two identical ss-RNA, molecules are present in HIV
BHD342

16. Which of the following is not a malignant tumour ?
(1) Adenocarcinoma (2) Osteoma
(3) Angiosarcoma (4) Melanoma
BHD343

17. Which of following gene is not related to normal body growth ?
(1) Protooncogene
(2) Oncogene
(3) Tumour suppresser gene
(4) Gene related to programmed cell death
BHD344

18. Oestrogen works in cancer as :-
(1) Tumour promoters
(2) Tumour marker
(3) Oncogenic transformers
(4) All the above
BHD345

19. Transformation of normal cells into cancerous neoplastic cells may be induced by:
(i) x-rays (ii) Chemical agents
(iii) g-rays (iv) Oncovirus
(v) Smoking (vi) U.V. rays
(1) i, iii, vi (2) i, iii, iv, vi
(3) i, ii, iii, iv, v, vi (4) i, ii, iii, iv, vi
BHD346

20. Which of the following is not the property of cancer cells ?
(1) Very high karyoplasmic index
(2) Number of ribosomes increase
(3) Number of mitochondria increase
(4) Nucleus is always small and regular in shape
BHD347

21. Which of the following virus is sexually transmitted which cause cancer ?
(1) HPV (2) Pox virus
(3) HIV (4) Influenza virus
BHD348

22. **Assertion:** Uncontrolled abnormal excessive mitotic cell division is cancer
Reason : Cancer is caused by X-rays
(1) **A** and **R** are right but **R** is not correct explanation of **A**
(2) **A** is right and **R** is explanation of **A**
(3) **A** is right **R** is wrong
(4) **A** and **R** both are wrong
BHD349

23. Mark the odd one
(1) Tuberculosis (2) Typhoid
(3) AIDS (4) Pneumonia
BHD350

24. STD caused by bacteria is
(1) Syphilis (2) Genital herpes
(3) Trichomoniasis (4) Genital warts
BHD351

25. **Assertion:** Non infected cells can be protected from virus by interferons.
Reason: Non infected cells secrete protein called interferons.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.
BHD352

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26. **Assertion:** Our body elicits highly intensified secondary response for same pathogens.
Reason: Our body contain memory T - cells.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.
BHD353

27. **Assertion:** Number of disease like polio, diphtheria, pneumonia and tetanus have been controlled to large extent
Reason: Vaccines are prepared against these diseases.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.
BHD354

28. **Assertion:** In tetanus preformed antibodies are given to patient.
Reason: Quick immune response is required in tetanus.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.
BHD355

29. **Assertion:** Body is able to reject organ taken from other recipient.
Reason: Our body posses cell mediated immune response responsible for graft rejection.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.
BHD356

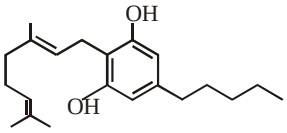
30. **Assertion:** Monoclonal antibodies are designed to attach specific proteins in cancer cells.
Reason: Immunotherapy helps in early detection of cancer.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.
BHD357

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	1	3	3	4	1	3	1	4	2	1	4	2	2	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	2	1	3	4	1	1	3	1	3	1	1	1	1	2

Human Health & Disease # 04

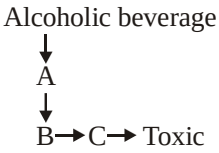
1. Cannabinoid receptors present principally in:
(1) Heart (2) Liver
(3) Brain (4) GIT
BHD358

2. 

is the structural formula of :-
(1) Opioid molecule
(2) Cannabionoid molecule
(3) Morphine
(4) LSD
BHD359

3. Which of the following is not cannabinoid ?
(1) Heroin (2) Hashish
(3) Marijuana (4) Ganja
BHD360

4. Which of the following drug interferes with the transport of the neuro-transmitter dopamine ?
(1) Morphine (2) Smack
(3) Cocaine (4) Marijuana
BHD361

5. 

In the above diagram what is A, B and C respectively ?

	A	B	C
(1)	Stomach	Ethyl alcohol	Acetaldehyde
(2)	Ethyl alcohol	Acetaldehyde	Liver
(3)	Ethyl alcohol	Liver	Acetaldehyde
(4)	Liver	Acetaldehyde	Ethyl alcohol

BHD362

6. Smoke of tobacco contains :-
(1) Carcinogen (2) Nicotine
(3) CO (4) All
BHD363

7. Which of the following combination is not a cannabinoids ?
(1) Charas, Ganja (2) Datura, LSD
(3) Charas, Hashish (4) Marijuana, Bhang
BHD364

8. In malaria, fever and chills occur every 3 to 4 days due to rupture of RBCs and release of which toxic substance?
(1) Haemophilia (2) Hematocrit
(3) Hemozoin (4) Haemoglobin
BHD365

9. Opium is not much clinically used because :-
(1) Less potent drug
(2) Shows physical dependency
(3) Toxicity present in our body
(4) Prolonged used as medicine
BHD366

10. (a) Smack
(b) Diacetylmorphine
(c) White
(d) Odourless
(e) Bitter crystalline compound
Above statements/information are correct for:-
(1) Morphine (2) Heroin
(3) Coccain (4) Barbiturates
BHD367

11. Mark the correctly matched options :-

(A)	Leaves of *Cannabis sativa*	
(B)	Opium poppy	
(C)	Flowing branch of *Datura*	

(1) A and B (2) B and C
(3) A, B and C (4) Only C
BHD368

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12. The period between.....of age may be thought of as adolescence period

(1) 18-21 yrs. (2) 12-21 yrs.

(3) 12-18 yrs. (4) 18-25 yrs.

BHD369

13. Which disorder is not related with smoking?

(a) Lung cancer

(b) Bronchitis

(c) Emphysema

(d) Coronary heart disease

(e) Gastric ulcer

(f) Urinary bladder cancer

(g) Throat cancer

(1) a, b, e, g

(2) a, b, c, f, g

(3) c, d, f

(4) None of these (all are related)

BHD370

14. Withdrawal syndrome is characterised by:-

(a) Anxiety (b) Shakiness

(c) Nausea (d) Sweating

(1) a, c (2) b, c

(3) a, c, d (4) a, b, c, d

BHD371

15. Which measure would be particularly useful for prevention and control of alcohol and drug abuse among adolescents ?

(a) Avoid undue peer pressure

(b) Seeking professional and medical help

(c) Looking for danger sign

(d) Education and counselling

(e) Seeking help from parents and peers

(1) a, b, d (2) a, c, d, e

(3) c, e (4) a, b, c, d, e

BHD372

16. Match the column-I with column-II and select the correct one :-

	Column-I		**Column-II**
(A)	Charas	(i)	*Claviceps purpuria*
(B)	Cocaine	(ii)	Euphoria
(C)	L.S.D.	(iii)	Stimulant
(D)	Amphetamine	(iv)	*Cannabis sativa*

	A	**B**	**C**	**D**
(1)	iv	ii	iii	i
(2)	iv	ii	i	iii
(3)	iv	i	ii	iii
(4)	iii	i	iv	ii

BHD373

17. **Assertion:** AIDS is not a congenital disease.

Reason: In AIDS, its pathogen i.e. HIV attacks on T-helper cells.

(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.

(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.

(3) If the assertion is true but the reason is false.

(4) If both the assertion and reason are false.

BHD374

18. **Assertion:** A suitable donor is required for transplantation.

Reason: CMI response leads to graft rejection.

(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.

(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.

(3) If the assertion is true but the reason is false.

(4) If both the assertion and reason are false.

BHD375

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19. **Assertion:** Interferon are antiviral proteins, secreted by virus infected cells.

Reason: Interferon is given to AIDS patients to treat Kaposi Sarcoma, it acts as Biological Response Modifier.

(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.

(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.

(3) If the assertion is true but the reason is false.

(4) If both the assertion and reason are false.

BHD376

20. **Assertion:** Frequent and repeated intake of drugs leads to addiction.

Reason: Tolerance level of receptors present in body increases in vicious cycle.

(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.

(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.

(3) If the assertion is true but the reason is false.

(4) If both the assertion and reason are false.

BHD377

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	2	1	3	3	4	2	3	2	2	4	3	4	4	4
Question	16	17	18	19	20										
Answer	2	2	1	2	1										

Origin & Evolution # 1

1. Which theory can explain the origin of universe?
(1) Cosmic panspermia theory
(2) Theory of natural selection
(3) Big-bang theory
(4) Theory of special creation

BEV402

2. Who experimentally disproved the theory of spontaneous generation?
(1) Louis Pasteur
(2) Oparin & Haldane
(3) Stanley Miller
(4) Hugo deVries

BEV403

3. The life could have come from:
(1) Outer space with cosmic rays
(2) Dead and decaying matter
(3) Pre-existing non-living organic molecules
(4) Another planet in form of spores

BEV404

4. What is paramount necessity for origin of life?
(1) Light (2) Air
(3) Water (4) Oxygen

BEV405

5. What is the correct sequence of formation of various structures in order of origin of life?
I. Protobionts
II. Simple organic molecules
III. Eobionts
IV. Complex organic molecules
Options:
(1) I, II, III, IV (2) II, I, III, IV
(3) II, IV, I, III (4) III, II, I, IV

BEV406

6. The first living organisms on earth were:
(1) Chemoautotrophs, anaerobic
(2) Chemoheterotrophs, aerobic
(3) Photoautotrophs, aerobic
(4) Chemoheterotrophs, anaerobic

BEV407

7. Experimental evidences in favour of chemical evolution were provided by:
(1) Louis Pasteur (2) Haldane
(3) Stanley Miller (4) Oparin

BEV408

8. **Assertion :** Evolution is not a directed process in the sense of determinism.
Reason : Evolution is a stochastic process based on chance events in nature and chance mutation in the organism.
(1) Both (A) & (R) are correct but (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is not correct.
(3) (A) is not correct but (R) is correct.
(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

BEV409

9. Select the **correct** statement:
(1) Life originated on earth after 4 billion years of origin of earth
(2) Prokaryotic cells are formed after the formation of eukaryotic cells
(3) Water and oxygen are two essential components for the origin of life through chemosynthesis
(4) First non-cellular forms of life would have been giant molecules like RNA, protein, polysaccharides etc.

BEV410

10. Which statement is not true about the theory of big bang?
(1) It can explain the origin of universe
(2) It can explain the origin of life
(3) It can explain the origin of solar system
(4) It can explain the origin of galaxies

BEV411

11. The temperature maintained by Miller in his experiment to prove the chemical evolution:
(1) 25 °C (2) 800 °C
(3) 5000 °C (4) 100 °C

BEV412

12. Which is not a property of protobionts?
(1) Colloidal drops like structures
(2) Able to grow in size
(3) Able to reproduce
(4) Clusters of macromolecules

BEV413

13. Read two statements (a) and (b) carefully and select the correct option regarding them:
(a) The first life on earth was chemoautotrophic and anaerobic.
(b) Protobionts could exhibit simple metabolism and reproduction.

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Options:
(1) Statement (a) is correct but statement (b) is incorrect
(2) Statement (a) is incorrect but statement (b) is correct
(3) Both statements are correct
(4) Both statements are incorrect

BEV414

14. First inorganic and organic molecule formed on earth were respectively :
(1) Water and ammonia
(2) Water and methane
(3) Ammonia and amino acids
(4) Ammonia and carbon dioxide

BEV415

15. **Assertion:** Formation of life was preceded by chemical evolution.
Reason: Inorganic molecules are formed from disintegration of organic constituents initially.
(1) If the assertion is true but the reason is false.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(4) If both the assertion and reason are false.

BEV416

16. Mark out the incorrect statement in context of evolution:
(1) Evolution term was introduced by Herbert Spencer.
(2) Evolution is a slow and gradual process.
(3) Evolution is a reversible process.
(4) Evolution can be explained as 'descent with modification'.

BEV417

17. "When we look at stars we apparently are peeping into the past." Why it is said so? Because:
(1) Stars are very large in size
(2) Stars are very small in size
(3) Light coming from a star takes a long time to reach our eyes
(4) There is no life on stars just as in past of earth

BEV418

18. According to some early Greek philosophers life can transfer from planet to planet in form of a unit. These units are called as:
(1) Spores (2) Sperms
(3) Pangenes (4) Protobiont

BEV419

19. Identify the correct match from column I and column II:

	Column-I		Column-II
A	Origin of universe	(i)	3 bya
B	Origin of earth	(ii)	2 bya
C	Appearance of first life	(iii)	4 bya
D	First non-cellular form of life	(iv)	4.5 bya
E	First cellular form of life	(v)	20 bya

Options:

	A	B	C	D	E
(1)	i	ii	iv	iii	v
(2)	v	ii	i	iv	iii
(3)	ii	i	iii	v	iv
(4)	v	iv	iii	i	ii

BEV420

20. Theory of special creation doesn't believe in:
(1) Changeable fauna and flora
(2) Constant diversity
(3) Unchangeable forms of organisms
(4) 4000 years old earth

BEV421

21. Which fact can best support that the earth is very old?
(1) Changing climate on earth
(2) Fossilized earlier life forms
(3) Elliptical shape of earth
(4) Raising water level of oceans

BEV422

22. The swan neck flask experiment of Louis Pasteur proves that:
(1) Life can be originated from decaying and rotting matter
(2) Life cannot be originated from decaying and rotting matter
(3) RNA was the first genetic material
(4) Life originated from pre existed non living organic molecules

BEV423

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23. The first life on earth was originated by:
(1) Chemical synthesis
(2) Spontaneous generation
(3) Pre existed life
(4) Panspermia

BEV424

24. Identify the correct match from column I and column II:

	Column-I		Column-II
a	First inorganic compound	(i)	Methane
b	First organic compound	(ii)	Protocell
c	First sign of life	(iii)	Water
d	First form of life	(iv)	Nucleoprotein

(1) a-i, b-ii, c-iii, d-iv (2) a-ii, b-iii, c-iv, d-i
(3) a-iii, b-i, c-iv, d-ii (4) a-iv, b-ii, c-iii, d-i

BEV425

25. Identify the correct sequence of changes in modes of nutrition in organism after origin of life on earth:
(a) Non oxygenic photosynthesis
(b) Chemoheterotrophs
(c) Oxygenic photosynthesis
(d) Chemoautorophs
Options:
(1) (b), (a), (c), (d) (2) (d), (b), (c), (a)
(3) (d), (c), (a), (b) (4) (b), (d), (a), (c)

BEV426

26. **Assertion:** Chemical evolution was more or less accepted.
Reason: Miller in his experiment observed the formation of amino acids in similar condition which was their on earth at the time of origin.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.

BEV427

27. Which was not a consequence of liberation of oxygen by cyanobacteria on earth?
(1) Atmosphere became oxidizing
(2) Formation of ozone layer
(3) Aerobic mode of respiration started
(4) Chances of further chemical evolution increased

BEV428

28. Analysis of meteorite contents tells us that:
(1) Life came on earth due to meteorite hits
(2) Similar processes are occurring elsewhere in space that had occurred on earth
(3) All stars originated at same time
(4) Water is essential for origin of life

BEV429

29. Which theory states that life is immutable and has not changed ever since its origin?
(1) Cosmozoic theory
(2) Special creation theory
(3) Biogenesis theory
(4) Abiogenesis theory

BEV430

30. Which of the following conditions are needed for origin of life?
(1) Supply of replicators i.e. self producing molecules
(2) Copying of these replicators must have been subjected to error via mutation
(3) Perpetual supply of energy
(4) All of the above

BEV431

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	1	3	3	3	4	3	1	4	2	2	3	4	2	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	3	1	4	1	2	2	1	3	4	1	4	2	2	4

Origin & Evolution # 2

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1. Which two periods are correctly matched with the dominance of organisms in that particular period?
(1) (A) Jurassic – Amphibians
(B) Carboniferous – Fishes
(2) (A) Jurassic – Amphibians
(B) Devonian – Fishes
(3) (A) Devonian – Amphibians
(B) Jurassic – Dinosaurs
(4) (A) Devonian – Fishes
(B) Jurassic – Dinosaurs

BEV432

2. Which ancestor is shared by birds, crocodiles and dinosaurs?
(1) Thecodonts (2) Synapsids
(3) Pelycosaurs (4) *Archaeopteryx*

BEV433

3. **Assertion:** New life forms have arisen at different times in the history of earth.
Reason: Fossils study showed that life forms varied over time.
(1) If both the assertion and reason are false.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.

BEV434

4. Which option is not an example of analogous organs?
(1) Wings of butterfly and birds
(2) Eyes of octopus and mammals
(3) Potato and ginger
(4) Potato and sweet potato

BEV435

5. The most common fossils found in earth's crust are:
(1) Petrified fossils (2) Mould fossils
(3) Unaltered fossils (4) Coprolites

BEV436

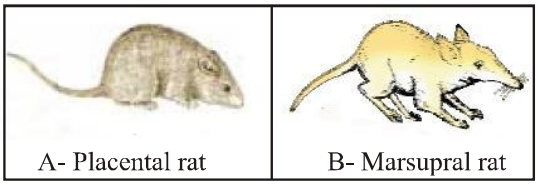
6. **Assertion:** The aquatic mammals like Dolphins, whales etc, do not contain gill slits.
Reason: their adaptation to aquatic life is secondary .
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.

BEV437

7. **Assertion:** Sweet potato and potato is an example for analogy.
Reason: They have different origin but perform same function that is storage of food.
(1) If both the assertion and reason are false.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.

BEV438

8. The organism A and B are example of:

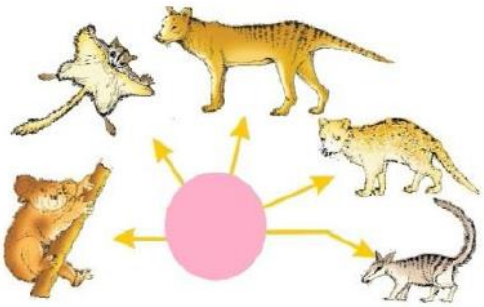


(1) Convergent evolution
(2) Divergent evolution
(3) Adaptive convergence
(4) Both (1) & (3)

BEV439

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9. What is not true about the organisms given below in the diagram?



(1) They all are example of convergent evolution
(2) They all are example of adaptive radiation
(3) They have evolved from a common ancestor
(4) They all are Australian marsupials

BEV440

10. Some statements are given for wings of bat and butterfly:
(a) They are anatomically similar
(b) They are homologous structure
(c) They are result of divergent evolution
(d) They perform different functions
(e) They are example of adaptive divergence
(f) They indicates common ancestry
How many statements are correct?
(1) Zero (2) One
(3) Two (4) Three

BEV441

11. Pouched mammals survived in Australia but not in other continents, because:
(1) Climate of Australia was cold
(2) Pouched mammals have high adaptive power
(3) They lack competition from other mammals
(4) They have high reproductive capacity

BEV442

12. When organism of one ancestral stock shows similarities with the organism of another ancestral stock based on the habitat and feeding habit, than this phenomenon can be called as:
(1) Convergent evolution
(2) Divergent evolution
(3) Adaptive divergence
(4) Adaptive radiation

BEV443

13. Some examples are given below:
(a) Forelimbs of mammals
(b) Thorn of Bougainvillea and tendril of cucurbita
(c) Eyes of the octopus and and mammals
(d) Flippers of penguins and dolphins
(e) Sweet potato and potato
(f) Wings of birds and butterfly
How many examples among these are the examples of analogous structures in animals?
Options:
(1) One (2) Two
(3) Three (4) Four

BEV444

14. Which is correctly matched?
(1) 500 mya – Invertebrates formed
(2) 320 mya – Jawless fishes evolved
(3) 3000 mya – First cellular form of life appeared
(4) 65 mya – Dinosaurs appeared

BEV445

15. **Assertion :** Comparative biochemistry provides strong evidence in favour of common ancestry of living beings.
Reason: Genetic code in universal.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If the assertion is true but the reason is false.
(3) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(4) If both the assertion and reason are false.

BEV446

16. What is the correct sequence of origin of mammals?
(1) Sauropsids → Pelycosaurs → Therapsids → Mammals
(2) Synapsids → Pelycosaurs → Therapsids → Mammals
(3) Sauropsids → Synapsids → Thecodonts → Mammals
(4) Synapsids → Sauropsids → Therapsids → Mammals

BEV447

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17. The similarities in composition and structure of protoplasm, enzymes, hormones, DNA, blood etc indicates that:
(1) Organisms did not share ancestors in recent past
(2) Organisms did not share ancestors in distant past
(3) Organisms have shared ancestors in recent or distant past
(4) Organisms never shared ancestors

BEV448

18. **Assertion:** Homologous organ represent divergent evolution.
Reason: Homology indicate common ancestry.
(1) If both the assertion and the reason are true and the reason is not a correct explanation of the assertion.
(2)If both the assertion and reason are true but the reason is a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.

BEV449

19. What is parallelism?
(1) Adaptive divergence
(2) Adaptive divergence of widely separated species
(3) Adaptive convergence in widely separated species
(4) Adaptive convergence in closely related species

BEV450

20. Evolution of different mammals residing in different habitats like air, land and water is an example of:
(1) Adaptive radiation
(2) Convergent evolution
(3) Micro evolution
(4) Adaptive convergence

BEV451

21. **Assertion :** The first cells used RNA as their hereditary molecule.
Reason : DNA evolved from RNA.
(1) Both (A) & (R) are correct but (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is not correct.
(3) (A) is not correct but (R) is correct.
(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

BEV452

22. The recapitulation theory means that:
(1) All organisms start as an egg
(2) Life history of an organism reflects its evolutionary history
(3) Non living matters form life
(4) Progeny resembles parents

BEV453

23. Which one is flying reptiles among the following?
(1) *Archaeopteryx*
(2) *Pteranodon*
(3) *Stegosaurus*
(4) *Tyrannosaurus*

BEV454

24. Before how many years dinosaurs suddenly disappeared from earth?
(1) 320 mya (2) 350 mya
(3) 200 mya (4) 65 mya

BEV455

25. **Assertion :** Darwin finches are classical example of natural selection.
Reason: Darwin explains that beak sizes and shape in finches varied according to food habit of different finches.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.

BEV456

26. Which pair of mammals is not an example of convergent evolution?
(1) Mole - Marsupial mole
(2) Anteater - Numbat
(3) Lemur - Spotted cuscus
(4) Bobcat - Wolf

BEV457

27. Identify the correct match from column I and column II:

Column I		Column II	
(a)	Devonian period	(i)	Golden age of amphibians
(b)	Carboniferous period	(ii)	Age of birds and mammals
(c)	Jurassic period	(iii)	Golden age of fishes
(d)	Coenozoic era	(iv)	Golden age of dinosaurs

Options:
(1) a-i, b-ii, c-iii, d-iv
(2) a-ii, b-iii, c-iv, d-i
(3) a-iii, b-i, c-iv, d-ii
(4) a-iv, b-ii, c-iii, d-i

BEV458

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	1	4	3	1	1	4	4	1	1	3	1	3	1	3
Question	16	17	18	19	20	21	22	23	24	25	26	27			
Answer	2	3	2	4	1	1	2	2	4	1	4	3			

Origin & Evolution # 3

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1. Only two statements are correct among the given five statements. Identify them:
(A) Fitness is the end result of ability to adapt and get selected by nature.
(B) Members of a population don't vary in characteristics.
(C) Mutation is a continuous source of variations.
(D) Evolution is a single step process called saltation.
(E) Evolution is not a directed process in the sense of determinism.
Options:
(1) A, E (2) A, B (3) C, D (4) D, E

BEV459

2. Which conclusion can be made on the basis of experiment of Lederberg?
(1) Already existed adaptive mutations provide selection advantage to the individuals of a population
(2) Already existed harmful mutations provide selection advantage to the individuals of a population
(3) Individuals of a population develop new characters in response to changed environment which helps in their selection
(4) Any change in environment doesn't affect the individuals of a population

BEV460

3. The common and continuous source of variations in a population is :
(1) gene recombination (2) mutation
(3) genetic drift (4) gene flow

BEV461

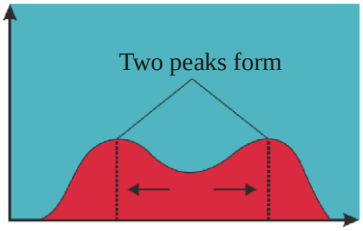
4. Which is not true for directional selection?
(1) Only one extreme value is selected
(2) Mean value never change
(3) Operates in changing environment
(4) Peak shifts in one direction

BEV462

5. Weismann's theory of continuity of germplasm disproves the theory of:
(1) Theory of Lamarck
(2) Theory of biogenesis
(3) Theory of Darwin
(4) Theory of mutation

BEV463

6. Which type of natural selection is shown in given graphical representation?



(1) Stabilizing selection
(2) Directional selection
(3) Disruptive selection
(4) Artificial selection

BEV464

7. Read four statements (a-d) carefully:
(a) Natural resources are unlimited.
(b) Populations are stable in size except for seasonal fluctuations.
(c) Members of a population vary in characteristics.
(d) There is no competition for resources among the individuals of a population.
How many statements are correct among these?
(1) One (2) Two (3) Three (4) Four

BEV465

8. Mutations are ___(A)___ and ___(B)___ while Darwinian variations are ___(C)___ and ___(D)___. Fill in the blanks for A, B, C and D from the options given in the table below:

	A	B	C	D
(1)	Small	Directional	Random	Directionless
(2)	Directional	Large	Small	Directionless
(3)	Random	Directionless	Small	Directional
(4)	Large	Directional	Small	Directional

BEV466

9. **Assertion:** Herbicide resistant varieties can be selected in lesser time scale.
Reason: It can be possible by anthropogenic action that is excessive use of herbicides
(1) If both the assertion and the reason are true and the reason is not a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is a correct explanation of the assertion.

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(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.

BEV467

10. Production of improved varieties of domesticated animals is an example of:
(1) Natural selection
(2) Artificial selection
(3) Sexual selection
(4) All of these

BEV468

11. The observations about the number of white winged and dark winged moths in England during different environmental conditions shows that:
(1) Predators can't spot a moth against contrasting background.
(2) No variant is completely wiped out.
(3) The areas where industrialization did not occur, the count of melanic moths was high.
(4) Moths that were able to camouflage themselves, didn't survive.

BEV469

12. The source of variations in an asexually reproducing population:
(1) Mutation
(2) Gene recombination
(3) Both (1) & (2)
(4) Hybridization

BEV470

13. The different varieties of beaks in black birds observed by Darwin were present in:
(1) Galapagos island (2) Australia
(3) England (4) Africa

BEV471

14. When a species splits into two or more geographically isolated populations and these populations finally form new species, then it is called as:
(1) Sympatric speciation
(2) Allopatric speciation
(3) Phyletic speciation
(4) Quantum speciation

BEV472

15. Presence of long neck and high forelimbs in giraffe can be best explained by:
(1) Lamarckism (2) Neodarwinism
(3) Neolamarckism (4) Bottleneck effect

BEV473

16. Identify the correct match from column I and column II:

Column I		Column II	
A	Principles of population	(i)	Darwin
B	Principles of geology	(ii)	Lamarck
C	Genetics & Origin of species	(iii)	Malthus
D	Origin of species	(iv)	Stebbins
E	Origin of life	(v)	Dobzhansky
F	Philosophie zoologique	(vi)	Oparin
G	Process of organic evolution	(vii)	Charles Lyell

Options:

	A	B	C	D	E	F	G
(1)	i	ii	iv	iii	v	vii	vi
(2)	ii	iii	i	iv	vi	v	vii
(3)	iii	vii	v	i	vi	ii	iv
(4)	vii	iii	v	i	vi	ii	iv

BEV474

17. According to Hugo DeVries what causes evolution?
(1) Small and continuous variations
(2) Small and discontinuous variations
(3) Large and discontinuous variations
(4) large and continuous variations

BEV475

18. The change in gene frequency occurred in small population by chance is called as:
(1) Mutation (2) Genetic drift
(3) Gene migration (4) Gene flow

BEV476

19. **Assertion:** Single step large mutation is called saltation.
Reason: Mutation is responsible for speciation.
(1) If both the assertion and the reason are true and the reason is not a correct explanation of the assertion.
(2) If both the assertion and reason are true but the reason is a correct explanation of the assertion.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are false.

BEV477

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20. Which of the following can contribute to genetic drift?
(1) Small population size
(2) Bottleneck effect
(3) Founder effect
(4) All of the above

BEV478

21. Balancing selection is connected with the successful reproduction of:
(1) Homozygous recessives
(2) Homozygous dominants
(3) Heterozygous individuals
(4) All of the above

BEV479

22. Which phenomenon is essential for the formation of new species along with natural selection ?
(1) Mutation
(2) Reproductive isolation
(3) Genetic drift
(4) Gene flow

BEV480

23. Which is not true for genetic drift?
(1) It is the change in gene frequency by chance
(2) It operates rapidly in large population
(3) It can happened due to habitat fragmentation
(4) It is also known as Sewall-Wright effect

BEV481

24. Unit of natural selection is:
(1) Individual (2) Population
(3) Species (4) Gene

BEV482

25. Unit of evolution is:
(1) Population (2) Gene
(3) Species (4) Individual

BEV483

26. A change in medium composition in a bacterial culture, results into growth of a specific section of that population. This section survived because:
(1) These bacteria can convert themselves according to changing medium
(2) These bacteria already had the gene that provides them selective advantage in changing medium
(3) These bacteria survived by chance
(4) Exact reason can't be predicted

BEV484

27. Lederberg's replica plate experiment proves that:
(1) Pre-existed advantageous mutations provide the selective advantage to individuals in the changed environment
(2) Mutations appear after change in environment and provide selective advantage
(3) Both conclusions are correct
(4) Change in environment doesn't affect the populations

BEV485

28. The improvement in varieties of animals and plants for human advantage is a process of:
(1) Natural selection
(2) Artificial selection
(3) Sexual selection
(4) Stabilizing selection

BEV486

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	1	1	2	1	3	2	3	2	2	2	1	1	2	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28		
Answer	3	3	2	1	4	3	2	2	1	1	2	1	2		

Origin & Evolution # 4

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1. The only true statement about the prehistoric man *Homo habilis* is:
(1) It was the first man who used fire
(2) It was the first man who made tools
(3) It was the first man who painted cave walls
(4) It was the first man who used hides to protect body

BEV487

2. Identify the correct match from column – I, II and III :

	Prehistoric man		Cranial capacity		Prehistoric man
A	Java man	i	1400 cc	a	China
B	Peking man	ii	1650 cc	b	Java
C	Neanderthal man	iii	1050 cc	c	Germany
D	Cro-Magnon man	iv	900 cc	D	France

Options:
(1) A-i-b, B-ii-c, C-iii-d, D-iv-a
(2) A-ii-c, B-iii-d, C-iv-a, D-i-b
(3) A-i-d, B-ii-a, C-iii-b, D-iv-c
(4) A-iv-b, B-iii-a, C-i-c, D-ii-d

BEV488

3. About how many years ago *Dryopithecus* and *Ramapithecus* were existed?
(1) 15 million years ago
(2) 50,000 years ago
(3) 1,40,000 years ago
(4) 18,000 years ago

BEV489

4. Which is not correctly matched among the following options?
(1) *Homo habilis* – Brain capacity 650 – 800 cc
(2) *Homo erectus* – Fossils discovered in Java
(3) Neanderthal man – First human being like creature
(4) *Homo sapiens* – Arose in Africa

BEV490

5. The first man who used hides to protect their body and buried their dead:
(1) *Homo sapiens* (2) *Neanderthal man*
(3) *Homo erectus* (4) *Homo habilis*

BEV491

6. Pre historic cave arts were developed by:
(1) *Homo erectus*
(2) *Homo sapiens*
(3) *Australopithecus*
(4) *Homo habilis*

BEV492

7. Which statement is not true about Australopithecus?
(1) They lived approx 2 million years ago
(2) They probably lived in east African grasslands
(3) They hunted with stones
(4) They didn't eat fruit

BEV493

8. Who is the direct ancestor of modern man?
(1) Cro-Magnon man (2) Java man
(3) Peking man (4) Neanderthal man

BEV494

9. Which is not correctly matched?
(1) Modern man – First man who started agriculture
(2) Neanderthal man – First man who started to buried dead bodies
(3) Peking man – First man who started use of fire
(4) Cro-Magnon man – First man who painted walls of caves

BEV495

10. Which statement is not true about *Ramapithecus* and *Dryopithecus*?
(1) They existed about 15 million years ago
(2) They were hairy and walked like chimpanzee and gorilla
(3) *Ramapithecus* was more man like
(4) *Dryopithecus* was more man like

BEV496

11. The correct sequence of evolution of man:
I. *Homo habilis* II. Java Man
III. Modern Man IV. Cro-magnon man
Options:
(1) II. I, III, IV (2) II, I, IV, III
(3) I, II, III, IV (4) I, II, IV, III

BEV497

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12. **Assertion:** Homo erectus probably ate meat.
Reason: Homo erectus had a large brain around 900 cc.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are false
(3) If the assertion is true but the reason is false.
(4) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.

BEV498

13. **Assertion:** Agriculture came around 18,000 years ago.
Reason: Pre-historic cave art developed about 10,000 years ago.
(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(2) If both the assertion and reason are false.
(3) If the assertion is true but the reason is false.
(4) If both the assertion and the reason are true and the reason is not a correct explanation of the assertion.

BEV499

14. In which of the following the cranial capacity is minimum?
(1) *Australopithecus*
(2) Cro-Magnon man
(3) Peking man
(4) Gibbon

BEV500

15. Identify the correct match from column I and column II:

	Column I		Column II
(a)	*Australopithecus*	(i)	800-1000 cc
(b)	*Homo habilis*	(ii)	1400-1600 cc
(c)	*Homo erectus*	(iii)	650-800 cc
(d)	*Homo sapiens*	(iv)	600 cc

Options:
(1) a-i, b-ii, c-iii, d-iv
(2) a-ii, b-iii, c-iv, d-i
(3) a-iii, b-i, c-iv, d-ii
(4) a-iv, b-iii, c-i, d-ii

BEV501

16. Which is not correctly matched?
(1) *Australopithecus* - Carnivorous
(2) *Homo habilis* - Tool maker man
(3) Cro - Magnon man - Cave paintings
(4) Modern man - Agriculture

BEV502

17. Some characters of pre-historic hominid species are given below:
(i) Brain capacity 900 cc
(ii) Lived near east and central Asia
(iii) First human being like
(iv) Used hides to protect their bodies
(v) Buried their dead
(vi) Existed around 10000 years back
Which characters are shared by Neanderthal man among these?
(1) (i), (ii), (iv), (vi)
(2) (ii), (iv), (v)
(3) (i), (ii), (iv), (v)
(4) (i), (ii), (iii), (iv), (v)

BEV503

18. Read four statements about hominid species:
(a) Agriculture and human settlements started about 18000 years ago.
(b) Cave art developed approximately 10000 years ago.
(c) *Homo sapiens* arose in Africa.
(d) *Homo erectus* existed about 1.5 million years ago.
how many statements are correct among these?
(1) One (2) Two
(3) Three (4) Four

BEV504

19. Select the incorrect match:
(1) Java man - first man who stood erect
(2) Cro-Magnon man - man who started domestication of animals
(3) Neanderthal man - man who believed in immortality of soul
(4) *Homo habilis* - first tool maker man

BEV505

20. Which is a character of human?
(1) Longer forelimbs than hind limbs
(2) 'U' shaped jaw
(3) Parallel thumb to palm
(4) Opposable thumb

BEV506

21. Who is considered as the connecting link between apes and man?
(1) *Dryopithecus*
(2) *Ramapithecus*
(3) *Australopithecus*
(4) *Shivapithecus*

BEV507

22. DNA sequences in many human genes are very similar to the sequences of corresponding genes in chimpanzees. The most likely explanation for this result is that:
(1) Humans and chimpanzees share a relatively recent common ancestor.
(2) Humans evolved from chimpanzees.
(3) Chimpanzees evolved from humans.
(4) Humans and chimpanzees are not closely related.

BEV508

23. **Assertion:** Humans and apes have a common ancestry.
Reason: Man and chimpanzee have similar banding of pattern of chromosome.
(1) If both the assertion and reason are false.
(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.
(3) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.
(4) If the assertion is true but the reason is false.

BEV509

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	4	1	3	2	2	4	1	3	4	4	4	2	4	4
Question	16	17	18	19	20	21	22	23							
Answer	1	2	2	1	4	3	1	3							

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